Combinatorics

Course Notes – MATH 502

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Preface

Counting is innate to man.

History of India
Abū Rayān Muammad ibn Amad Al-Bīrūnī

This are lecture notes I prepared for a graduate Combinatorics course which I taught in Fall 2016 at Colorado State University.

They are based extensively on

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- Peter Cameron, J.H. van Lint, Designs, Graphs, Codes and their Links [CvL91]
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- J.H. van Lint, W.M. Wilson, A Course in Combinatorics [vLW01]
- Marshall Hall, Combinatorial Theory [Hal86]
- Ronald Graham, Donald Knuth, Oren Patashnik, Concrete Mathematics [GKP94]
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Chapter

I Basic Counting

One of the basic tasks of combinatorics is to determine the cardinality of (finite) classes of objects. Beyond basic applicability of such a number – for example to estimate probabilities – the actual process of counting may be of interest, as it gives further insight into the problem:

- If we cannot count a class of objects, we cannot claim that we know it.
- The process of enumeration might – for example by giving a bijection between classes of different sets – uncover a relation between different classes of objects.
- The process of counting might lend itself to become constructive, that is allow an actual construction of (or iteration through) all objects in the class.

The class of objects might have a clear mathematical description – e.g. all subsets of the numbers $\{1, \ldots, 5\}$. In other situations the description itself needs to be translated into proper mathematical language:

Definition I.1 (Derangements): Given n letters and n addressed envelopes, a *derangement* is an assignment of letters to envelopes that no letter is in the correct envelope. How many derangement of n exist?

This particular problem will be solved in section II.5

We will start in this chapter by considering the enumeration of some basic constructs – sets and sequences. More interesting problems, such as the derangements here, arise later if further conditions restrict to sub-classes, or if obvious descriptions could have multiple sequences describe the same object.

I.1 Basic Counting of Sequences and Sets

I am the sea of permutation.
I live beyond interpretation.
I scramble all the names and the combinations.
I penetrate the walls of explanation.

Lay My Love
Brian Eno

A sequence of length k is simply an element of the k-fold cartesian product. Entries are chosen independently, that is if the first entry has a choices and the second entry b, there are $a \cdot b$ possible choices for a length two sequence. Thus, if we consider sequences of length k, entries chosen from a set A of cardinality n, there are n^k such sequences.

This allows for duplication of entries, but in some cases – arranging objects in sequence – this is not desired. In this case we can still choose n entries in the first position, but in the second position need to avoid the entry already chosen in the first position, giving $n-1$ options. (The number of options is always the same, the actual set of options of course depends on the choice in the first position.) The number of sequences of length k thus is $(n)_k = n \cdot (n-1) \cdot (n-2) \cdot \cdots \cdot (n-k+1) = \frac{n!}{(n-k)!}$, called[1] "*n lower factorial k*".

This could be continued up to a sequence of length n (after which all n element choices have been exhausted). Such a sequence is called a *permutation* of A, there are $n! = n \cdot (n-1) \cdot \cdots \cdot 2 \cdot 1$ such permutations.

Next we consider sets. While every duplicate-free sequence describes a set, sequences that have the same elements arranged in different order describe the same set. Every set of k elements from n thus will be described by $k!$ different duplicate-free sequences. To enumerate sets, we therefore need to divide by this factor, and get for the number of k-element sets from n the count given by the *binomial coefficient*

$$\binom{n}{k} = \frac{(n)_k}{k!} = \frac{n!}{(n-k)!k!}$$

Note (using a convention of $0! = 1$) we get that $\binom{n}{0} = \binom{n}{n} = 1$. It also can be convenient to define $\binom{n}{k} = 0$ for $k < 0$ or $k > a$.

Binomial coefficients also allow us to count the number of *compositions* of a number n into k parts, that is to write n as a sum of exactly k positive integers with ordering being relevant. For example $4 = 2 + 2 = 1 + 3 + 3 + 1$ are the 3 possible compositions of 4 into 2 parts.

To get a formula of the number of possibilities, write the maximum composition

$$n = 1 + 1 + 1 + \cdots + 1$$

[1] Warning: The notation $(n)_k$ has different meaning in other areas of mathematics!

which has $n-1$ plus-signs. We obtain the possible compositions into k parts by grouping summands together to only have k summands. That is, we designate $k-1$ plus signs from the given $n-1$ possible ones as giving us the separation. The number of possibilities thus is $\binom{n-1}{k-1}$.

If we also want to allow summands of 0 when writing n as a sum of k terms, we can simply assume that we temporarily add 1 to each summand. This guarantees that each summand is positive, but adds k to the sum. We thus count the number of ways to express $n+k$ as a sum of k summands which is by the previous formula $\binom{n+k-1}{k-1} = \binom{n+k-1}{n}$.

Example: Check this for $n = 4$ and $k = 2$.

This count is particular relevant as it also counts multisets, that is sets of n elements from k, in which we allow the same element to appear multiple times. Each such set is described as a sum of k terms expressing n, the i-th term indicating how often the i-th element is in the set.

Swapping the role of n and k, We denote by $\left(\binom{n}{k}\right) = \binom{n+k-1}{k}$ the number of k-element multisets chosen from n possibilities.

The results of the previous paragraphs are summarized in the following theorem:

THEOREM I.2: The number of ways to select k objects from a set of n is given by the following table:

	Repetition	No Repetition
Order significant (sequences)	n^k	$(n)_k$
Order not significant (sets)	$\binom{n+k-1}{k} = \left(\binom{n}{k}\right)$	$\binom{n}{k}$

NOTE I.3: Instead of using the word *significant* some books talk about *ordered* or *unordered* sequences. I find this confusing, as the use of ordered is opposite that of the word *sorted* which has the same meaning in common language. We therefore use the language of *significant*.

I.2 Bijections and Double Counting

> As long as there is no double counting, Section 3(a) adopts the principle of the recent cases allowing recovery of both a complainants actual losses and a misappropriator's unjust benefit…
>
> Draft of Uniform Trade Secrets Act
> AMERICAN BAR ASSOCIATION

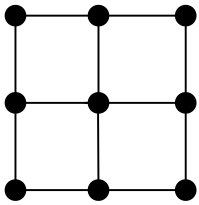


Figure I.1: A 3×3 grid

In this section we consider two further important counting principles that can be used to build on the basic constructs.

Instead of counting a set A of objects directly, it might be easiest to establish a bijection to another set B, that is a function $f\colon A \to B$ which is one-to-one (also called *injective*) and onto (also called *surjective*). Once such a function has been established we know that $|A| = |B|$ and if we know $|B|$ we thus have counted $|A|$.

As an example, consider the following problem: We have an $n \times n$ grid of points with horizontal and vertical connections (depicted in figure I.1 for 3×3) and want to count the number of different paths from the bottom left, to the top right corner, that only go right or up.

Each such path thus has exactly $n-1$ right steps, and $n-1$ up steps. We thus (this already could be considered as one bijection) could count instead 0/1 sequences (0 is right, 1 is up) of length $2n-2$ that contain exactly $n-1$ ones (and zeros). Denote the set of such sequences by A.

To determine $|A|$, we observe that each sequence is determined uniquely by the *positions* of the ones and there are exactly $n-1$ of them. Thus let B the set of all $n-1$ element subsets of $\{1, \ldots, 2n-2\}$.

We define $f\colon A \to B$ to assign to a sequence the positions of the ones, $f\colon a_1, \ldots, a_{2n-2} \mapsto \{i \mid a_i = 1\}$.

As every sequence has exactly $n-1$ ones, indeed f goes from A to B. As the positions of the ones define the sequence, f is injective. And as we clearly can construct a sequence which has ones in exactly $n-1$ given positions, f is surjective as well. Thus f is a bijection.

We know that $|B| = \binom{2n-2}{n-1}$, this is also the cardinality of A.

Example: Check this for $n = 2$, $n = 3$.

The second useful technique is to "double counting", counting the same set in two different ways (which is a bijection from a set to itself). Both counts must give the same result, which can often give rise to interesting identities. The following lemma is an example of this paradigm:

Lemma I.4 (Handshaking Lemma): At a convention (where not everyone greets everyone else but no pair greets twice), the number of delegates who shake hands an odd number of times is even.

Proof: we assume without loss of generality that the delegates are given by $\{1, \ldots, n\}$. Consider the set of handshakes

$$S = \{(i, j) \mid i \text{ and } j \text{ shake hands.}\}.$$

We know that if (i, j) is in S, so is (j, i). This means that $|S|$ is even, $|S| = 2y$ where y is the total number of handshakes occurring.

On the other hand, let x_i be the number of pairs with i in the first position. We thus get $\sum x_i = |S| = 2y$. If a sum of numbers is even, there must be an even number of odd summands.

But x_i is also the number of times that i shakes hands, proving the result. □

> About binomial theorem I'm teeming with a lot o' news,
> With many cheerful facts about the square of the hypotenuse.
>
> The Pirates of Penzance
> W.S. Gilbert

The combinatorial interpretation of binomial coefficients and double counting allows us to easily prove some identities for binomial coefficients (which typically are proven by induction in undergraduate classes):

Proposition I.5: Let n, k be nonnegative integers with $k \leq n$. Then:

a) $\binom{n}{k} = \binom{n}{n-k}$.

b) $k\binom{n}{k} = n\binom{n-1}{k-1}$.

c) $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$ (Pascal's triangle).

d) $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.

e) $\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$.

f) $(1+t)^n = \sum_{k=0}^{n} \binom{n}{k} t^k$ (Binomial Theorem).

Proof: a) Instead of selecting a subset of k elements we could select the $n-k$ elements not in the set.
b) Suppose we want to count committees of k people (out of n) with a designated chair. We can do so by either choosing first the $\binom{n}{k}$ committees and then for each

team the k possible chairs out of the committee members. Or we choose first the n possible chairs and then the remaining $k-1$ committee members out of the $n-1$ remaining persons.
c) Suppose that I am part of a group that contains $n+1$ persons and we want to determine subsets of this group that contain k people. These either include me (and $k-1$ further persons from the n others), or do not include me and thus k from the n other people.
d) We count the total number of subsets of a set with n elements. Each subset can be described by a 0/1 sequence of length n, indicating whether the i-th element is in the set.
e) Suppose we have n men and n women and we want to select groups of n persons. This is the right hand side. The left hand side enumerates separately the options with exactly k men, which is $\binom{n}{k}\binom{n}{n-k} = \binom{n}{k}^2$ by a).
f) Clearly $(1+t)^n$ is a polynomial of degree n. The coefficient for t^k gives the number of possibilities to choose the t-summand when multiplying out the product

$$(1+t)(1+t)\cdot\cdots\cdot(1+t)$$

of n factors so that there are k such summands overall. This is simply the number of k-subsets, $\binom{n}{k}$. □

Example: Prove the theorem using induction. Compare the effort. Which method gives you more insight?

I.3 Stirling's Estimate

Since the factorial function is somewhat unhandy in estimates, it can be useful to have an approximation in terms of elementary functions. The most prominent of such estimates if probably given by Stirling's formula:

Theorem I.6:

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \left(1 + O\left(\frac{1}{n}\right)\right). \tag{I.7}$$

Here the factor $1 + O(1/n)$ means that the quotient of estimate and real value is bounded by $1 \pm 1/n$. Figure I.2 shows a plot of the ratio of the estimate to $n!$.
Proof: Consider the natural[2] logarithm of the factorial:

$$\log(n!) = \log 1 + \log 2 + \cdots + \log n$$

If we define a step function $L(x) = \log(\lfloor x + 1/2 \rfloor)$, we thus have that $\log(n!)$ is an integral of $L(x)$. We also know that

$$I(x) = \int_0^x \log(t)dt = x\log x - x.$$

[2] all logarithms in this book are natural, unless stated differently

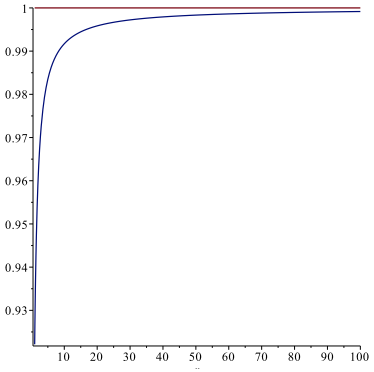


Figure I.2: Plot of $\sqrt{2\pi n}\left(\frac{n}{e}\right)^n/n!$

We thus need to consider the integral over $L(x) - \log(x)$. To avoid divergence issues, we consider this in two parts. Let

$$a_k = \frac{1}{2}\log k - \int_{k-\frac{1}{2}}^{k} \log x dx = \int_{k-\frac{1}{2}}^{k} \log(k/x)dx,$$

and

$$b_k = \int_{k}^{k+\frac{1}{2}} \log x dx - \frac{1}{2} logk = \int_{k}^{l+\frac{1}{2}} \log(x/k)dx.$$

Then

$$S_n = a_1 - b_1 + a_2 - b_2 + \cdots + a_n = \log n! - \frac{1}{2}\log n + I(n) - I(\frac{1}{2}).$$

A substitution gives

$$a_k = \int_0^{\frac{1}{2}} \log \frac{1}{1-(t/k)} dt, b_k = \int_0^{\frac{1}{2}} \log(1+t/k)dt,$$

from which we see that $a_k > b_k > a_{k+1} > 0$. By the Leibniz' criterion thus S_n converges to a value S and we have that

$$\log n! - (n+\frac{1}{2})\log n + n \to S - I(\frac{1}{2}).$$

Taking the exponential function we get that

$$n! \to e^C \sqrt{n}\left(\frac{n}{e}\right)^n$$

with $C = \sum_{k=1}^{\infty}(a_k - b_k) - I(\frac{1}{2})$.

Using standard analysis techniques, one can show that $e^C = \sqrt{2\pi}$, see [Fel67]. Alternatively for concrete calculations, we could simply approximate the value to any accuracy desired. □

The appearance of e might seem surprising, but the following example shows that it arises naturally in this context:

PROPOSITION I.8: The number s_n of all sequences (of arbitrary length) without repetition that can be formed from n objects is $\lfloor e \cdot n! \rfloor$.

Proof: If we has a sequence of length k, there are $\binom{n}{k}$ choices of objects each in $k!$ possible arrangements, thus $k!\binom{n}{k} = \frac{n!}{(n-k)!}$ such sequences. Summing over all values of $n - k$ we get

$$s_n = \sum_{k=0}^{n} \frac{n!}{k!} = n! \sum_{k=0}^{n} \frac{1}{k!}.$$

Using the Taylor series for e^x we see that

$$\begin{aligned} e \cdot n! - s_n &= \frac{1}{n+1} + \frac{1}{(n+1)(n+2)} + \cdots \\ &< \frac{1}{n+1} + \frac{1}{(n+1)^2} + \cdots \\ &= \frac{1}{n} < 1. \end{aligned}$$

□

This is an example (if we ignore language meaning) of the popular problem how many words could be formed from the letters of a given word, *if no letters are duplicate*.

Example: If we allow duplicate letters the situation gets harder. For example, consider words (ignoring meaning) made from the letters of the word COLORADO. The letter O occurs thrice, the other five letters only once. If a word contains at most one O, the formula from above gives $\sum_{k=0}^{6} \frac{6!}{k!} = 720+720+360+120+30+5+1 = 1957$ such words.

If the word contains two O, and k other letters there are $\binom{5}{k}$ options to select these letters and $(k+2)!/2$ possibility to arrange the letters (the denominator 2 making up for the fact that both O cannot be distinguished). Thus we get

$$\frac{2!}{2} + \frac{5 \cdot 3!}{2} + \frac{10 \cdot 4!}{2} + \frac{10 \cdot 5!}{2} + \frac{5 \cdot 6!}{2} + \frac{7!}{2} = 1 + 15 + 120 + 600 + 1800 + 2520 = 5056$$

such words.

If the word contains three O, and k other letters we get a similar formula, but with a cofactor $(k+3)!/6$, that is

$$\frac{3!}{6} + \frac{5 \cdot 4!}{6} + \frac{10 \cdot 5!}{6} + \frac{10 \cdot 6!}{6} + \frac{5 \cdot 7!}{6} + \frac{8!}{6} = 1 + 20 + 200 + 1200 + 4200 + 6720 = 12341$$

possibilities, summing up to 19354 possibilities in total.

Lucky we do not live in MISSISSIPPI!

I.4 The Twelvefold Way

A generalization of counting sets and sequences is given by considering functions between finite sets. We shall consider functions $f\colon N \to X$ with $|N| = n$ and $|X| = x$. These functions could be arbitrary, injective, or surjective. Furthermore, we could consider the elements of N and of X as distinguishable or as indistinguishable.

Note that "indistinguishable" does not mean equal. It just means that if we exchange the role of two elements the result is considered the same.

What does indistinguishable mean formally: In this case we actually count equivalence classes of functions. Two functions $f, g\colon N \to X$ are N-equivalent if there is a bijection $u\colon N \to N$ such that $f(u(a)) = g(a)$ for all $a \in N$. If we say the elements of N are indistinguishable, we are counting functions up to this N-equivalence.

Similarly, we define an X-equivalence of f and g if there is $v\colon X \to X$ such that $v(f(a)) = g(a)$ for all $a \in N$. If we say the elements of X are indistinguishable, we count functions up to X-equivalence.

We can combine both equivalences to get even larger equivalence classes, which are the case of elements of both N and X being indistinguishable.

(The reader might feel this to be insufficiently stringent, or wonder about the case of different classes of equivalent objects. We will treat such situations in Section VI.6 under the framework of group actions.)

This gives in total $3 \cdot 2 \cdot 2 = 12$ possible classes of functions. This set of counting problems is called the *Twelvefold Way* in [Sta12, Section 1.9].

Example: For each of the 12 categories, give an example of a concrete counting problem in common language.

Determining explicit formulae for the different counting problems is not even easy, indeed we will be able to do so after substantial further work in the following chapters.

Some further counting tasks

A *partition* of a set A is a collection $\{A_i\}$ of subsets (called *parts* or *cells*) $\varnothing \neq A_i \subset A$ such that for all i:

- $\bigcup_i A_i = A$
- $A_i \cap A_j = \varnothing$ for $j \neq i$.

Note that a partition of a set gives an equivalence relation and that any equivalence relation on a set defines a partition into equivalence classes.

DEFINITION I.9: We denote the number of partitions of $\{1,\dots,n\}$ into k (non-empty) parts by $S(n,k)$. It is called the *Stirling number of the second kind*[3] OEIS A0082h7. The total number of partitions of $\{1,\dots,n\}$ is given by the *Bell number* OEIS A000110

$$B_n = \sum_{k=1}^{n} S(n,k).$$

Example: There are $B_3 = 5$ partitions of the set $\{1,2,3\}$.

Again we might want to set $S(n,k) = 0$ unless $1 \le k \le n$.

We will give a formula for $S(n,k)$ in Lemma II.5 and study $B(n)$ in section II.5.

In some cases we shall care not which numbers are in which parts of a partition, but only the size of the parts. We denote the number of partitions of n into k parts by $p_k(n)$, the total number of partitions by $p(n)$ OEIS A000041.

Again we study the function $p(n)$ later, however here we shall not achieve a closed formula for the value.

The twelvefold way theorem

We now extend the table of theorem I.2:

THEOREM I.10: if N, X are finite sets with $|N| = n$ and $|X| = x$, the number of (equivalence classes of) functions $f: N \to X$ is given by the following table. (In the first two columns, d/i indicates whether elements are considered distinguishable or indistinguishable. The boxed numbers refer to the explanations in the proof.):

N	X	f arbitrary	f injective	f surjective
d	d	1 x^n	2 $(x)_n$	3 $x!S(n,x)$
i	d	4 $\left(\!\binom{x}{n}\!\right)$	5 $\binom{x}{n}$	6 $\binom{n-1}{x-1} = \left(\!\binom{x}{n-x}\!\right)$
d	i	7 $\sum_{k=0}^{x} S(n,k)$	8 $\begin{cases} 1 & \text{if } n \le x \\ 0 & \text{if } n > x \end{cases}$	9 $S(n,x)$
i	i	10 $\sum_{k=0}^{x} p_k(n)$	11 $\begin{cases} 1 & \text{if } n \le x \\ 0 & \text{if } n > x \end{cases}$	12 $p_x(n)$

Proof: If N is distinguishable, we can simply write the elements of N in a row and consider a function on N as a sequence of values. In 1), we thus have sequences

[3] There also is a Stirling number of the first kind

of length n with x possible values, in 2) such sequences without repetition, both formulas we already know.

If such a sequence takes exactly x values, each value $f(n)$ can be taken to indicate the cell of a partition into x parts, into which n is placed. As we consider a partition as a set of parts, it does not distinguish the elements of X, that shows that the value in 9) has to be $S(n,x)$. If we distinguish the elements of X we need to account for the $x!$ possible arrangements of cells, yielding the value in 3).

Similarly to 9), if we do not require f to be surjective, the number of different values of f gives us the number of parts. Up to x different parts are possible, thus we need to add the values of the Stirling numbers.

To get 12) from 9) and 10) from 7) we notice that making the elements of N indistinguishable simply means that we only care about the sizes of the parts, not which number is in which part. This means that the Stirling number $S(n,x)$ gets replaced by the partition (shape) count $p_x(n)$.

If we again start at 1) but now consider the elements of N as indistinguishable, we go from sequences to sets. If f is injective we have ordinary sets, in the general case multisets, and have already established the results of 4) and 5).

For 6), we interpret the x distinct values of f to separate the elements of N into x parts. This is a composition of n into x parts, for which the count has been established.

In 8) and 11), finally, injectivity demands that we assign all elements of n to different values which is only possible if $n \leq x$. As we do not distinguish the elements of X it does not matter what the actual values are, thus there is only one such function up to equivalence. □

Chapter

II Recurrence and Generating Functions

Finding a close formula for a combinatorial counting function can be hard. It often is much easier to establish a recursion, based on a reduction of the problem. Such a reduction often is the principal tool when constructing all objects in the respective class.

An easy example of such a situation is given by the number of partitions of n, given by the Bell number B_n:

Lemma II.1: For $n \geq 1$ we have:

$$B_n = \sum_{k=1}^{n} \binom{n-1}{k-1} B_{n-k}$$

Proof: Consider a partition of $\{1, \ldots, n\}$. Being a partition, it must have 1 in one cell. We group the partitions according to how many points are in the cell containing 1. If there are k elements in this cell, there are $\binom{n-1}{k-1}$ options for the other points in this cell. And the rest of the partition is simply a partition of the remaining $n-k$ points. □

II.1 Power Series – A Review of Calculus

A powerful technique for working with recurrence relations is that of generating functions. The definition is easy, for a counting function $f: \mathbb{Z}_{\geq 0} \to \mathbb{Q}$ defined on the nonnegative integers we define the associated generating function as the power series

$$F(t) = \sum_{n=0}^{\infty} f(n) t^n \in \mathbb{R}[[t]]$$

Here $\mathbb{R}[[t]]$ is the ring of formal power series in t, that is the set of all formal sums $\sum_{n=0}^{\infty} a_n t^n$i with $a_n \in \mathbb{R}$.

> *When writing down such objects, we ignore the question whether the series converges, i.e. whether F can be interpreted as a function on a subset of the real numbers.*

Addition and multiplication are as one would expect: If $F(t) = \sum_{n\geq 0} f(n)t^n$ and $G(t) = \sum_{n\geq 0} g(n)t^n$ we define new power series $F + G$ and $F \cdot G$ by:

$$\begin{aligned}(F+G)(t) &:= \sum_{n\geq 0}(f(n)+g(n))t^n \\ (F\cdot G)(t) &:= \sum_{n\geq 0}\left(\sum_{a+b=n} f(a)g(b)\right)t^n.\end{aligned}$$

With this arithmetic $R[[t]]$ becomes a commutative ring. (There is no convergence issue, as the sum over $a + b = n$ is always finite.)

We also define two operators, called *differentiation* and *integration* on $\mathbb{R}[[t]]$ by

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\sum_{n\geq 0} a_n t^n\right) = \sum_{n\geq 1}(na_n)t^{n-1}$$

and

$$\int\left(\sum_{n\geq 0} a_n t^n\right) = \sum_{n\geq 0}\frac{a_n}{n+1}t^{n+1},$$

Two power series are equal if and only if all coefficients are equal, an identity involving a power series as a variable is called a *functional equation*.

As a shorthand we define the following names for particular power series:

$$\begin{aligned}\exp(t) &= \sum_{n\geq 0}\frac{t^n}{n!} \\ \log(1+t) &= \sum_{n\geq 1}\frac{(-1)^{n-1}t^n}{n}\end{aligned}$$

and note[1] that the usual functional identities $\exp(a+b) = \exp(a)\exp(b)$, $\log(ab) = \log(a) + \log(b)$, $\log(\exp(t)) = t$ and differential identities $\frac{\mathrm{d}}{\mathrm{d}t}\exp(t) = \exp(t)$, $\frac{\mathrm{d}}{\mathrm{d}t}\log(t) = 1/t$ hold.

[1] All of this can either be proven directly for power series, or we choose t in the interval of convergence, use results from Calculus, and obtain the result from the uniqueness of power series representations.

A telescoping argument shows that for $r \in \mathbb{R}$ we have that $(1-rt)\left(\sum_n r^n t^n\right) = 1$, that is we can use the *geometric series* identity

$$\sum_n r^n t^n = \frac{1}{1-rt}$$

to embed rational functions (whose denominators factor completely; this can be taken as given if we allow for complex coefficients) into the ring of power series.

For a real number r, we *define*[2]

$$\binom{r}{n} = \frac{r(r-1)\cdots(r-n+1)}{n!}\, t^n$$

as well as

$$(1+t)^r = \sum_{n\geq 0} \binom{r}{n}.$$

For obvious reasons we call this definition the *binomial formula*. Note that for integral r this agrees with the usual definition of exponents and binomial coefficients, so there is no conflict with the traditional definition of exponents.

We also notice that for arbitrary r, s we have that $(1+t)^r(1+t)^s = (1+t)^{r+s}$ and $\frac{\mathrm{d}}{\mathrm{d}t}(1+t)^r = r(1+t)^{r-1}$.

A power series may (if t is chosen in the domain of convergence, but that is not something for us to worry about) be used to define a function on the real numbers and an identity amongst power series, involving differentiation, then could be interpreted as a differential equation. The uniqueness of solutions for differential equations then implies that the power series for a solution of the differential equation must be the unique power series satisfying this equation. Or, in other words:

> *We can use differential equations and their tabulated solutions as a canned tool for equations amongst power series.*

Generating Functions and Recursion

Up to this point, generating functions seem to be a formality for formalities sake. The crucial observation now is that an index shift can be accomplished by multiplication with t. A recurrence relation thus becomes a functional equation for the generating function. In many important cases we can solve this functional equation using knowledge from Analysis, and then determine a formula for the coefficients of the generating function using Taylor's theorem.

A recursion with coefficients depending on the index, similarly leads to a differential equation, and is treated in the same way.

Instead of defining this formally, it is probably easiest to consider an example to describe a general method:

[2] Unless r is an integer, this is a definition in the ring of formal power series.

We define a function recursively by setting $f(0) = 1$ and $f(n+1) = 2f(n)$. (This recursion comes from the number of subsets of a set of cardinality n – fix one element and distinguish between subsets containing this element and those that don't.) The associated generating function is $F(t) = \sum_{n\geq 0} f(n)t^n$. We now observe that

$$2tF(t) = \sum_{n\geq 0} 2f(n)t^{n+1} = \sum_{n\geq 0} f(n+1)t^{n+1} = F(t) - 1.$$

We solve this functional equation as

$$F(t) = \frac{1}{1-2t}$$

and (geometric series!) obtain the power series representation

$$F(t) = \sum_{n\geq 0} 2^n t^n.$$

this allows us to conclude that $f(n) = 2^n$, solving the recursion (and giving us the combinatorial result we knew already that a set of cardinality n has 2^n subsets.)

II.2 Linear recursion with constant coefficients

Then, at the age of forty, you sit,
theologians without Jehovah,
hairless and sick of altitude,
in weathered suits,
before an empty desk,
burned out, oh Fibonacci,
oh Kummer[1]*, oh Gödel, oh Mandelbrot*
in the purgatory of recursion.

Dann, mit vierzig, sitzt ihr,
o Theologen ohne Jehova,
haarlos und höhenkrank
in verwitterten Anzügen
vor dem leeren Schreibtisch,
ausgebrannt, o Fibonacci,
o Kummer, o Gödel, o Mandelbrot,
im Fegefeuer der Rekursion.

[1] also means: "grief"

Die Mathematiker
Hans Magnus Enzensberger

Suppose that $f(n)$ satisfies a recursion of the form

$$f(n) = a_1 f(n-1) + a_2 f(n-2) + \cdots + a_k f(n-k).$$

That is, there is a fixed number of recursion terms and each term is just a scalar multiple of a prior value (we shall see below that easy cases of index dependence also can be treated this way). We also assume that k initial values $f(0), \cdots, f(k-1)$ have been established.

The most prominent case of this are clearly the Fibonacci numbers OEIS A000045 with $k = 2$, $f(0) = f(1) = 1$, and we shall use these as an example.

Step 1: Get the functional equation Using the recursion, expand the coefficient $f(n)$ in the generating function $F(t) = \sum_n f(n)t^n$ with terms of lower index. Note that for $n < k$ the recursion does not hold, you will need to look at the initial values to see whether the given formula suffices, or if you need to add explicit multiples of powers of t to get equality.

Separate summands into different sums, factor out powers of t to get $f(n)$ combined with t^n.

Replace all expressions $\sum f(n)t^n$ back with the generating function $F(t)$. The whole expression also must be equal to $F(t)$, this is the functional equation.
Example: in the case of the Fibonacci numbers, the recursion is $f(n) = f(n-1) + f(n-2)$ for $n > 1$. Thus we get (using the initial values $f(0) = f(1) = 1$ that

$$\begin{aligned}
\sum_{n\geq 0} f(n)t^n &= \sum_{n>1}(f(n-1)t^n + f(n-2)t^n) + f(1)t + f(0)\\
&= \sum_{n>1} f(n-1)t^n + \sum_{n>1} f(n-2)t^n + t + 1\\
&= t\sum_{n>1} f(n-1)t^{n-1} + t^2\sum_{n>1} f(n-2)t^{n-2} + t + 1\\
&= t\sum_{n>0} f(n)t^n + t^2\sum_{n\geq 0} f(n)t^n + t + 1\\
&= t(\sum_{n\geq 0} f(n)t^n - f(0)) + t^2\sum_{n\geq 0} f(n)t^n + t + 1\\
&= t\cdot F(t) - t\cdot f(0) + t^2F(t) + t + 1 = t\cdot F(t) + t^2F(t) + 1.
\end{aligned}$$

The functional equation is thus $F(t) = t\cdot F(t) + t^2F(t) + 1$.

Step 2: Partial Fractions We can solve the functional equation to express $F(t)$ as a rational function in t. Using partial fractions (Calculus 2) we can write this as a sum of terms of the form $\frac{a_i}{(t-\alpha_i)^{e_i}}$.

Example: We solve the functional equation to give us $F(t) = \frac{1}{1-t-t^2}$. For a partial fraction decomposition, let $\alpha = \frac{-1+\sqrt{5}}{2}$, $\beta = \frac{-1-\sqrt{5}}{2}$ the roots of the equation $1 - t - t^2 = 0$. Then

$$F(t) = \frac{1}{1-t-t^2} = \frac{a}{t-\alpha} + \frac{b}{t-\beta}$$

We solve this as $a = -1/\sqrt{5}$, $b = 1/\sqrt{(5)}$.

Step 3: Use known power series to express each summand as a power series The geometric series gives us that

$$\frac{a}{t-\alpha} = \sum_{n\geq 0} \frac{-a}{\alpha^{n+1}} t^n$$

If there are multiple roots, denominators could arise in powers. For this we notice that

$$\frac{1}{(t-\alpha)^2} = \sum_{n\geq 0} \frac{(n+1)}{\alpha^{n+2}} t^n$$

and for integral $c \geq 1$ that

$$\frac{1}{(t-1)^c} = -1^c \sum_{n\geq 0} \binom{c+n-1}{n} t^n$$

Using these formulae, we can write each summand of the partial fraction decomposition as an infinite series.

Example: In the example we get

$$\begin{aligned} F(t) &= \frac{-1}{\sqrt{5}} \frac{1}{t-\alpha} + \frac{1}{\sqrt{5}} \frac{1}{t-\beta} \\ &= \frac{-1}{\sqrt{5}} \sum_{n\geq 0} \frac{-1}{\alpha^{n+1}} t^n + \frac{1}{\sqrt{5}} \sum_{n\geq 0} \frac{-1}{\beta^{n+1}} t^n \end{aligned}$$

Step 4: Combine to one sum, and read off coefficients We now take this (unique!) power series expression and read off the coefficients. The coefficient of t^n will be $f(n)$, which gives us an explicit formula.

Example: Continuing the calculation above, we get

$$F(t) = \sum_{n\geq 0} \left(\frac{1}{\sqrt{5} \cdot \alpha^{n+1}} + \frac{-1}{\sqrt{5} \cdot \beta^{n+1}} \right) t^n$$

and thus a closed formula for the Fibonacci numbers:

$$\begin{aligned} f(n) &= \frac{1}{\sqrt{5} \cdot \alpha^{n+1}} + \frac{-1}{\sqrt{5} \cdot \beta^{n+1}} \\ &= \frac{1}{\sqrt{5} \cdot \left(\frac{-1+\sqrt{5}}{2}\right)^{n+1}} + \frac{-1}{\sqrt{5} \cdot \left(\frac{-1-\sqrt{5}}{2}\right)^{n+1}} \\ &= \frac{1}{\sqrt{5}} \left(\left(\frac{2}{\sqrt{5}-1}\right)^{n+1} + \left(\frac{2}{\sqrt{5}+1}\right)^{n+1} \right) \end{aligned}$$

We notice that $\frac{2}{\sqrt{5}-1} > \frac{2}{\sqrt{5}+1} > 0$, thus asymptotically

$$f(n+1)/f(n) = \frac{2}{\sqrt{5}-1} = \phi \approx 1.618$$

the value of the golden ratio.

Another example

We try another example. Take the (somewhat random) recursion given by

$$\begin{aligned} g(0) &= g(1) = 1 \\ g(n) &= g(n-1) + 2 * g(n-2) + (-1)^n, \qquad \text{for } n \geq 2 \end{aligned}$$

We get for the generating function

$$\begin{aligned} G(t) &= \sum_n g(n)t^n = \sum_{n\geq 1} g(n-1)t^n + 2\sum_{n\geq 2} g(n-2)t^n + \sum_n \geq 0(-1)^n z^n + z \\ &= tG(t) + 2t^2G(t) + \frac{1}{1+t} + t. \end{aligned}$$

(you should verify that the addition of z was all that was required to resolve the initial value settings.)

We solve this functional equation as

$$G(t) = \frac{1+t+t^2}{(1-2t)(1+t)^2}$$

and get the partial fraction decomposition

$$G(t) = \frac{-7}{18(t-\frac{1}{2})} - \frac{1}{9(t+1)} + \frac{1}{3(t+1)^2}.$$

We can read off the power series representation

$$\begin{aligned} G(t) &= \frac{7}{18}\sum_n 2^{n+1}t^n + \frac{1}{9}\sum_n (-1)^{n+1}t^n + \frac{1}{3}\sum_n -1^n(n+1)t^n \\ &= \sum_n \left(\frac{7}{9}2^n - \frac{1}{9}(-1)^n + \frac{1}{3}(-1)^n(n+1)\right)t^n \\ &= \sum_n \left(\frac{7}{9}2^n + \left(\frac{1}{3}n + \frac{2}{9}\right)(-1)^n\right)t^n, \end{aligned}$$

solving the recursion as $g(n) = \frac{7}{9}2^n + \left(\frac{1}{3}n + \frac{2}{9}\right)(-1)^n$.

II.3 Nested Recursions: Domino Tilings

> The Domino Theory had become conventional wisdom and was rarely challenged.
>
> Diplomacy
> Henry Kissinger

We apply the method of generating functions to some counting problems.

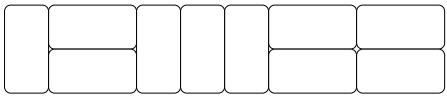


Figure II.1: A 2×10 domino tiling

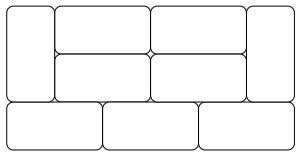


Figure II.2: A tiling pattern that has no vertical cut

Suppose we have tiles that have dimensions 1×2 (in your favorite units) and we want to tile a corridor. Let $f(n)$ the number of possible tilings of a corridor that has dimensions $2 \times n$. We can start on the left with a vertical domino (thus leaving to the right of it a tiling of a corridor of length $n-1$) or with two horizontal dominoes (leaving to the right of it a corridor of length $n-2$). this gives us the recursion

$$d(n) = d(n-1) + d(n-2), \qquad n > 1$$

with $d(1) = 1$ and $d(2) = 2$ (and thus $d(0) = 1$ to fit the recursion). This is just again the Fibonacci numbers we have already worked out.

If we assume that the tiles are not symmetric, there are actually 2 ways to place a horizontal tile and 2 ways to place a vertical tile. We thus get a recursion with different coefficients,

$$d(n) = 2d(n-1) + 4d(n-2), \qquad n > 1$$

with $d(1) = 2$, $d(2) = 8$ (and thus $d(0) = 1$).

If we expand the corridor to dimensions $3 \times n$, the recursion seems to be problematic – it is possible to have patterns or arbitrary length that do not reduce to a shorter length, see figure II.2.

We thus instead look only at the left end of the tiling and persuade ourselves (going back to the assumption of symmetric tiles) that every tiling has to start with one of the patterns depicted in the top row of figure II.3. We thus consider corridors that not only start with a straight line, but also corridors that start with the top left (respectively bottom left) tile missing. The count of these tilings will be given by a function $e(n)$. (Note that both have the same counting function because we can flip top/bottom.)

This gives us the recursion

$$d(n) = d(n-2) + 2 \cdot e(n-1)$$

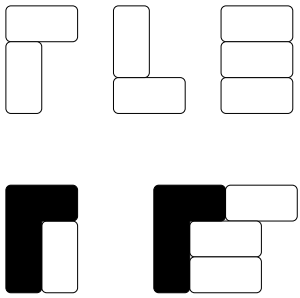


Figure II.3: Variants for a $3 \times n$ tiling

In the bottom row of figure II.3, we enumerate the possible ways how a tiling with missing corner can start, giving us the recursion

$$e(n) = d(n-1) + e(n-2)$$

We use the initial values $d(0) = 1$, $d(1) = 0$, $d(2) = 3$, $e(0) = 0$, $e(1) = 1$, $e(2) = 3$, and thus get the following identities for the associate generating functions

$$\begin{aligned} D(t) &= \sum_n d(n)t^n = \sum_{n>1} \left(d(n-2) + 2e(n-1)\right) t^n + d(0) + d(1)t \\ &= t^2 \sum_{n>1} d(n-2)t^{n-2}n + 2t \sum_{n>1} e(n-1)t^{n-1} + 1 \\ &= t^2 \sum_{n\geq 0} d(n)t^n + 2t\Big(\sum_{n\geq 0} e(n)t^n - e(0)t^0\Big) + 1 \\ &= t^2 D(t) + 2tE(t) + 1 \end{aligned}$$

and

$$\begin{aligned} E(t) &= \sum_n e(n)t^n = \sum_{n>1} \left(d(n-1) + e(n-2)\right) t^n + e(0) + e(1)t \\ &= t(D(t) - d(0)) + t^2 E(t) + t = tD(t) + t^2 E(t) \end{aligned}$$

We solve this second equation as

$$E(t) = \frac{t}{1-t^2} D(t)$$

and substitute into the first equation, obtaining the functional equation

$$D(t) = t^2 D(t) + \frac{2t^2}{1-t^2} D(t) + 1$$

which we solve for

$$D(t) = \frac{1-t^2}{1-4t^2+t^4}$$

This is a function in t^2, indicating that $d(n) = 0$ for odd n (indeed, this must be, as in this case $3 \times n$ is odd and cannot be tiled). We thus can consider instead the function

$$R(t) = \frac{1-t}{1-4t+t^2} = \sum_n r(n)t^n$$

with $d(2n) = r(n)$. Partial fraction decomposition, geometric series, and final collection of coefficients gives us the formula

$$d(2n) = r(n) = \frac{(2+\sqrt{3})^n}{3-\sqrt{3}} + \frac{(2-\sqrt{3})^n}{3+\sqrt{3}}$$

and the sequence of $r(n)$ given by OEIS A001835

$$1, 3, 11, 41, 153, 571, 2131, 7953, 29681, 110771, 413403, \ldots$$

II.4 Catalan Numbers

The induction I used was pretty tedious, but I do not doubt that this result could be obtained much easier. Concerning the progression of the numbers 1, 2, 5, 14, 42, 132, etc. …

Die Induction aber, so ich gebraucht, war ziemlich mühsam, doch zweifle ich nicht, dass diese Sach nicht sollte weit leichter entwickelt werden können. Ueber die Progression der Zahlen 1, 2, 5, 14, 42, 132, etc. …

Letter to Goldbach
September 4, 1751
Leonard Euler

We next look at an example of a recursion which is not just linear.

Definition II.2: The n-th *Catalan number*[3] C_n OEIS A000108 is defined[4] as the number of ways a sum of $n+1$ variables can be evaluated by inserting parentheses.

Example II.3: We have $C_0 = C_1 = 1$, $C_2 = 2$: $(a+b)+c$ and $a+(b+c)$, and $C_3 = 5$:

$$\begin{array}{l} ((a+b)+c)+d \\ (a+(b+c))+d \\ a+((b+c)+d) \\ a+(b+(c+d)) \\ (a+b)+(c+d) \end{array}$$

[3] Named in honor of Eugène Charles Catalan (1814-1894) who first stated the standard formula. The naming after Catalan only stems from a 1968 book, see `http://www.math.ucla.edu/~pak/papers/cathist4.pdf`. Catalan himself attributed them to Segner, though Euler's work is even earlier.

[4] Careful, some books use a shifted index, starting at 1 only!

To get a recursion, we consider the "outermost" addition, assume thus comes after $k+1$ of the variables. This addition combines two proper parenthetical expressions, the one on the left on $k+1$ variables, the one on the right on $(n+1)-(k+1) = n-k$ variables. We thus get the recursion

$$C_n = \sum_{k=0}^{n-1} C_k C_{n-k-1}, \qquad \text{if } n > 0,\ C_0 = 1.$$

in which we sum over the products of lower terms.

This pattern is in fact typical for the way we reduced:

We classify the set of objects according to a parameter (the outermost addition at k), and then recurse to two parts. More generally, if we had a count $d(n)$ for which we can split the objects into two parts, according to a smaller parameter k, where one is counted by $a(n)$ and the other by $b(n)$, we would get a formula

$$d(n+1) = \sum_{k=0}^{n} a(k)b(n-k)$$

This looks suspiciously like the formula for the multiplication of polynomials (or power series) and indeed, if the associated generating functions are $D(t)$, $A(t)$ and $B(t)$, we would get that $D(t) = A(t)B(t)$. (Initial values might make the recursion only work starting from a minimum value of n, and so there might be fudge factors to consider.)

Indeed this is what happens here – we get the functional equation

$$\mathcal{C}(t) = t \cdot \mathcal{C}(t)^2 + 1.$$

(which is easiest seen by writing out the expression for $t\mathcal{C}(t)^2$ and collecting terms). The factor t is due to the way we index and 1 is due to initial values.

This is a quadratic equation in the variable $\mathcal{C}(t)$ and we get:

$$\mathcal{C}(t) = \frac{1-\sqrt{1-4t}}{2t},$$

where we choose the solution for which for $t = 0$ we get the desired value of C_0.

The binomial series then gives us that

$$\sqrt{1-4t} = \sum_{k\geq 0} \binom{1/2}{k} (-4t)^k = 1 + \sum_{k\geq 1} \frac{1}{2k} \binom{-1/2}{k-1} (-4t)^k.$$

We conclude that

$$\begin{aligned} \frac{1-\sqrt{1-4t}}{2t} &= \sum_{k\geq 1} \frac{1}{k} \binom{-1/2}{k-1} (-4t)^{k-1} \\ &= \sum_{n\geq 0} \binom{2n}{n} \frac{t^n}{n+1}, \end{aligned}$$

in other words:

$$C_n = \binom{2n}{n}\frac{1}{n+1}.$$

Catalan numbers have many other combinatorial interpretations, and we shall encounter some in the exercises. Exercise 6.19 in [Sta99] (and its algebraic continuation 6.25, as well as an online supplement) contain hundreds of combinatorial interpretations.

II.5 Index-dependent coefficients and Exponential generating functions

A recursion does not necessarily have constant coefficient, but might have a coefficient that is a polynomial in n. In this situation we can use (formal) differentiation, which will convert a term $f(n)t^n$ into $nf(n)t^{n-1}$. The second derivative will give a term $n(n-1)f(n)t^{n-2}$; first and second derivative thus allow us to construct a coefficient $n^2 f(n)$ and so on for higher order polynomials.

The functional equation for the generating function then becomes a differential equation, and we might hope that a solution for it can be found in the extensive literature on differential equations.

Alternatively, (that is we use the special form of derivatives for a typical summand), such a situation often can be translated immediately to a generating function by using the power series

$$\frac{1}{(1-t)^{k+1}} = \sum_n \binom{n+k}{n} t^n.$$

For an example of variable coefficients, we take the case of counting *derangements* OEIS A000166, that is permutations that leave no point fixed. We denote by $d(n)$ the number of derangements on $\{1,\dots,n\}$.

To build a recursion formula, suppose that π is a derangement on $\{1,\dots,n\}$. Then $n^\pi = i < n$. We now distinguish two cases, depending on how i is mapped by n:

a) If $i^\pi = n$, then π swaps i and n and is a derangement of the remaining $n-2$ points, thus there are $d(n-2)$ derangements that swap i and n. As there are $n-1$ choices for i, there are $(n-1)d(n-2)$ derangements that swap n with a smaller point.

b) Suppose there is $j \neq i$ such that $j^\pi = n$. In this case we can "bend" π into another permutation ψ, by setting

$$k^\psi = \begin{cases} k^\pi & \text{if } k \neq j \\ i & \text{if } k = j \end{cases}.$$

We notice that ψ is a derangement of the points $\{1,\dots,n-1\}$.

Vice versa, if ψ is a derangement on $\{1,\dots,n-1\}$, and we choose a point $j<n$, we can define π on $\{1,\dots,n\}$ by

$$k^\pi = \begin{cases} k^\psi & \text{if } k \neq i,n \\ n & \text{if } k=j \\ j^\psi & \text{if } k=n \end{cases}.$$

We again notice that π is a derangement, and that different choices of ψ and i result in different π's. Furthermore, the two constructions are mutually inverse, that is every derangement that is not in class a) is obtained by this construction.

There are $d(n-1)$ possible derangements ψ and $n-1$ choices for j, so there are $(n-1)d(n-1)$ derangements in this second class.

We thus obtain a recursion

$$d(n) = (n-1)\left(d(n-1)+d(n-2)\right)$$

and hand-calculate[5] the initial values $d(0)=1$ and $d(1)=0$.

We now could build a differential equation for the generating function, but there is a problem: Because of the factor n in the recursion, the coefficients $d(n)$ grow roughly like $n!$, but a series whose coefficients grow that quickly will have convergence radius 0, making it unlikely that the solution exists in the literature on differential equations.

We therefore introduce the *exponential generating function* which is defined simply by dividing the i-th coefficient by a factor of $i!$, thus keeping coefficient growth bounded.

In our example, we get

$$D(t) = \sum_n \frac{d(n)}{n!}t^n$$

and thus

$$\frac{\mathrm{d}}{\mathrm{d}t}D(t) = \sum_{n\geq 1}\frac{n\cdot d(n)}{n!}t^{n-1} = \sum_{n\geq 1}\frac{d(n)}{(n-1)!}t^{n-1} = \sum_{n\geq 0}\frac{d(n+1)}{n!}t^n.$$

We also have

$$\begin{aligned} t\cdot D(t) &= \sum_{n\geq 0} d(n)\frac{t^{n+1}}{n!} = \sum_{n\geq 0}(n+1)d(n)\frac{t^{n+1}}{(n+1)!} \\ &= \sum_{n\geq 1} n\cdot d(n-1)\frac{t^n}{n!} = \sum_{n\geq 0} n\cdot d(n-1)\frac{t^n}{n!} \\ t\cdot D'(t) &= \sum_{n\geq 0}\frac{n\cdot d(n)}{n!}t^n \end{aligned}$$

[5] The reader might have an issue with the choice of $d(0)=1$, as it is unclear what a derangement on no points is. By going backwards to the recursion (using $d(2)$ and $d(1)$ to calculate $d(0)$), it turns out that this is the right number to make the recursion work in case $d(0)$ is referred to.

From this, the recursion (in the form $d(n+1) = n(d(n) + d(n-1))$) gives us:

$$\begin{aligned} t \cdot D(t) + t \cdot D'(t) &= \sum_{n \geq 0} (n \cdot d(n-1) + n \cdot d(n)) \frac{t^n}{n!} \\ &= \sum_{n \geq 0} d(n+1) \frac{t^n}{n!} = D'(t), \end{aligned}$$

and thus the separable differential equation

$$\frac{D'(t)}{D(t)} = \frac{t}{1-t}.$$

with $D(0) = d(0) = 1$.

Standard techniques from Calculus give us the solution

$$D(t) = \frac{e^{-t}}{1-t}.$$

Looking up this function for Taylor coefficients (respectively determining the formula by induction) shows that

$$D(t) = \sum_{n \geq 0} \left(\sum_{i=0}^{n} \frac{(-1)^i}{i!} \right) t^n$$

and thus (introducing a factor $n!$ to make up for the denominator in the generating function) that

$$d(n) = n! \left(\sum_{i=0}^{n} \frac{(-1)^i}{i!} \right).$$

This is simply $n!$ multiplied with a Taylor approximation of e^{-1}. Indeed, if we consider the difference, the theory for alternating series gives us for $n \geq 1$ that:

$$\begin{aligned} \left| d(n) - \frac{n!}{e} \right| &= n! \left| \sum_{i=n+1}^{\infty} \frac{(-1)^i}{i!} \right| \\ &< n! \left| \frac{(-1)^{n+1}}{(n+1)!} \right| = \frac{1}{n+1} \leq \frac{1}{2} \end{aligned}$$

We have proven:

Lemma II.4: $d(n)$ is the integer closest to $n!/e$.

That is asymptotically, if we put letters into envelopes, the probability is $1/e$ that no letter is in the correct envelope.

Bell Numbers

$$\int_1^{\sqrt[3]{3}} z^2\,dz \times \cos\frac{3\pi}{9} = \log(\sqrt[3]{e})$$

The integral z-squared dz,
From one to the cube root of three,
Times the cosine,
Of three pi over nine
Equals log of the cube root of e.

Anon.

A second use of generating functions is that in some cases it might be possible to give a reasonable (exponential) generating function for a counting function which itself has no closed form expression.

We go back to the Bell numbers B_n, defined as giving the total number of partitions of $\{1,\dots,n\}$. The recursion is built by what now is a standard argument:

Given a partition X, consider the cell containing 1. It has k elements in total, these are chosen from the $n-1$ numbers >1. The remaining cells form a partition of the $n-k$ numbers outside the cell. Thus, for $n \geq 1$, we have:

$$B_n = \sum_{k=1}^{n} \binom{n-1}{k-1} B_{n-k}$$

Let $F(t) = \sum_n B_n t^n/n!$ be the exponential generating function of the B_n. Then

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}F(t) &= \sum_{n\geq 1} \frac{B_n t^{n-1}}{(n-1)!} = \sum_{n\geq 1}\left(\sum_{k=1}^{n}\binom{n-1}{k-1}B_{n-k}\right)\frac{t^{n-1}}{(n-1)!} \\
&= \sum_{n\geq 1}\sum_{k=1}^{n} \frac{t^{k-1}}{(k-1)!}\frac{B_{n-k}t^{n-k}}{(n-k)!} \\
&= \left(\sum_{i\geq 0}\frac{t^i}{i!}\right)\cdot\left(\sum_{j\geq 0}\frac{B_j t^j}{j!}\right) = \exp(t)F(t).
\end{aligned}$$

(using a variable substitution $i = k-1$, $j = n-k$ and switching the summation order).

This is again a separable differential equation, its solution is

$$F(t) = c\cdot\exp(\exp(t))$$

for some constant c. As $F(0) = 1$ we solve for $c = \exp(-1)$ and thus get the exponential generating function

$$\sum_n \frac{B_n t^n}{n!} = \exp(\exp(t)-1).$$

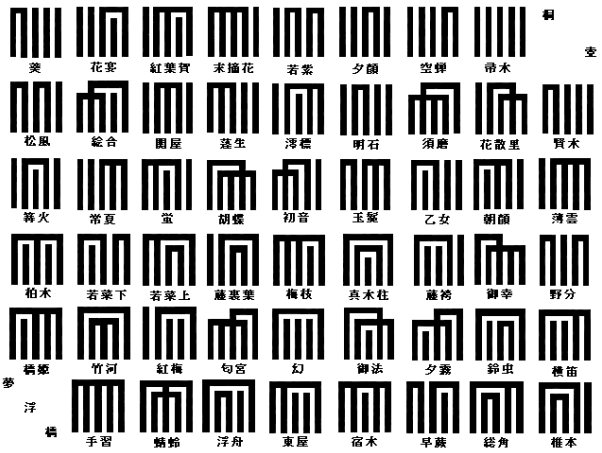


Figure II.4: The Genji-mon

ja.wikipedia,org

There is no nice way to express the power series coefficients of this function in closed form, a Taylor approximation is (with denominators being deliberately kept in the form of $n!$ to allow reading off the Bell numbers):

$$1 + t + \frac{2t^2}{2!} + \frac{5t^3}{3!} + \frac{15t^4}{4!} + \frac{52t^5}{5!} + \frac{203t^6}{6!} + \frac{877t^7}{7!} + \frac{4140t^8}{8!} + \frac{21147t^9}{9!} + \frac{115975t^{10}}{10!}.$$

One, somewhat surprising application of Bell numbers is to consider rhyme schemes. Given a sequence of n lines, the lines which rhyme form the cells of a partition of $\{1, \ldots, n\}$. For example, the partition $\{\{1, 2, 5\}, \{3, 4\}\}$ is the scheme used by Limericks.

We can read off that $B_5 = 52$. The classic 11th century Japanese novel *Genji monogatari* (The Tale of Genji) has 54 chapters of which first and last are considered "extra". The remaining 52 chapters are introduced each with a poem in one of the 52 possible rhyme schemes and a symbol illustrating the scheme. These symbols, see figure II.4, the *Genji-mon*, have been used extensively in Art. See `http://viewingjapaneseprints.net/texts/topictexts/faq/faq_genjimon.html`.

Splitting and Products

Exponential generating functions have one more trick up their sleeve, and arguably this is their most important characteristic. Think back to the case of the Catalan numbers, where a split to to smaller parts gave a product of generating functions.

Suppose we are counting objects where parts have identifying labels (these would be called *labelled* objects). In the example of the Catalan numbers this would be if

we cared not only about the parentheses placement, but also about the symbols, that is $(a + b) + c$ would be different from $(c + a) + b$.

The recursion formula then will include a binomial factor for choosing k out of n objects for the left side, that is

$$d(n) = \sum_{k=0}^{n} \binom{n}{k} a(k)b(n-k).$$

We can write this as

$$\frac{d(n)}{n!} = \sum_{k=0}^{n} \frac{a(k)}{k!} \frac{b(n-k)}{(n-k)!},$$

that is the *exponential* generating functions multiply!

Lets look at this in a pathetic example, the number of functions from $N = \{1, \ldots, n\}$ to $\{1, \ldots, r\}$ (which we know already well as r^n).

Let $a(n)$ count the number of constant functions on an n-element set, that is $a(n) = 1$. The associated exponential generating function thus is

$$A(t) = \sum_{n} \frac{t^n}{n!} = \exp(t)$$

(which, incidentally, shows why these are called "exponential" generating functions).

If we take an arbitrary function f on N, we can partition N into r (possibly empty) sets $N_1, \ldots, N_r$, such that f is constant on N_i and the N_i are maximal with this property.

We get all possible functions f by combining constant functions on the possible N_i's for all possible partitions of N. Note that the ordering of the partitions is significant – they indicate the actual values.

We are thus exactly in the situation described, and get as exponential generating function (start with $r = 2$, then use induction for larger r) the r-fold product of the exponential generating functions for the number of constant functions:

$$D(t) = \underbrace{\exp(t) \cdot \cdots \exp(t)}_{r \text{ factors}} = \exp(rt)$$

The coefficient for t^n in the power series for $\exp(rt)$ of course is simply $\frac{r^n}{n!}$, that is the counting function is r^n, as expected.

Indeed, using this approach we could have determined the exponential generating function for the Bell numbers without the need to use recursion and differential equations.

For this, lets consider first the Stirling numbers of the second kind, $S(n, k)$ denoting the number of partitions of n into k (non-empty) parts. According to I.10, part 3) there are $k!S(n, k)$ order-significant partitions of n into k parts.

We denote the associated exponential generating function (for order-significant partitions) by

$$\mathcal{S}_k(t) = \sum_n k!S(n,k)\frac{t^n}{n!}.$$

We also know that there is – apart from the empty set – exactly one partition into one cell. That is

$$\mathcal{S}_1(t) = t + \frac{t^2}{2!} + \frac{t^3}{3!} + \cdots = \exp(t) - 1$$

If we have a partition into k-parts, we can fix the first cell and then partition the rest further. Thus, for $k > 1$ we have that

$$\mathcal{S}_k(t) = \mathcal{S}_1(t) \cdot \mathcal{S}_{k-1}(t),$$

which immediately gives that

$$\mathcal{S}_k(t) = (\exp(t) - 1)^k$$

We deduce that the Stirling numbers of the second kind have the exponential generating function

$$\sum_n S(n,k)\frac{t^n}{n!} = \frac{(\exp(t)-1)^k}{k!}.$$

Using the fact that $B_n = \sum_{k=1}^n S(n,k)$, we thus get the exponential generating function for the Bell numbers as

$$\begin{aligned}\sum_n B_n \frac{t^n}{n!} &= \sum_k \sum_n S(n,k)\frac{t^n}{n!} \\ &= \sum_k \frac{(\exp(t)-1)^k}{k!} = \exp(\exp(t)-1)\end{aligned}$$

in agreement with the above result.

As we have the exponential generating function for the Stirling numbers, we might as well deduce a coefficient formula for them:

Lemma II.5:

$$S(n,k) = \frac{1}{k!}\sum_{i=1}^{k}(-1)^{k-i}\binom{k}{i}i^n.$$

Proof: We first note that – as $0^n = 0$ – the i-sum could start at 1 or 0 alternatively without changing the result.

Then multiplying through with $k!$, we know by the binomial formula that

$$\begin{aligned} k!\sum_n S(n,k)\frac{t^n}{n!} &= (\exp(t)-1)^k = \sum_{i=0}^k \binom{k}{i}\exp(t)^i(-1)^{k-i} \\ &= \sum_{i=0}^k \binom{k}{i}\exp(i\cdot t)(-1)^{k-i} = \sum_{i=0}^k \binom{k}{i}\left(\sum_n \frac{(it)^n}{n!}\right)(-1)^{k-i} \\ &= \sum_n \left(\sum_{i=0}^k \binom{k}{i}(-1)^{k-i}i^n\right)\frac{t^n}{n!}. \end{aligned}$$

and we read off the coefficients. □

Involutions

Finally, lets use this in a new situation:

DEFINITION II.6: A permutation π on $\{1,\dots,n\}$ is called[6] an *involution* OEIS A000085 if $\pi^2 = 1$, that is $(i^\pi)^\pi = i$ for all i.

We want to determine the number $s(n)$ of involutions on n points.

The reduction we shall use is to consider the number of cycles of the involution, also counting 1-cycles. Let $s_r(n)$ the number of involutions that have exactly r cycles. Clearly $s_0(0) = 1$, $s_1(0) = 0$, $s_1(1) = 1$, $s_1(2) = 1$ which gives the exponential generating function $S_1(t) = t + \frac{t^2}{2}$.

When considering an arbitrary involution, we can split off a cycle, seemingly leading to a formula

$$\left(t + \frac{t^2}{2}\right)^r$$

for $S_r(t)$. However doing so distinguishes the cycles in the order they were split off, that is we need to divide by $r!$ to avoid overcounting. Thus

$$S_r(t) = \frac{1}{r!}\left(t + \frac{t^2}{2}\right)^r$$

and thus for the exponential generating function of the number of involutions a sum over all possible r:

$$S(t) = \sum_{r=0}^\infty S_r(t) = \sum_r \frac{1}{r!}\left(t + \frac{t^2}{2}\right)^r = \exp(t + \frac{t^2}{2}) = \exp(t)\exp\left(\frac{t^2}{2}\right).$$

[6] Group theorists often exclude the identity, but it is convenient to allow it here.

We easily write down power series for the two factors

$$\begin{aligned} \exp(t) &= \sum_n \frac{t^n}{n!} \\ \exp\left(\frac{t^2}{2}\right) &= \sum_n \frac{t^{2n}}{2^n \cdot n!} \end{aligned}$$

and multiply out, yielding

$$\begin{aligned} S(t) &= \left(\sum_n \frac{t^{2n}}{2^n \cdot n!}\right) \cdot \left(\sum_n \frac{t^n}{n!}\right) \\ &= \sum_m \sum_{k=0}^{\lfloor m/2 \rfloor} \frac{t^{2k} t^{m-2k}}{2^k k!(m-2k)!} \end{aligned}$$

and thus (again introducing a factor of $n!$ for making up for the exponential generating function)

$$s(n) = \sum_{k=0}^{\lfloor m/2 \rfloor} \frac{n!}{2^k k!(n-2k)!}.$$

Chapter

III

Inclusion, Incidence, and Inversion

It is calculus exam week and, as every time, a number of students were reported to require alternate accommodations:

- 14 students are sick.
- 12 students are scheduled to play for the university Quiddich team at the county tournament.
- 12 students are planning to go on the restaurant excursion for the food appreciation class.
- 5 students on the Quiddich team are sick (having been hit by balls).
- 4 students are scheduled for the excursion and the tournament.
- 3 students of the food appreciation class are sick (with food poisoning), and
- 2 of these students also planned to go to the tournament, i.e. have all three excuses.

The course coordinator wonders how many alternate exams need to be provided.

Using a Venn diagram and some trial-and-error, it is not hard to come up with the diagram in figure III.1, showing that there are 28 alternate exams to offer.

III.1 The Principle of Inclusion and Exclusion

> By the method of exclusion, I had arrived at this result, for no other hypothesis would meet the facts.
>
> A Study in Scarlet
> Arthur Conan Doyle

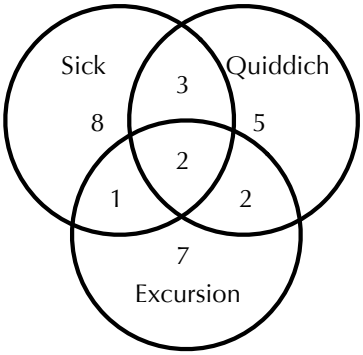


Figure III.1: An example of Inclusion/Exclusion

The *Principle of Inclusion and Exclusion* (PIE) formalizes this process for an arbitrary number of sets:

Let X be a set and $\{A_1, \ldots, A_n\}$ a family of subsets. For any subset $I \subset \{1, \ldots, n\}$ we define

$$A_I = \bigcap_{i \in I} A_i,$$

using that $A_\varnothing = X$. Then

Lemma III.1: The number of elements that lie in none of the subsets A_i is given by

$$\sum_{I \subset \{1,\ldots,n\}} (-1)^{|I|} |A_I| .$$

Proof: Take $x \in X$ and consider the contribution of this element to the given sum. If $x \notin A_i$ for any i, it only is counted for $I = \varnothing$, that is contributes 1.

Otherwise let $J = \{1 \le a \le n \mid x \in A_a\}$ and let $j = |J|$. We have that $x \in A_I$ if and only if $I \subset J$. Thus x contributes

$$\sum_{I \subset J} (-1)^{|I|} = \sum_{i=0}^{j} \binom{j}{i} (-1)^i = (1-1)^j = 0$$

□

As a first application we determine the number of derangements on n points in an different way:

Let $X = S_n$ be the set of all permutations of degree n, and let A_i be the set of all permutations π with $i^\pi = i$. Then $S_n \setminus \bigcup_i A_i$ is exactly the set of derangements, there are $\binom{n}{i}$ possibilities to intersect i of the A_i's, and the formula gives us:

$$d(n) = \sum_{i=0}^{n} (-1)^i \binom{n}{i} (n-i)! = n! \sum_{i=0}^{n} \frac{(-1)^i}{i!}.$$

For a second example, we calculate the number of surjective mappings from an n-set to a k-set (which we know already from I.10 to be $k!S(n,k)$):

Let X be the set of all mappings from $\{1,\dots,n\}$ to $\{1,\dots,k\}$, then $|X| = k^n$. Let A_i be the set of those mappings f, such that i is not in the image of f, so $|A_i| = (k-1)^n$. More generally, if $I \subset \{1,\dots k\}$ we have that $|A_I| = (k-|I|)^n$. The surjective mappings are exactly those in X outside any of the A_i, thus the formula gives us the count

$$\sum_{i=0}^{k}(-1)^i\binom{k}{i}(k-i)^n,$$

using again that there are $\binom{k}{i}$ possible sets I of cardinality i.

The factor $(-1)^i$ in a formula often is a good indication that inclusion/exclusion is to be used.

Lemma III.2:

$$\sum_{i=0}^{n}(-1)^i\binom{n}{i}\binom{m+n-i}{k-i} = \begin{cases}\binom{m}{k} & \text{if } m \geq k,\\ 0 & \text{if } m < k.\end{cases}$$

Proof: To use PIE, the sets A_i need to involve choosing from an n, set, and after choosing i of these we must choose from a set of size $m+n-i$.

Consider a bucket filled with n blue balls, labeled with $1,\dots,n$, and m red balls. How many selections of k balls only involve red balls? Clearly the answer is the right hand side of the formula.

Let X be the set of all k-subsets of balls and A_i those subsets that contain blue ball number i, then PIE gives the left side of the formula. □

We finish this section with an application from number theory. The *Euler function* $\varphi(n)$ counts the number of integers $1 \leq k \leq n$ with $\gcd(k,n) = 1$.

Suppose that $n = \prod_{i=1}^{r} p_i^{e_i}$, $X = \{1,\dots,n\}$ and A_i the integers in X that are multiples of p_i. Then (inclusion/exclusion)

$$\varphi(n) = n - \sum_{i=1}^{r}\frac{n}{p_i} + \sum_{1\leq i,j\leq r}\frac{n}{p_ip_j} - \cdots = n\prod\left(1-\frac{1}{p_i}\right).$$

with the second identity obtained by multiplying out the product,

We also note – exercise **??** – that $\sum_{d\mid n}\varphi(d) = n$. That is, the sum over one function over a nice index set is another (easier) function. We will put this into larger context in later sections.

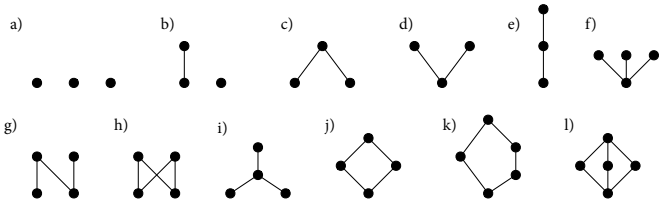


Figure III.2: Hasse Diagrams of Small Posets and Lattices

III.2 Partially Ordered Sets and Lattices

> The doors are open; and the surfeited grooms
> Do mock their charge with snores:
> I have drugg'd their possets,
> That death and nature do contend about them
>
> Macbeth, Act II, Scene II
> William Shakespeare

A *poset* or *partially ordered set* is a set A with a relation $R \subset A \times A$ on the elements of A which we will typically write as $a \leq b$ instead of $(a, b) \in R$, such that for all $a, b, c \in A$:

(reflexive) $a \leq a$.

(antisymmetric) $a \leq b$ and $b \leq a$ imply that $a = b$.

(transitive) $a \leq b$ and $b \leq c$ imply that $a \leq c$.

For example, A could be the set of subsets of a particular set, and $\leq$ with be the "subset or equal" relation.

A convenient way do describe a poset for a finite set A is by its *Hasse-diagram*. Say that a *covers* b if $a \geq b$, $a \neq b$ and there is no $a \neq c \neq b$ with $a \geq c \geq b$. The Hasse diagram of the poset is a graph in the plane which connects two vertices a and b only if a covers b, and in this case the edge from b to a goes upwards.

Because of transitivity, we have that $a \leq b$ if and only if one can go up along edges from a to reach b.

Figure III.2 gives a number of examples of posets, given by their Hasse diagrams, including all posets on 3 elements.

An isomorphism of posets is a bijection that preserves the $\leq$ relation.

An element in a poset is called *maximal* if there is no larger (wrt. $\leq$) element, *minimal* is defined in the same way. Posets might have multiple maximal and minimal elements.

Linear extension

My scheme of Order gave me the most trouble

Autobiography
Benjamin Franklin

A partial order is called a *total order*, if for every pair $a, b \in A$ of elements we have that $a \leq b$ or $b \leq a$.

While this is not part of our definition, we can always embed a partial order into a total order.

Proposition III.3: Let $R \subset X \times X$ be a partial order on X. Then there exists a total order (called a *linear extension*) $T \subset X \times X$ such that $R \subset T$.

To avoid set acrobatics we shall prove this only in the case of a finite set X. Note that in Computer science the process of finding such an embedding is called a *topological sorting*.

Proof: We proceed by induction over the number of pairs a, b that are incomparable. In the base case we already have a total order.

Otherwise, let a, b such an incomparable pair. We set (arbitrarily) that $a < b$. Now let

$$L = \{x \in X \mid x \leq_R a\}, U = \{x \in X \mid b \leq_R x\}$$

We claim that $S = R \cup \{(l, u) \mid l \in L, u \in u\}$ is a partial order. As $(a, b) \in S$ it has fewer incomparable pairs, this shows by induction that there exists a total order $T \supset S \supset R$, proving the theorem.

Since R is reflexive, S is. For antisymmetry, suppose that for $x \neq y$ we have that $(x, y), (y, x) \in S$. Since R is a partial order, not both can be in R. Suppose that

$$(x, y), (y, x) \in S \setminus R = \{(l, u) \mid l \in L, u \in u\}.$$

This implies that $x \leq_R a$ and $b \leq_R x$, thus by transitivity $b \leq_R a$, contradicting the incomparability.

If $(x, y) \in R$, $(y, x) \in S \setminus R$, we have that $b \leq_R x \leq_R y \leq_R a$, again contradicting incomparability.

For transitivity, suppose that $(x, y) \in S \setminus R$ and $(y, z) \in S$. Then $(y, z) \in R$, as otherwise $b \leq_R y \leq_R a$. But then $b \leq_R y \leq_R z$, implying that $(x, z) \in S$. The other case is analog. □

This theorem implies that we can always label the elements of a countable poset with positive integers, such that the poset ordering implies the integer ordering (but this is in general not unique).

Lattices

Definition III.4: Let A be a poset and $a, b \in A$.

- A *greatest lower bound* of a and b is an element $c \leq a, b$ which is maximal in the set of elements with this property.

- A *least upper bound* of a and b is an element $c \geq a, b$ which is minimal in the set of elements with this property.

A is a *lattice* if any pair $a, b \in A$ have a unique greatest lower bound, called the *meet* and denoted by $a \wedge b$; as well as unique least upper bound, called the *join* and denoted by $a \vee b$.

Amongst the Hasse diagrams in figure III.2, e,j,k,l) are lattices, while the others are not. Lattices always have unique maximal and minimal elements, sometimes denoted by 0 (minimal) and 1 (maximal).

Other examples of lattices are:

1. Given a set X, let $A = \mathcal{P}(X) = \{Y \subseteq X\}$ the power set of X with $\leq$ defined by inclusion. Meet is the intersection, join the union of subsets.

2. Given an integer n, let A be the set of divisors of n with $\leq$ given by "divides". Meet and join are gcd, respectively lcm.

3. For an algebraic structure S, let A be the set of all substructures (e.g. group and subgroups) of S and $\leq$ given by inclusion. Meet is the intersection, join the substructure spanned by the two constituents.

4. For particular algebraic structures there might be classes of substructures that are closed under meet and join, e.g. normal subgroups. These then form a (sub)lattice.

Using meet and join as binary operations, we can axiomatize the structure of a lattice:

Proposition III.5: Let X be a set with two binary operations $\wedge$ and $\vee$ and two distinguished elements $0, 1 \in X$. Then $(X, \wedge, \vee, 0, 1)$ is a lattice if and only if the following axioms are satisfies for all $x, y, z \in X$:

Associativity: $x \wedge (y \wedge z) = (x \wedge y) \wedge z$ and $x \vee (y \vee z) = (x \vee y) \vee z$;

Commutativity: $x \wedge y = y \wedge x$ and $x \vee y = y \vee x$;

Idempotence: $x \wedge x = x$ and $x \vee x = x$;

Inclusion: $(x \vee y) \wedge x = x = (x \wedge y) \vee x$;

Maximality: $x \wedge 0 = 0$ and $x \vee 1 = 1$.

Proof: The verification that these axioms hold for a lattice is left as exercise to the reader.

Vice versa, assume that these axioms hold. We need to produce a poset structure and thus define that $x \leq y$ iff $x \wedge y = x$. Using commutativity and inclusion this implies the dual property that $x \vee y = (x \wedge y) \vee y = y$.

To show that $\leq$ is a partial order, idempotence shows reflexivity. If $x \leq y$ and $y \leq x$ then $x = x \wedge y = y \wedge x = y$ and thus antisymmetry. Finally suppose that $x \leq y$ and $y \leq z$, that is $x = x \wedge y$ and $y = y \wedge z$. Then

$$x \wedge z = (x \wedge y) \wedge z) = x \wedge (y \wedge z) = x \wedge y = x$$

and thus $x \leq z$. Associativity gives us that $x \wedge y \leq x, y$ if also $z \leq x, y$ then

$$z \wedge (x \wedge y) = (z \wedge x) \wedge y = z \wedge y = z$$

and thus $z \leq x \wedge y$, thus $x \wedge y$ is the unique greatest lower bound. The least upper bound is proven in the same way and the last axiom shows that 0 is the unique minimal and 1 the unique maximal element. □

Definition III.6: An element x of a lattice L is *join-irreducible* (JI) if $x \neq 0$ and if $x = y \vee z$ implies that $x = y$ or $x = z$.

For example, figure III.3 shows a lattice in which the black vertices are JI, the others not.

When representing elements of a finite lattices, it is possible to do so by storing the JI elements once and representing every element based on the JI elements that are below. This is used for example in one of the algorithms for calculating the subgroups of a group.

Product of posets

The cartesian product provides a way to construct new posets (or lattices) from old ones: Suppose that X, Y are posets with orderings $\leq_X, \leq_Y$, we define a partial order on $X \times Y$ by setting

$$(x_1, y_1) \leq (x_2, y_2) \quad \text{if and only if} \quad x_1 \leq_X x_2 \quad \text{and} \quad y_1 \leq_Y \leq y_2.$$

Proposition III.7: This is a partial ordering, so $X \times Y$ is a poset. If furthermore both X and Y are lattices, then so is $X \times Y$.

The proof of this is exercise **??**.

This allows us to describe two familiar lattices as constructed from smaller pieces (with a proof also delegated to the exercises):

Proposition III.8: a) Let $|A| = n$ and $\mathcal{P}(A)$ the power-set lattice (that is the subsets of A, sorted by inclusion). Then $\mathcal{P}(A)$ is (isomorphic to) the direct product of n copies of the two element lattice $\{0,1\}$.
b) For an integer $n = \prod_{i=1}^{r} p_i^{e_i} > 1$ written as a product of powers of distinct primes, let $\mathcal{D}(n)$ be the lattice of divisors of n. Then $\mathcal{D}(n) \cong \mathcal{D}(p_1^{e_1}) \times \cdots \times \mathcal{D}(p_r^{e_r})$.

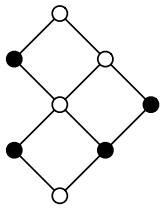


Figure III.3: Order-ideal lattice for the "N" poset.

III.3 Distributive Lattices

A lattice L is *distributive*, if for any $x, y, z \in L$ one (and thus also the other) of the two following laws hold:

$$\begin{aligned} x \vee (y \wedge z) &= (x \vee y) \wedge (x \vee z) \\ x \wedge (y \vee z) &= (x \wedge y) \vee (x \wedge z) \end{aligned}$$

Example: These laws clearly hold for the lattice of subsets of a set or the lattice of divisors of an integer n.

Lattices of substructures of algebraic structures are typically not distributive, the easiest example (diagram l) in figure III.2) is the lattice of subgroups of $C_2 \times C_2$ which also is the lattice of subspaces of F_2^2.

If $P = (X, \leq)$ is a poset, a subset $Y \leq X$ is an *order ideal*, if for any $y \in Y$ and $z \in X$ we have that $z \leq y$ implies $z \in Y$.

Lemma III.9: The set of order ideals is closed under union and intersection.

Proof: Let A, B be order ideals and $y \in A \cup B$ and $z \leq y$. Then $y \in A$ or $y \in B$. In the first case we have that $z \in A$, in the second case that $z \in B$, and thus always $z \in A \cup B$. The same argument also works for intersections. □

This implies:

Lemma III.10: The set of order ideals of P, denoted by $J(P)$ is a lattice under intersection and union.

As a sublattice of the lattice of subsets, $J(P)$ is clearly distributive.

For example, if P is the poset on 4 elements with a Hasse diagram given by the letter N (figure III.2, g) then figure III.3 describes the lattice $J(P)$.

In fact, any finite distributive lattice can be obtained this way

THEOREM III.11 (Fundamental Theorem for Finite Distributive Lattices, BIRKHOFF): Let L be a finite distributive lattice. Then there is a unique (up to isomorphism) finite poset P, such that $L \cong J(P)$.

To prove this theorem we use the following definition:

DEFINITION III.12: For any element x of the poset P, let $\downarrow x = \{y \mid y \leq x\}$ be the principal order ideal generated by x.

LEMMA III.13: An order ideal of a finite poset P is join irreducible in $J(P)$ if and only it is principal.

Proof: First consider a principal order ideal $\downarrow x$ and suppose that $\downarrow x = b \vee c$ with b and c being order ideals. Then $x \in b$ or $x \in c$, which by the order ideal property implies that $\downarrow x \subset b$ or $\leq x \subset c$.

Vice versa, suppose that a is a join irreducible order ideal and assume that a is not principal. Then for any $x \in a$, $\downarrow x$ is a proper subset of a. But clearly $a = \bigcup_{x \in a} \downarrow x$. □

COROLLARY III.14: Given a finite poset P, the set of join-irreducibles of $J(P)$, considered as a subposet of $J(P)$, is isomorphic to P.

Proof: Consider the map that maps x to $\downarrow x$. It maps P bijective to the set of join-irreducibles, and clearly preserves inclusion. □

We now can prove theorem III.11

Proof: Given a distributive lattice L, let X be the set of join-irreducible elements of L and P be the subposet of L formed by them. By corollary III.14, this is the only option for P up to isomorphism, which will show uniqueness.

Let $\phi: L \to J(P)$ be defined by $\phi(a) = \{x \in X \mid x \leq a\}$, that is it assigns to every element of L the JI elements below it. (Note that indeed $\phi(a)$ is an order ideal.). We want to show that ϕ is an isomorphism of lattices.

Step1: Clearly we have that $a = \bigvee_{x \in \phi(a)} x$ for any $a \in L$ (using the join over the empty set equal to 0). Thus ϕ is injective.

Step 2: To show that ϕ is surjective, let $Y \in J(P)$ be an order ideal of P, and let $a = \bigvee_{y \in Y} y$. We aim to show that $\phi(a) = Y$: Clearly every $y \in Y$ also has $y \leq a$, so $Y \subset \phi(a)$. Next take a join irreducible $x \in \phi(a)$, that is $x \leq a$. Then $x \leq \bigvee_{y \in Y} y$ and thus

$$x = x \wedge \left(\bigvee_{y \in Y} y \right) = \bigvee_{y \in Y} (x \wedge y)$$

by the distributive law. Because x is JI, we must have that $x = x \wedge y$ for some $y \in Y$, implying that $x \leq y$. But as Y is an order ideal this implies that $x \in Y$. Thus $\phi(a) \subset Y$ and thus equality, showing that ϕ is surjective.

Step 3: We finally need to show that ϕ maps the lattice operations: Let $x \in X$. Then $x \le a \wedge b$ if and only if $x \le a$ and $x \le b$. Thus $\phi(a \wedge b) = \phi(a) \cap \phi(b)$.

For the join, take $x \in \phi(a) \cup \phi(b)$. Then $x \in \phi(a)$, implying $x \le a$, or (same argument) $x \le b$; therefore $x \le a \vee b$. Vice versa, suppose that $x \in \phi(a \vee b)$, so $x \le a \vee b$ and thus

$$x = x \wedge (a \vee b) = (x \wedge a) \vee (x \wedge b).$$

Because x is JI that implies $x = x \wedge a$, respectively $x = x \wedge b$.

In the first case this gives $x \le a$ and thus $x \in \phi(a)$; the second case similarly gives $x \in \phi(b)$. □

Example: If we take the lattice of subsets of a set, the join-irreducibles are the 1-element sets. If we take divisors of n, the join-irreducibles are prime powers.

III.4 Chains and Extremal Set Theory

> Man is born free;
> and everywhere he is in chains
>
> The Social Contract
> Jean-Jacques Rousseau

Definition III.15: A *chain* in a poset is a subset such that any two elements of it are comparable. (That is, restricted to the chain the order is total.)

An *antichain* is a subset, such that any two (different) elements are incomparable.

We shall talk about a partition of a poset into a collection of chains (or antichains) if the set of elements is partitioned.

Clearly a chain C and antichain A can intersect in at most one element. This gives the following duality:

Lemma III.16: Let P be a poset.
a) If P has a chain of size r, then it cannot be partitioned in fewer than r antichains.
b) If P has an antchain of size r, then it cannot be partitioned in fewer than r chains.

A stronger version of this goes usually under the name of Dilworth's theorem[1].

Theorem III.17 (Dilworth, 1950): The minimum number m of chains in a partition of a finite poset P is equal to the maximum number M of elements in an antichain.

[1] proven earlier by Gallai and Milgram

Proof: The previous lemma shows that $m \geq M$, so we only need to show that we can partition P into M chains. We use induction on $|P|$, in the base case $|P| = 1$ nothing needs to be shown.

Consider a chain C in P of maximal size. If every antichain in $P \setminus C$ contains at most $M - 1$ elements, we apply induction and partition $P \setminus C$ into $M - 1$ chains and are done.

Thus assume now that $\{a_1, \ldots, a_M\}$ was an antichain in $P \setminus C$. Let

$$\begin{aligned} S^- &= \{x \in P \mid x \leq a_i \text{ for some } i\} \\ S^+ &= \{x \in P \mid x \geq a_i \text{ for some } i\} \end{aligned}$$

Then $S^- \cup S^+ = P$, as there otherwise would be an element we could add to the antichain and increase its size.

As C is of maximal size, the largest element of C cannot be in S^-, and thus we can apply induction to S^-. As there is an antichain of cardinality M in S^-, we partition S^- into M disjoint chains.

Similarly we partition S^+ into M disjoint chains. But each a_i is maximal element of exactly one chain in S^- and minimal element of exactly one chain of S^+. We can combine these chains at the a_i's and thus partition P into M chains. □

Corollary III.18: If P is a poset with $nm + 1$ elements, it has a chain of size $n + 1$ or an antichain of size $m + 1$.

Proof: Suppose not, then every antichain has at most m elements and by Dilworth's theorem we can partition P into m chains of size $\leq n$ each, so $|P| \leq mn$. □

Corollary III.19 (Erdős-Szekeres, 1935): Every sequence of $nm + 1$ distinct integers contains an increasing subsequence of length at least $n + 1$, or a decreasing subsequence of at length least $m + 1$.

Proof: Suppose the sequence is $a_1, \ldots, a_N$ with $N = nm + 1$. We construct a poset on N elements $x_1, \ldots, x_N$ by defining $x_i \leq x_j$ if and only if $i \leq j$ and $a_i \leq a_j$. (Verify that it is a partial order!) □

The theorems of this section in fact belong into a bigger context that has its own chapter, chapter IV, devoted to.

A similar argument applies in the following two famous theorema:

Theorem III.20 (Sperner, 1928): Let $N = \{1, \ldots, n\}$ and $A_1, \ldots A_m \subset N$, such that $A_i \not\subset A_j$ if $i \neq j$. Then $m \leq \binom{n}{\lfloor n/2 \rfloor}$.

Proof: Consider the poset of subsets of N and let $\mathcal{A} = \{A_1, \ldots, A_m\}$. Then $\mathcal{A}$ is an antichain.

A maximal chain $\mathcal{C}$ in this poset will consist of sets that iteratively add one new point, so there are $n!$ maximal chains, and $k!(n-k)!$ maximal chains that involve a particular k-subset of N.

We now count the pairs $(A, \mathcal{C})$ such that $A \in \mathcal{A}$ and $\mathcal{C}$ is a maximal chain with $A \in \mathcal{C}$. As a chain can contain at most one element of an antichain this is at most $n!$.

On the other hand, denoting by a_k the number of sets A_i with $|A_i| = k$, we know there are

$$n! \geq \sum_{k=0}^{n} k!(n-k)!a_k = n!\left(\sum_{k=0}^{n} \frac{a_k}{\binom{n}{k}}\right)$$

such pairs. As $\binom{n}{k}$ is maximal for $k = \lfloor n/2 \rfloor$ we get

$$\binom{n}{\lfloor n/2 \rfloor} \geq \binom{n}{\lfloor n/2 \rfloor} \sum_{k=0}^{n} \frac{a_k}{\binom{n}{k}} \geq \sum_{k=0}^{n} a_k = m.$$

□

We note that equality is achieved if $\mathcal{A}$ is the set of all $\lfloor n/2 \rfloor$-subsets of N.

THEOREM III.21 (ERDŐS-KO-RADO, 1961): Let $\mathcal{A} = \{A_1, \ldots, A_m\}$ a collection of m distinct k-subsets of $N = \{1, \ldots, n\}$, where $k \leq n/2$, such that any two subsets have nonempty intersection. Then $m \leq \binom{n-1}{k-1}$.

Proof: Consider "cyclic k-sequences" $\mathcal{F} = \{F_1, \ldots, F_n\}$ with $F_i = \{i, i+1, \ldots, i+k-1\}$, taken "modulo n" (that is each number should be $((x-1) \bmod n) + 1$).

Note that $\mathcal{A} \cap \mathcal{F} \leq k$, since if some F_i equals A_j, then any only other $F_l \in \mathcal{A}$ must intersect F_i, so we only need to consider (again considering indices modulo n) F_l for $i - k + 1 \leq l \leq i + k - 1$. But F_l will not intersect F_{l+k}, allowing at most for a set of k subsequent F_i's to be in $\mathcal{A}$.

As this holds for an arbitrary $\mathcal{A}$, the result remains true after applying any arbitrary permutation π to the numbers in $\mathcal{F}$. Thus

$$z := \sum_{\pi \in S_n} |\mathcal{A} \cap \mathcal{F}^{\pi}| \leq k \cdot n!$$

We now calculate the sum z by fixing $A_j \in \mathcal{A}$, $F_i \in \mathcal{F}$ and observe that there are $k!(n-k)!$ permutations π such that $F_i^{\pi} = A_j$. Thus $z = m \cdot n \cdot k!(n-k)!$, proving the theorem. □

III.5 Incidence Algebras and Möbius Functions

A common tool in mathematics is to consider instead of a set S the set of functions defined on S. To use this paradigm for (finite) posets, define an *interval* on a poset

P as a set of elements z such that $x \le z \le y$ for a given pair $x \le y$, and denote by $\mathrm{Int}(P)$ the set of all intervals.

For a field K, we shall consider the set of functions on the intervals:

$$I(P) = I(P, K) = \{f : \mathrm{Int(P)} \to K\}$$

and call it the *incidence algebra* of P. This set of functions is obviously a K-vector space under pointwise operations. We shall denote intervals by their end points x, y and thus write $f(x, y) \in I(P)$.

We also define a multiplication on $I(P)$ by defining, for $f, g \in I(P)$ a function fg by

$$(fg)(x, y) = \sum_{x \le z \le y} f(x, z)g(z, y)$$

In exercise **??** we will show that with this definition $I(P)$ becomes an associative K-algebra[2] with a one, given by

$$\delta(x, y) = \begin{cases} 1 & x = y \\ 0 & x \neq y \end{cases} .$$

We could consider $I(P)$ as the set of formal K-linear combinations of intervals $[x, y]$ and a product defined by

$$[x, y][a, b] = \begin{cases} [x, b] & a = y \\ 0 & a \neq y \end{cases} ,$$

and extended bilinearily.

If P is finite, we can, by theorem III.3, arrange the elements of P as $x_1, \ldots, x_n$ where $x_i \le x_j$ implies that $i \le j$. Then $I(P)$ is, by exercise **??** isomorphic to the algebra of upper triangular matrices $M = (m_{i,j})$ where $m_{i,j} = 0$ if $x_i \not\le x_j$.

Lemma III.22: Let $f \in I(P)$. Then f has a (two-sided) inverse if and only if $f(x, x) \neq 0$ for all $x \in P$.

Proof: The property $fg = \delta$ is equivalent to:

$$f(x, x)g(x, x) = 1 \qquad \text{for all } x \in P,$$

(implying the necessity of $f(x, x) \neq 0$) and

$$g(x, y) = -f(x, x)^{-1} \sum_{x < z \le y} f(x, z)g(z, y).$$

If $f(x, x) \neq 0$ the second formula will define the values f^{-1} uniquely, depending only on the interval $[x, y]$. Reverting the roles of f and g shows the existence of a

[2] An *algebra* is a structure that is both a vector space and a ring, such that vector space and ring operations interact as one would expect. The prototype is the set of matrices over a field.

left inverse and standard algebra shows that both have to be equal. □

The *zeta function* of P is the characteristic function of the underlying relation, that is $\zeta(x, y) = 1$ if and only if $x \leq y$ (and 0 otherwise).

This implies that

$$\zeta^2(x, y) = \sum_{x \leq z \leq y} \zeta(x, z)\zeta(z, y) = \sum_{x \leq z \leq y} 1 = \left|\{z \mid z \leq z \leq y\}\right|.$$

By Lemma III.22, ζ is invertible. The inverse $\mu = \zeta^{-1}$ is called the *Möbius function* of the lattice P. The identities

$$\mu(x, x) = 1 \tag{III.23}$$

$$\mu(x, y) = -\sum_{x \leq z < y} \mu(x, z) \tag{III.24}$$

follow from $\mu\zeta = \delta$ and allow for a recursive computation of values of μ and imply that μ is integer-valued.

For illustration, we shall compute the Möbius function for a number of common posets.

Lemma III.25: Let P be the total order on the numbers $\{1, \ldots, n\}$. Then for any $x, y \in P$ we have:

$$\mu(x, y) = \begin{cases} 1 & \text{if } x = y \\ -1 & \text{if } x + 1 = y \\ 0 & \text{otherwise} \end{cases}$$

Proof: The case of $x = y$ is trivial. If $x + 1 = y$, the sum in III.24 has only one summand, and the result follows. Thus assume that $x \leq y$ but $y \notin \{x, x + 1\}$. Then

$$\mu(x, y) = -\mu(x, x) - \mu(x, x + 1) - \sum_{x+2 \leq z < y} \mu(x, z) = -\sum_{x+2 \leq z < y} \mu(x, z)$$

and the result follows by induction on $y - x$. □

Lemma III.26: If P, Q are posets, the Möbius function on $P \times Q$ satisfies

$$\mu((x_1, y_1), (x_2, y_2)) = \mu_P(x_1, x_2)\mu_Q(y_1, y_2)$$

Proof: It is sufficient to verify that the right hand side of the equation satisfies III.24. □

Together with Theorem III.8 and Lemma III.25 we get

Corollary III.27: a) For $X, Y \in \mathcal{P}(A)$, we have that $\mu(X, Y) = -1^{|Y|-|X|}$ if $X \subseteq Y$, and 0 otherwise.
b) If x, y are divisors of n, then in $\mathcal{D}(n)$ we have that $\mu(x, y) = (-1)^d$ if y/x is the product of d different primes, and 0 otherwise.

Part b) explains the name: $\mu(1, n)$ is the value of the classical number theoretic Möbius function.

Part a) connects us back to section III.1: The Möbius function gives the coefficients for inclusion/exclusion over an arbitrary poset. We will investigate and clarify this further in the rest of this section.

Möbius inversion

The property of being inverse of the incidence function can be used to invert summation formulas with the aid of the Möbius function:

Theorem III.28 (Möbius inversion formula): Let P be a finite poset, and $f, g: P \to K$, where K is a field. Then

$$g(t) = \sum_{s \leq t} f(s) \qquad \text{for all } t \in P$$

is equivalent to

$$f(t) = \sum_{s \leq t} g(s)\mu(s, t) \qquad \text{for all } t \in P.$$

Proof: Let K^P be the K-vector space of functions $P \to K$. The incidence algebra $I(P)$ acts linearly on this vector space by

$$(f\xi)(t) = \sum_{s \leq t} f(s)\xi(s, t).$$

The two equations thus become

$$g = f\zeta, \qquad \text{respectively} \qquad f = g\mu,$$

and their equivalence follows from the fact that μ is the inverse of ζ. □

The classical Möbius inversion formula from Number Theory follows as a special case of this.

If we consider the linear poset $\{1, \ldots, n\}$ with total ordering, Lemma III.25 gives us that (assuming $f(0) = g(0)$)

$$g(n) = \sum_{i=0}^{n} f(i)$$

is (unsurprisingly) equivalent to

$$f(n) = g(n) - g(n-1),$$

the finite difference analog of the fundamental theorem of Calculus!

Going back to the start of the chapter, consider n subsets A_i of a set X. We take the the poset $\mathcal{P}(\{1,\ldots,n\})$ and define, for $I \subset \{1,\ldots,n\}$

$$\begin{aligned} f(I) &= -1^{|I|}\left|X \setminus \bigcup_{i\in I} A_i\right| \\ g(I) &= \left|\bigcap_{i\in I} A_i\right|. \end{aligned}$$

Then $g(I) = \sum_{J\subset I} f(J)$ by Inclusion/Exclusion over the complements $X \setminus A_i$, while

$$f(I) = \sum_{J\subset I} g(J)\mu(J,I) = \sum_{J} -1^{|I|-|J|}\left|\bigcap_{j\in J} A_j\right| = -1^{|I|}\,|X \setminus \bigcup A_i|$$

is the ordinary inclusion/exclusion formula.

Chapter

IV Connections

Mathematicians are like Frenchmen.
When you talk to them, they translate
it into their own language,
and then it is something quite different.

Die Mathematiker sind eine Art Franzosen;
redet man zu ihnen, so übersetzen sie es
in ihre Sprache, und dann ist es
alsobald ganz etwas anders.

Maximen und Reflexionen:
Über Natur und Naturwissenschaft
Johann Wolfgang von Goethe

With so many joints and connections, leaks were plentiful. As the magazine *The Builder* remarked, in 1856: "The fate of a theater is to be burned. It seems simply a question of time."

Connections
James Burke

In this chapter we will look at connections – both in an applied sense of modeling situations of connected objects, in the abstract sense of connecting mathematical theorems that initially seem to be unrelated, and in connecting concepts that might seem to be hopelessly abstract to practical applications. One of the joys of mathematics is to discover such connections and see the unity of the discipline.

This practical relevance of the results places much of this chapter also in close contact to the realm of (discrete) optimization.

We shall investigate a gaggle of theorems (which each in their area are fundamental), but which turn out to be equivalent in the sense that we can deduce each

from each other. Furthermore this kind of derivation is often easier than to derive the theorem from scratch. The description owes much to [Rei84]

Many theorems in this chapter are formulated in the language of graphs: We typically denote a graph by $\Gamma = (V, E)$ with vertioces V and edges E being 2-element sets of vertices. A *digraph* is a graph with directed edges (that is we consider edges as elements of $V \times V$ instead of 2-element subsets of V). In a digraph, we would allow distinct edges (x, y) and (y, x). Weighted edges means we have a weight function $w\colon E \to \mathbb{R}$.

Our start will be Dilworth's Theorem III.17 that we have proven already:

> The minimum number m of chains in a part ion of a finite poset P is equal to the maximum number M of elements in an antichain.

IV.1 Halls' Marriage Theorem

He was, methinks, an understanding fellow who said, 'twas a happy marriage betwixt a blind wife and a deaf husband.

Celuy là s'y entendoit, ce me semble, qui dict qu'un bon mariage se dressoit d'une femme aveugle avec un mary sourd.

Essais, Livre III
Michel de Montaigne

Consider a family of subsets $A_1, \dots, A_n \subseteq X$. A *system of distinct representatives* (SDR) for these sets is an n-tuple of *distinct* elements $x_1, \dots, x_n$ such that $x_i \in A_i$. Example: SDRs do not have to exist, for example consider $A_1 = A_2 = \{1\}$.

We define, for a subset $J \subseteq \{1, \dots, n\}$ of indices, a set

$$A(J) = \bigcup_{j \in J} A_j.$$

We immediately see a reason for the above example failing: For an SDR to exist, by necessity $|A(J)| \geq |J|$ for any such subset J, since there are otherwise not sufficiently many elements available to pick distinct representatives. The following Theorem[1] shows this condition is not only necessary, but also sufficient.

Theorem IV.1 (Halls' Marriage Theorem): The family $\{A_1, \dots, A_n\}$ of finite sets has a system of distinct representatives if and only if

$$|A(J)| \geq |J| \quad \text{for all} \quad J \subseteq \{1, \dots, n\}. \tag{IV.2}$$

The name "Marriage Theorem" comes from the following interpretation: Suppose we have a set of m men and n women. We let A_i be the set of men that women

[1] Proven by the British Mathematician Philip Hall and extended by the unrelated American Mathematician Marshall Hall for the infinite case. Thus the apostrophe placement in the section title.

i would consider for a potential marriage[2]. Then every women can marry a suitable man if and only if every group of k women together considers at least k men as suitable.

Proof:[Dilworth⇒Hall] As the necessity of the condition is trivial, we show sufficiency:

Given sets A_i satisfying condition (IV.2), Let $Y = \{y_1, \dots, y_n\}$ be n symbols representing the sets. We create a poset P in the following way: The elements of P is the disjoint union $X \cup Y$. The only relations are that $x \le y_i$ iff $x \in A_i$.

Clearly X is an antichain in P. Suppose S is another arbitrary antichain and let $J = \{1 \le j \le n \mid y_j \in S\}$. The antichain criterion imposes that $A(J) \cap S = \varnothing$, so

$$|S| \le |J| + (|X| - |A(J)|) \le |X|$$

because of (IV.2). That means X is a maximal antichain, and by Dilworth's theorem, P can be partitioned into $|X|$ chains. As a chain cannot contain more than one point from X or more than one point from Y, the pigeonhole principle implies that each chain contains exactly one point from X and at most one point from Y. Suppose that $x_i \in X$ is the element that is together with $y_i \in Y$ in a chain. Then $x_i \le y_i$, and thus $x_i \in A_i$, so $\{x_1, \dots, x_n\}$ is a SDR. □

As an example of the use of this theorem we consider the following theorem due to G. Birkhoff. A matrix $M \in \mathbb{Z}_{\ge 0}^{n \times n}$ with nonnegative entries is called *doubly statistic* if every row and every column of M has the same row/column sum.

Corollary IV.3: A doubly statistic matrix $M = (m_{ij})$ with row/column sum l can be written as the sum of l permutation matrices.

Proof: We define sets A_i, corresponding to the row indices by $A_i = \{j \mid m_{ij} > 0\}$. For any k-tuple K of indices, the sum of the corresponding rows of M is kl. As every column of M has column sum l, this means that these k rows must have nonzero entries in at least k columns, that is $|\bigcup_{i \in K} A_i| \ge k$. Thus IV.2 is satisfied and there is a SDR. We set $p_{i,j} = 1$ if j is the chosen representative for A_i (and $p_{i,j} = 0$ otherwise). Then $P = (p_{i,j})$ is a permutation matrix and $M - P$ is doubly statistic with sum $l - 1$. The statement follows by induction on l. □

IV.2 Kőnig's Theorems – Matchings

> Eventually everything connects - people, ideas, objects…the quality of the connections is the key to quality per se.
>
> Charles Eames

[2]The theorem goes back to times of more restrictive social mores. The concerned reader might want to consider k applicants for n jobs and A_i being the set of candidates that satisfy the conditions for job i.

Let A be an $m \times n$ matrix with 0/1 entries. A *line* of A is a row or column. A set of lines *covers* A, if every nonzero entry of A lies on at least one of the lines. Nonzero entries are called *independent* if no two lie on a common line. The *term rank* of A is the maximum number of independent entries of A.

The following theorem is reminiscent of the properties of a basis in linear algebra:

THEOREM IV.4 (KŐNIG-EGERVÁRY): The minimum number of lines covering A equals the term rank of A.

Example: Let

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

Then A can be covered with 3 lines, and has an independent set of cardinality 3.

Proof:[HALL⇒KŐNIG-EGERVÁRY] Given $A \in \{0,1\}^{m\times n}$, let p be the term rank of A and q the minimum number of lines covering A. We have to show that $p = q$.

We first show that $p \le q$: A cover of l lines can cover at most l independent 1's (no line covers more than one) but we can cover *all* ones with q lines, so $p \le q$.

To see that $p \ge q$, without loss of generality permute the rows and columns of A such that a minimal cover involves the first r rows and the last s columns of the matrix. We aim to find an independent set that has r entries in the first r rows and in columns $1, \ldots, n-s$, and s entries in the last s columns in rows $r+1, \ldots, m$:

For a row index $1 \le i \le r$ let $N_i = \{1 \le j \le n-s \mid A_{i,j} = 1\}$. Then the union of any k of these N_i's contains at least k column indices – otherwise we could replace these k rows with $< k$ columns in a minimal cover. By Hall's theorem we thus have an SDR $\{x_1, \ldots, x_r\}$. By definition $x_i \in N_i$ implies that $A_{i,x_i} = 1$. Let $S = \{(i, x_i) \mid i = 1, \ldots, r\}$. Since the x_i are distinct, this is an independent set.

A dual argument for the last s columns gives an independent set T of positions in the last s columns and rows $r+1, \ldots, n$. Since no position in S shares row or column index with any position in T, $S \cup T$ also is an independent set, of cardinality $r + s = q$. □

We reformulate this theorem in the language of graphs. A graph is called *bipartite*, if its vertex V set can be written as a disjoint union $V = V_1 \cup V_2$, so that no vertex in V_i has a neighbor in V_i (but only neighbors in V_{3-i}). A *vertex cover* in a graph is a set of vertices that every edge is incident to at least one vertex in the cover. A *matching* in a graph is a set of edges such that no two edges are incident to the same vertex. A matching is called maximal, if no matching with a larger number of edges exists.

We assume that the graphs we consider here have no isolated vertices.

THEOREM IV.5 (Kőnig): In a bipartite graph without isolated vertices, the number of edges in a maximum matching equals the number of vertices in a minimum

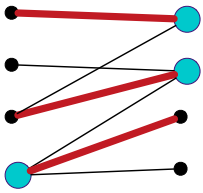


Figure IV.1: A bipartite graph with maximum matching and minimum vertex cover.

vertex cover.

Example: Figure IV.1 shows a bipartite graph with a maximum matching (bold).
Proof:[Kőnig-Egerváry⇒Kőnig] Let $W, U \subset V$ be the two parts of the bipartition of vertices. Suppose $|W| = m$, $|U| = n$. We describe the adjacency in an $m \times n$ matrix A with 0/1 entries, $A_{i,j} = 1$ iff w_i is adjacent to u_j. (Note that the examples for this and the previous theorem are illustrating such a situation.)

A matching in the graph corresponds to an independent set — edges sharing no common vertices. A vertex cover — vertices such that every edge is incident — correspond to a line cover of the matrix. The result follows immediately from the Kőnig-Egerváry theorem. □

Matchings in bipartite graphs can be interpreted as creating assignments between tasks and operators, between customers and producers, between men and women, etc., and thus clearly have practical implications (Compare the Marriage Theorem interpretation of Hall's theorem!)

Due to this practical relevance, there is obvious interest in an algorithm for finding maximum matchings in a bipartite graph, the *Assignment problem*. The first algorithm published uses Kőnig's theorem as a criterion whether an existing matching can be improved. Due to the origin of the theorem authors, it has been named the *Hungarian Method*. (It turns out that this method had been discovered before independently by Jacobi [Jac90].)

Let $M \subset E$ be a (not necessarily maximal) matching in a graph $\Gamma = (V, E)$. We call an edge *matched* if it is in M, and *free* otherwise. Similarly, vertices incident to matched edges are called matched, vertices not incident to any matched edge are free.

An *augmenting path* for M is a path whose end points are free and whose edges are alternatingly matched and free. This means all vertices but the end points are matched.

If the cardinality of M is that of a vertex cover C, no augmenting paths can exist: Every edge of M will be incident with exactly one vertex in C, leaving one of the two free edges at the end of the path without cover. Otherwise there must be

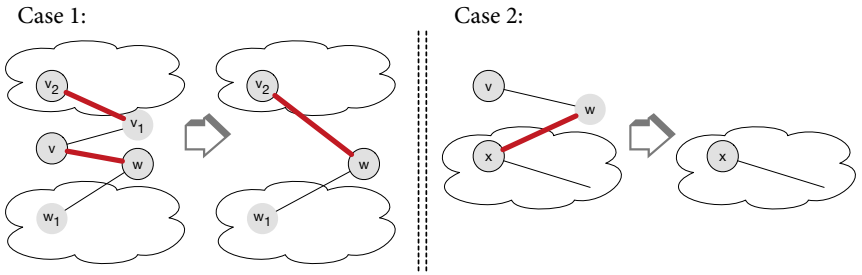


Figure IV.2: The two reduction in Corollary IV.6.

an augmenting path:

COROLLARY IV.6: If $|M| < |C|$ for a vertex cover C, there either is a vertex cover of smaller size, or there is an augmenting path for M.

Proof: We prove the statement by induction over the number of vertices $|V|$, the base case being trivial.

Case 1: Assume that two vertices $v, w \in C$ are connected by an edge in M. Then both v and w must be incident to other edges, as we otherwise could drop one of them from C and obtain a strictly smaller matching.

Furthermore one such edge $\{v, v_1\}$ must be to a vertex $v_1 \notin C$ (as we otherwise could drop v from the vertex cover). Clearly $\{v, v_1\}$ is free. We similarly find a free edge $\{w, w_1\}$ to a vertex $w_1 \notin C$. If both v_1 and w_1 are free, then v_1, v, w, w_1 is an augmenting path. Otherwise at least one vertex, WLOG v_1, is matched thus incident to an edge $\{v_1, v_2\} \in M$. As we assumed that $v_1 \notin C$ this implies that $v_2 \in C$, as the graph is bipartite we know that $v_2 \neq w$. Now consider the smaller graph Γ' obtained from Γ by deleting v and v_1 (and the adjacent edges) and adding an edge $\{v_2, w\}$ if it did not yet exist. Then $M' = (M - \{\{v, w\}, \{v_1, v_2\}\}) \cup \{\{v_2, w\}$ is a matching in Γ' and $C' = C - \{v\}$ a vertex cover in Γ' (see Figure IV.2, left). By induction, we can either obtain a strictly smaller vertex cover D for Γ' (in which case $D \cup \{v\}$ or $D \cup \{v_1\}$ is a strictly smaller cover for Γ), or an augmenting path for Γ'. If this path uses the new matching edge $\{v_2, w\}$, we replace this edge by the edge sequence $\{v_2, v_1\}, \{v_1, v\}, \{v, w\}$ and obtain an augmenting path in Γ. This completes case 1.

In Case 2, we now may assume that no two vertices of the cover are connected by an edge in the matching, as $|M| \leq |C|$ there must be an unmatched vertex $v \in C$.

If v is adjacent to any other unmatched vertex this is an augmented path of length 1.

Otherwise v is adjacent to a matched vertex $w \notin C$ (if all neighbors are in C, we could remove v from C). By a matched edge, w is adjacent to another vertex x, and to cover that edge we must have $x \in C$. We furthermore may assume that w was

chosen so that x is also incident to an edge not in the matching, as we otherwise could replace all x's with the respective w's in the cover and remove v from C, reducing the cover size.

Now consider the graph Γ', obtained by removing the vertices v and w and their incident edges and set $M' = M \setminus \{\{x, w\}\}$ and $C' = C \setminus \{v\}$ (Figure IV.2, right).

Then C' is a vertex cover of Γ' (all edges that v covered are gone). Also M' is a matching with $|M'| < |C'|$. By induction, either Γ' has a smaller cover D (in which case $D \cup \{v\}$ is a smaller cover for Γ), or there is an augmenting path. If this path imvolves x, we extend it by xwv and thus obtain an augmenting path for Γ. □

The existence of an augmenting path allows us to replace M with a larger matching by replacing the matched edges of an augmenting path in M with the free edges in the path.

To find a maximal matching thus simply start with some matching and searches for augmenting paths.

It is not hard to generalize this approach to (complete) bipartite graphs with weighted edges to obtain a (perfect: all vertices are matched) matching of maximal weight (profit):

Given an arbitrary matching with weighted edges, replace an edge $\{a, b\}$ with (assume: integral) weight w by a set of edges $\{a, b_1\}, \{b_1, a_2\}, \{a_2, b_2\}, \ldots, \{a_w, b\}$ with newly introduced vertices a_i, b_i that are not connected in any other way. Selecting the edge $\{a, b\}$ in the original graph thus now allows selection of w edges. Thus the maximum weight of a matching in the original graph corresponds to the cardinality of a maximum matching in the new graph.

Stable Matchings

> …to Alvin E. Roth and Lloyd S. Shapley "for the theory of stable allocations and the practice of market design".
>
> Citation for the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2012

Also of practical interest is the concept of a stable matching (or stable marriage). Imagine a matching in a bipartite graph represents an assignment of students to college places. Every student has a personal preference ranking of colleges, every college (place – assume for simplicity that every college just has one student) has a personal ranking of students. A matching is *instable* (otherwise: *stable*), if there is a student s and a college c such that s ranks c higher than their current college, and c ranks s higher than their current student.

A priori it is unclear that stable matchings exist, the following algorithm not only proves that but also gives a way of producing one:

ALGORITHM IV.7 (Deferred Acceptance, GALE, SHAPLEY): Every student has a temporarily assigned college, colleges may have a temporarily assigned student. Once a college has been assigned a student, it will replace it only by one it ranks higher. A student may be moved down from higher ranked to lower ranked colleges.

For bookkeeping, students carry a *rejected* label that will change status as the algorithm goes on, and also and carry for each college an indicator whether they have been rejected by this college.

1. [Initialize] Label every student as rejected

2. [Complete] If no student is rejected, terminate.

3. [Student Choice] Every student marked as rejected applies to the college she ranks highest amongst those who have not yet rejected her.

4. [College Choice] Every college tentatively picks from the students who chose it the one it ranks highest (and removes that students rejection status). It rejects all other students who have selected it,even if they had been picked tentatively before.

5. Go back to step 2.

Proof: The only way the process terminates is if no student is rejected, that is they were tentatively accepted by a college and this college has no student applying it would rank higher. All colleges which the student ranks higher have rejected her because they were able to accept a student they rank higher, so the matching is stable.

In every round, students select from colleges and are either tentatively accepted, or are left with colleges of lower preference. As there is a finite set of students and preferences, this process must terminate. □

This algorithm is used for example in the US to match graduating medical doctors to hospital training positions, `www.nrmp.org`. It also was instrumental for the award of the 2012 Nobel prize in Economics to SHAPLEY and ROTH.

IV.3 Menger's theorem

> Live in fragments no longer. Only connect, and the beast and the monk, robbed of the isolation that is life to either, will die.
>
> Howards End
> E. M. FORSTER

Kőnig's aim in proving the Theorem IV.5 was as a tool towards a proof for a more general theorem by Menger that deals with connections in a graph (and gets us in statement and proof back close to Dilworth's theorem), but whose proof, when first published, turned out to not work in the case of bipartite graphs.

Let u, w be nonadjacent vertices in a graph Γ. A *uw-vertex path* in Γ is a path from u to w. A collection of uw-vertex paths is called *independent*, if the paths are pairwise disjoint except for u and w. A set S of vertices, excluding u and w, is *uw-vertex separating*, if every uw path must contain a vertex of S. Clearly the minimum cardinality of a uw vertex separating set must be at least as large as that of a number of uw independent paths. It turns out that they are equal:

Theorem IV.8 (Menger): Let u, w be nonadjacent vertices in a graph. Then the maximum number of independent uw vertex paths equals the minimum cardinality of a uw vertex separating set.

As Kőnig's proof is comparatively long, we will instead give a direct proof due to G. Dirac. Note the similarity to the proof of Dilworth's theorem!

Proof: Let m be the minimum cardinality of a separating set, and M the maximum number of independent paths. We have seen that $m \geq M$.

Assume that the theorem is false and $\Gamma = (V, E)$ be a graph with a minimum number of edges for which the theorem fails, that is we have two nonadjacent vertices u and w and fewer than m independent vertex paths.

If e is an edge not incident to u or w, we can remove e and obtain a smaller graph Δ for which the theorem holds. Since Γ, and thus Δ has at most $m-1$ uw-independent paths this means that in Δ there is an uw-separating set T of size $m-1$. If we add one of the vertices incident to e to this set T, we obtain a set S of cardinality m which is uw separating – any path connecting in Γ but not in Δ clearly would have to use the edge e. We thus may assume that S contains a vertex not adjacent to u or w.

Next, we note that there is no path usw with $s \in S$ of length 2 – if there was we could remove s and have a graph with fewer edges and $m-1$ separating vertices and less than $m-1$ independent paths, contradicting the minimality of Γ.

For the final contradiction, denote by P_u the set of vertex paths between u and exactly one vertex in S and P_w ditto for w. The paths in P_u and P_w have only vertices of S in common (otherwise we could circumvent S, contradicting that it is separating).

We claim that all vertices of S are adjacent to u, or that all vertices in S are adjacent to w. This will contradict the choice of S and prove the theorem:

Consider the graph consisting of P_u, w and edges sw for $s \in S$. If w is not adjacent to all vertices in s, this graph has fewer edges than G, so by assumption has m independent paths from u to w. If we leave out the sw-step from these paths we obtain a set R_u of m independent paths from u to the m elements of S. Similarly, we get a set of independent paths R_w from w to the elements of S. Combining these paths at the elements of s yields m independent paths from u to w, contradiction. □

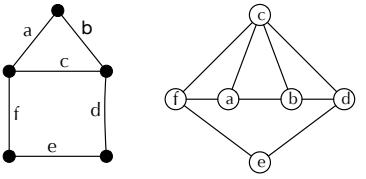


Figure IV.3: A graph and its line graph

A dual version of the theorem holds for disjoint paths:

THEOREM IV.9 (MENGER's theorem, edge version): The maximum number of edge-disjoint paths from u to w equals the minimum number of edges, whose removal would put u and v into disjoint components.

The proof of the theorem is by the ordinary version, applied to the *line graph* $L(\Gamma)$ of Γ. The vertices of $L(\Gamma)$ are the edges of Γ, two vertices are adjacent if the edges in Γ are incident to the same vertex. See figure IV.3 for an example.

MENGER's theorem also generalizes in the obvious way to directed graphs, though we shall not give a statement or proof here.

IV.4 Max-flow/Min-cut

> And he thought of himself floating on his back down a river, or striking out from one island to another, and he felt that that was really the life for a Tigger.
>
> The house at Pooh corner
> A. A. MILNE

A *network* is a digraph (V, E) with weighted edges with two distinguished vertices, the *source* s and the *target* t. We assume that the weights (which we call *capacities*) $c\colon E \to \mathbb{R}_{\geq 0}$ are nonnegative. (Below we shall assume that weights are in fact rational – clearly we can aproximate real weights to arbitrary precision. We shall investigate the issue of irrational capacities in the exercises.) An example is given in Figure IV.4. For a (directed) edge $e = (a, b)$ we denote by $\iota(e) = a$ the initial, and by $\tau(e) = b$ the terminal vertex of this edge.

The reader might want to consider this as a network of roads with weights being the maximal capacity. (It is possible to have different capacities in different directions on the same road.) We want to transport vehicles from s to t (and implicitly ask how many we can transport through the network).

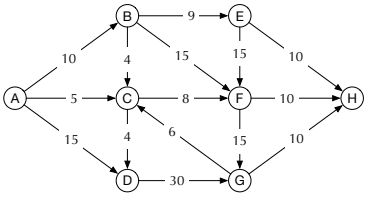


Figure IV.4: A network

Note that we are looking for a steady state in a contnuous transport situation, that is we do not want to transport once, but forever and ask for the transport per time unit.

A *flow* is a function f defined on the edges (indicating the number of units we transport along that edge) such that

- $0 \leq f(e) \leq c(e) \forall e \in E$ (do not exceed capacity).
- $\sum_{\iota(e)=v} f(e) = \sum_{\tau(e)=v} f(e) \forall v \in V \setminus \{s,t\}$ (No intermediate vertex can buffer capacity or create new flow. This is KIRCHHOFF's law in Physics.)

The *value of a flow* is the net transport out of the source:

$$\mathrm{val}(f) = \sum_{\iota(e)=s} f(e) - \sum_{\tau(e)=s} f(e).$$

This definition might seem asymmetric in focussing on the source, however Lemma IV.10 will show that this is a somewhat arbitrary choice.

The question we ask thus is for the maximal value of a flow. The main tool for this is to cosider the flow passing "by a point" in the network:

A *cut* $C \subset V$ in a network is a set of vertices such that $s \in C$, $t \notin C$. We do not require the cut to be connected, though in practice it usually will be.

We note that the value of any flow is equal to the net flow out of a cut. (This implies that the value of a flow also is equal to the net transport into the target):

LEMMA IV.10: Let C be a cut. We denote the edges crossing the cut by

$$\begin{aligned} C^o &= \{e \in E \mid \iota(e) \in C, \tau(e) \notin C\}, \\ C^i &= \{e \in E \mid \tau(e) \in C, \iota(e) \notin C\}. \end{aligned}$$

Then for any flow f we have that

$$\mathrm{val}(f) = \sum_{e \in C^o} f(e) - \sum_{e \in C^i} f(e).$$

Proof: We prove the statement by induction on the number of vertices in C. For $|C| = 1$ we have that $C = \{s\}$ and the statement is the definition of the value of a flow.

Now suppose that $|C| > 1$, let $s \neq v \in C$ and $D = C \setminus \{v\}$. By induction, the statement holds for D.

We now consider the following disjoint sets of edges:

$$\begin{aligned} P^o &= \{e \in E \mid \iota(e) \in D, \tau(e) \notin C\}, \\ P^i &= \{e \in E \mid \tau(e) \in D, \iota(e) \notin C\}, \\ Q^o &= \{e \in E \mid \iota(e) \in D, \tau(e) = v\}, \\ Q^i &= \{e \in E \mid \tau(e) \in D, \iota(e) = v\}, \\ R^o &= \{e \in E \mid \iota(e) = v, \tau(e) \notin C\}, \\ R^i &= \{e \in E \mid \tau(e) = v, \iota(e) \notin C\}. \end{aligned}$$

Any edge that starts at v must be in Q^i or in R^o, any edge ending at v must be in Q^o or in R^i. Kirchhoffs law thus gives us that

$$\sum_{e \in Q^i} f(e) + \sum_{e \in R^o} f(e) = \sum_{e \in Q^o} f(e) + \sum_{e \in R^i} f(e).$$

It is easily seen that $C^o = P^o \cup R^0$, $C^i = P^i \cup R^i$ and that $D^o = P^o \cup Q^o$ and $D^i = P^i \cup Q^i$. Therefore, using the previous expression to replace a difference of Q-sums by a difference of R-sums:

$$\begin{aligned} \operatorname{val}(f) &= \sum_{e \in D^o} f(e) - \sum_{e \in D^i} f(e) \\ &= \sum_{e \in P^o} f(e) - \sum_{e \in P^i} f(e) + \underbrace{\sum_{e \in Q^o} f(e) - \sum_{e \in Q^i} f(e)}_{= \sum_{e \in R^o} f(e) - \sum_{e \in R^i} f(e)} \\ &= \sum_{e \in C^o} f(e) - \sum_{e \in C^i} f(e). \end{aligned}$$

The claim follows by induction. □

We define the *capacity* of a cut as the sum of the capacity of the edges leaving the cut:

$$\operatorname{cap}(C) = \sum_{e \in C^o} c(e).$$

Lemma IV.10 thus implies that the value of *any* flow must be bounded by the capacity of *any* cut and thus (unsurprisingly) the maximal flow value is bounded by the minimum cut capacity – a chain is only as strong as its weakest link.

Similar to the dualities we considered before, equality is attained:

THEOREM IV.11 (Max-Flow Min-Cut, integer version): Suppose the capacity function c in integer valued. Then the maximum value of a flow in a network is equal to the minimum capacity of a cut. Furthermore, there is a maximum flow f that is integer valued.

NOTE IV.12: If the capacity function is rational valued, we can simply scale with the lcm of the denominators and obtain an integer valued capacity function. In the case of irrational capacities, the theorem holds by a boring approximation argument.

Proof:[MENGER⇒Max-flow] Given a directed network, replace every arc with integral weight c by c disjoint[3] directed edges.

Clearly, if one edge of a c-fold multi-edge set is in a minimum separating set, the other $c-1$ edges have to be.

Thus the cardinality of a minimum edge-separating set is a minimum cut, while k independent paths give a flow of k. By the edge version of MENGER's theorem, the minimum cardinality of an edge-separating set equals the maximum number of independent edge paths, proving the theorem. □

The algorithm of FORD and FULKERSON finds a maximal flow for a given network. It start with a valid flow (say $f(e) = 0$ for all edges) and then improves this flow if possible, using the following main step:

ALGORITHM IV.13 (increase flow): Given a flow f, return a flow with higher value, or show that no such flow exists.

1. Let $A := \{s\}$. *(A is a cut of vertices that can be supplied at higher rate from the source).* $I := \{\}$. *(I is a set of edges that are not yet at capacity.)* $R := \{\}$. *(R is a set of edges whose flow should be reduced.)*
2. If $t \in A$, go to step 6.
3. If there is an edge $e \in A^o$ with $f(e) < c(e)$, set $A := A \cup \{\tau(e)\}$, $I := I \cup \{e\}$. Go to step 2.
4. If there is an edge $e \in A^i$ with $f(e) > 0$ *(This is flow into A which we could reduce)*, set $A := A \cup \{\iota(e)\}$, $R := R \cup \{e\}$. Go to step 2.
5. If neither of these two properties hold, terminate with a message that the flow is maximal.
6. By tracing back how t got added to A, we construct an (undirected) path (the *augmenting path*) $P = (e_1, e_2, \ldots)$ from s to t. Let

$$d = \min\left(\min_{e \in P \cap I}(c(e) - f(e)),\ \min_{e \in P \cap R} f(e)\right).$$

Increase the flow on $P \cap I$ by d, reduce the flow on $P \cap R$ by d. *(This satisfies Kirchhoffs law.)* Return the larger flow.

[3] Or edges $\to \circ \to$ with intermediate vertices to ensure disjointness

Proof: We shall show termination in the case of integral capacities: Every time we adjust a flow, we may assume that the flow on one edge increases by an integral value. This can only happen a finite number of times.

Once the algorithm ends in step 5, we have a cut A such that $\sum_{e\in A^i} f(e) = 0$, $\sum_{e\in A^o} f(e) = \sum_{e\in A^o} c(e)$, thus $\mathrm{val}(f) = \mathrm{cap}(A)$ and the flow must be maximal by Theorem IV.11. □

Example: We illustrate the algorithm in the example from above. Figure IV.5, a) shows some flow (in blue). The total flow is 24.

We start building the list of undersupplied vertices and mark edges on which to increase, respectively reduce the flow:

Vertices	Increase	Reduce
s		
B	sB	
A		AB
D	AD	
t	DT	

We find an augmenting path sBADt (red in Figure b) and note that we can increase the flow by 2 (the limiting factor being edge sB) to a total of 26.

Starting again, we obtain (Figure c) the augmenting path sCFBAEt, on which we can increase the flow by 2 again, resulting is a flow of 28.

Finally we build the undersupplied cities as C,F,B and then cannot increase the set. Thus (Figure d), shaded) $\{s, C, F, B\}$ is a cut whose capacity of 28 is reached, proving that we have a maximal flow.

Note IV.14: As given, the algorithm is not necessarily in polynomial time, as there might be a huge number of small incremental steps. Take the network in Figure IV.6. Starting with a flow of 0, we choose the augmenting path sABt on which we can increase the flow by 1. Next, we take the augmenting path sBAt, reducing the flow on AB to 0 again and have a total flow of 2. We can iterate this, increasing the flow by 1 each step until we get the maximum flow of 200.

Polynomial time in this algoriuthm however can be acheived by a more careful choice of the augmenting path (e.g. a shortest path, Edmonds-Karp algorithm), that analysis however is beyond the scope of these notes.

Note IV.15: To close the circle, observe that the Max-flow/Min-cut theorem implies Dilworth's theorem: Given a poset, we model it as a directed network with flow from bottom to top and arc capacity one. Thus all theorems mentioned in this section (and in fact a few more we have not mentioned) are in a sense mutually equivalent and form the fundamental theorem of discrete optimization.

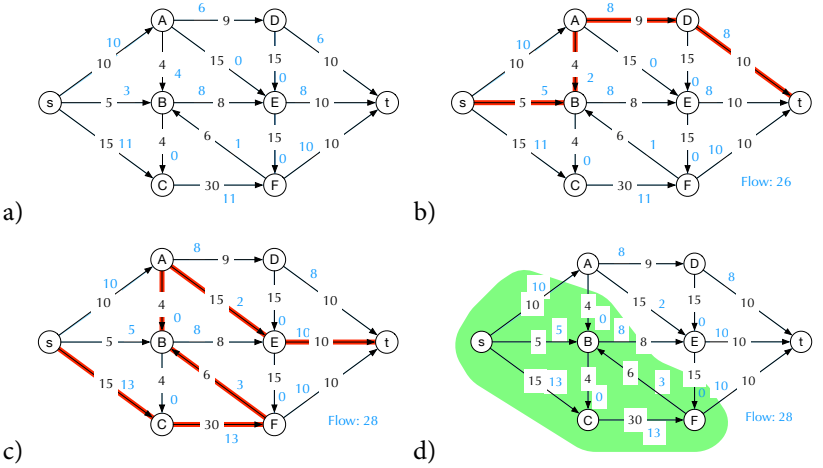


Figure IV.5: Increasing a given flow to maximum

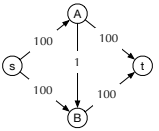


Figure IV.6: Example for bad augmenting paths

Braess' Paradox

On Earth Day this year, New York City's Transportation Commissioner decided to close 42d Street, which as every New Yorker knows is always congested. "Many predicted it would be doomsday," said the Commissioner [...] But to everyone's surprise [...] Traffic flow actually improved when 42d Street was closed.

New York Times, 12/25/1990, p.38
GINA KOLATA

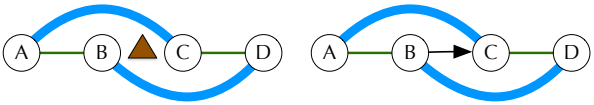


Figure IV.7: Braess' paradox: Before and after a new road is built.

> From the time of Say and Ricardo the classical economists have taught that supply creates its own demand
>
> The General Theory of Employment, Interest, and Money
> John Maynard Keynes

Often the question of maximal flow turns up not just for an existing network, but already at the stage of network design, aiming to maximize flow. Braess' paradox shows that this can be very nonintuitive.

While we shall give a theoretical example, concrete instances of this effect have been observed in the real world in several cities (New York; Stuttgart, Germany; Winnipeg, Canada;…) when roads had been temporarily blocked because of building work, or after new roads were built. The interested reader might observe obvious implications to society and politics.

Suppose we have four cities, A, B, C, D with connecting roads as shown in figure IV.7, left. The reader may observe that this configuration is similar to the one in Note IV.14.

Cities A and B as well as C and D are connected by a minor road which easily clogs up. The driving time thus depends strongly on the number of cars, driving time, with x thousand drivers on the road is

$$t_{AB} = t_{CD} = 10x.$$

An obstacle blocks direct connections from B to C.

High capacity roads have been build to connect A and C, as well as B and D. While overall time is longer, the impact for extra cars is less, time with x thousand drivers on the road is

$$t_{AC} = t_{BD} = 50 + x.$$

Assume 6000 drivers want to travel from A to D. A symmetry argument shows that half (3000) travel via B and half via C. Their individual travel time is $10 \cdot 3 + 50 + 3 = 83$. (It is not hard to show that this is a stable equilibrium, no traveler can gain by changing their route.)

After time, a new connection is built from B to C through the obstacle (right image). (For simplicity of the argument we shall assume that it is one-way, but that is not crucial for the paradox.) It has high capacity and is short, so travel time on this route is

$$t_{BC} = 10 + x.$$

Assuming perfect information, drivers will move towards a distribution in which travel time along all possible routes is equal. In the example this is 2000 drivers using ABD, 2000 drivers ACD and 2000 drivers using ABCD. (This means that 4000 drivers are traveling AB and CD, respectively, 2000 travel BC.) Their travel times are (again in a stable equilibrium)

$$t_{ABD} = 10 \cdot 4 + 10 + 2 + 10 \cdot 4 = 92 = 10 \cdot 4 + 50 + 2t_{ABD} = t_{ACD},$$

and thus more than before.

Chapter

V

Partitions and Tableaux

Some of the hardest, but also most intriguing, counting problems are those of partitions.

V.1 Partitions and their Diagrams

An *order-irrelevant partition* (from now on simply: partition) of n into k parts is an expression

$$n = x_1 + x_2 + \cdots + x_k, \qquad x_1 \geq x_2 \geq \cdots\cdots x_k \geq 1$$

with all x_i being integers. We will typically denote partitions by small Greek letters and write $\lambda \vdash n$ to mean that λ is a partition of n.

As in I.4, we denote by $p_k(n)$ the number of partitions of n into k parts and by $p(n)$ the total number of partitions of n (into any number of parts). OEIS A000041 For example

$$4 = 4 = 3 + 1 = 2 + 2 = 2 + 1 + 1 = 1 + 1 + 1 + 1,$$

so $p(4) = 5$ and

$$7 = 5 + 1 + 1 = 4 + 2 + 1 = 3 + 3 + 1 = 3 + 2 + 2,$$

giving $p_3(7) = 4$. Instead of writing a sum, we might give a count of the part sizes, that is instead of $7 = 3 + 2 + 2$ we would write $3^1 2^2$.

A convenient way to depict a partition is graphically by what is called a *Young diagram* (boxes) or *Ferrers' diagram* (dots): Each part is depicted by a row of dots or of empty squares, in descending[1] order. Figure V.1 gives two versions of the diagram for the partition $3^1 2^2 1^1$:.

[1] this is the convention in English speaking countries. The French build instead from the bottom up.

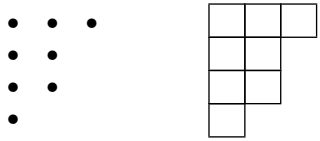


Figure V.1: The Ferrers and Young diagram for $3^1 2^2 1^1$

The *conjugate* λ^* of a partition λ is the partition with the "transposed" diagram, for example if $\lambda = 3^1 2^2 1^1$, then $\lambda^* = 4^1 3^1 1^1$. Clearly $\lambda^{**} = \lambda$.

V.2 Some partition counts

The number $p_k(n)$ is clearly equal to the number of ways to write $n-k = y_1+\cdots+y_k$ with $y_1 \geq \cdots \geq y_k \geq 0$. If exactly s of these integers y_i are nonzero, there will be $p_s(n-k)$ of this type. This gives the recursion

$$p_k(n) = \sum_{s=1}^{k} p_s(n-k)$$

which, together with the initial values $p_k(n) = 0$ for $n < k$ and $p_k(k) = 1$ gives us a way to determine $p_k(n)$ recursively.

We start by giving a generating function for the partition function $p(n)$:

Proposition V.1:

$$\sum_n p(n)t^n = \prod_{i\geq 1}(1-t^i)^{-1}$$

Proof: Expanding the right hand side, and using that $(1-t^i)^{-1} = (1+t^i+t^{2i}+\cdots$, we get

$$\prod_i (1-t^i)^{-1} = (1+t+t^2+\cdots)(1+t^2+t^4+\cdots)\cdots$$

In this product, t^n is obtained as a product of t^{x_1} from the first factor, t^{2x_2} from the second, and so on, such that $1^{x_1}2^{x_2}\cdots \vdash n$. Vice versa, every partition of n gives a different way to form t^n, that is $p(n)$ is the coefficient of t^n. □

This infinite product does not have a nice closed form, but we will use it below in Corollary V.6 to obtain a recursion formula for $p(n)$.

If we consider $p_k(n)$ for a fixed k, we only need to consider a partial product in the expansion. This can be used to calculate the value directly:

Lemma V.2: $p_3(n)$ is the integer closest to $\frac{n^2}{12}$.

Proof: Let $a(n) = p_3(n+3)$ the number of solutions of $n = x_1 + x_2 + x_3$ with $x_1 \geq x_2 \geq x_3 \geq 0$. We set $y_3 = x_3$, $y_2 = x_2 - x_3$ and $y_1 = x_1 - x_2$ and thus get that

$a(n)$ is the number of solutions of $n = y_1 + 2y_2 + 3y_3$. As in the previous proof, we get a generating function

$$\sum_n a(n)t^n = (1-t)^{-1}(1-t^2)^{-1}(1-t^3)^{-1}.$$

A partial fraction decomposition gives us

$$\begin{aligned}\sum_n a(n)t^n &= \frac{1}{6}(1-t)^{-3} + \frac{1}{4}(1-t)^{-2} + \frac{17}{72}(1-t)^{-1} \\ &\quad + \frac{1}{8}(1+t)^{-1} + \frac{1}{9}(1-\zeta t)^{-1} + \frac{1}{9}(t-\zeta^2 t)^{-1}\end{aligned}$$

with ζ a primitive 3-rd root of unity. Using the series expression

$$(1-t)^{-a-1} = \sum_j \binom{a+j}{j} t^j,$$

collecting and telescoping, we get

$$a(n) = \frac{1}{12}(n+3)^2 - \frac{7}{72} + \frac{(-1)^n}{8} + \frac{1}{9}(\zeta^n + \zeta^{2n}).$$

We write this in the form

$$\left| a(n) - \frac{1}{12}(n+3)^2 \right| \leq \frac{7}{72} + \frac{1}{8} + \frac{2}{9} < \frac{1}{2}$$

giving the desired result. □

More generally, we have the following approximation:

PROPOSITION V.3: For fixed k, we have that

$$p_k(n) \sim \frac{n^{k-1}}{k!(k-1)!} \qquad (n \to \infty).$$

Proof: Suppose that $n = x_1 + \cdots + x_k$ with $x_1 \geq \cdots \geq x_k \geq 1$. If we permute the variables x_i, we get (not necessarily different) compositions of n into k parts and each composition can be obtained by a permutation of a partition. By Section I.1 there are $\binom{n-1}{k-1}$ such compositions, giving us

$$k!p_k(n) \geq \binom{n-1}{k-1}.$$

We now set $y_i = x_i + (k-i)$ for $1 \leq i \leq k$. Then the y_i are all different, as

$$x_i + k - i = y_i = y_j = x_j + k - j$$

implies that $x_i + j = x_j + i$. Assuming WLOG that $i < j$ we have that $x_i \leq x_j$, contradiction. Clearly $y_1 \geq y_2 \geq \cdots y_k$.

The summation formula over i gives (after an index swap) that $y_1 + \cdots + y_k = n + \frac{k(k-1)}{2}$ is a partition of $n + \frac{k(k-1)}{2}$. As the parts are different, every permutation gives a different composition, but we do not necessarily obtain all compositions this way. Thus

$$k! p_k(n) \leq \binom{n + \frac{k(k-1)}{2} - 1}{k-1}.$$

Combining the two inequalities, and dividing by $k!$ gives (with $a_k = \frac{k(k-1)}{2}$ only dependent on k)

$$\frac{(n-1)(n-2)\cdots(n-k+1)}{k!(k-1)!} \leq p_k(n) \leq \frac{(n+a_k-1)\cdots(n+a_k-k+1)}{k!(k-1)!}.$$

As k (and thus a_k) are both fixed, both numerators approach n^{k-1} if $n \to \infty$. □

V.3 Pentagonal Numbers

> Our chief weapon is surprise...
> surprise and fear...
> fear and surprise...
> Our **two** weapons are fear and surprise...
> and ruthless efficiency...
>
> The Spanish Inquisition
> Monty Python

For reasons we shall see soon, we divert in a surprising way.

A number is *pentagonal* if it is of the form $k(3k-1)/2$ or generalized pentagonal OEIS A001318 if it is of the form $k(3k+1)/2$ for some positive integer k. Alternatively, one can simply use $k(3k-1)/2$ for arbitrary integers k. The first few pentagonal numbers for positive index are $1(3 \cdot 1 - 1)/2 = 1$, $2(3 \cdot 2 - 1)/2 = 5$, $3(3 \cdot 3 - 1)/2 = 12$, $4(3 \cdot 4 - 1)/2 = 22$ as depicted in Figure V.2, respectively $0(3 \cdot 0 - 1)/2 = 0$, $-1(3 \cdot (-1) - 1)/2 = 2$, $-2(3 \cdot (-2) - 1)/2 = 7$, $-3(3 \cdot (-3) - 1)/2 = 15$ for other indices.

The pentagonal numbers give the following extraordinary[2] theorem.

Theorem V.4 (Euler's Pentagonal Numbers Theorem): a) If n is not a pentagonal number, the number of partitions of n into an even and an odd number of distinct parts are equal.

[2] as in the title: *une loi tout extraordinaire*, [Eul15]

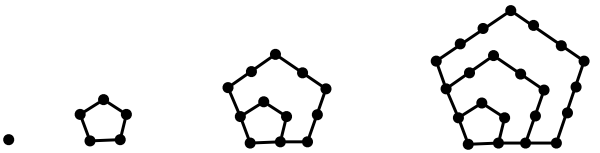


Figure V.2: The first few pentagonal numbers: 1,5,12,22

b) If $n = k(3k-1)/2$ for $k \in \mathbb{Z}$ even, then the number of partitions of n into an even number of distinct parts exceeds the number of partitions into an odd number of distinct parts by one.
c) If $n = k(3k-1)/2$ for $k \in \mathbb{Z}$ odd, then the number of partitions of n into an even number of distinct parts is one less than the number of partitions into an odd number of distinct parts.

Example: There are 2 partitions of 6 into an even number of distinct parts and 2 into an odd number of distinct parts. There are 3 partitions of 7 into an even number of distinct parts, and 2 into an odd number of distinct parts. There are 7 partitions of 12 into even parts, and 8 into odd parts. Proof: The proof will consist of constructing a bijection between partitions with an even number of distinct parts and partitions with an odd number of distinct parts. In the case that n is pentagonal this will have leave out exactly one partition.

For a partition $\lambda \vdash n$ into different parts, we consider two subsets of the cells, as depicted in Figure V.3:

The *base* is the bottom row of cells (the shortest row), the *slope* is the cells starting at the end of the top row and descending straight down-left as far as possible (this might be only of length 1).



Figure V.3: Base (dashed) and Slope of two partitions

Base and slope might intersect in one cell, we call those cells of the base that are not in the slope the *pure base*; similarly the *pure slope* are those slope cells that are not also in the base.

We now divide the partitions of n with distinct parts into three classes:

Class 1 are those partitions for which the pure base contains more cells than the slope.

Class 2 are those partitions for which the pure slope is at least as long as the base.

Class 3 are all remaining partitions[3].

For a partition λ in Class 1, we create a new partition μ by removing the slope (possibly including the base point) and placing these cells in a new row at the bottom of the diagram. As the pure base is larger than the slope this results in a partition into different parts. Furthermore, the slope of μ is at least as long as the slope of λ (which is the base of μ) if base and slope of μ are distinct; if they intersect the slope of μ is strictly larger. This means μ is in Class 2.

Vice versa, if μ is in Class 2, we remove the base (possibly including the lowest point of the slope) and attach it from top right as new slope. This results in a partition λ into different parts which lies in Class 1.

For example, the left partition in Figure V.3 is in Class 2 and will be transformed into the partition on the right side in Class 1, and vice versa.

It is easily seen that these two operations are mutually inverse, that is Class 1 and Class 2 contain the same number of partitions. Furthermore the operations change the number of parts by exactly one. That means that in Class 1 $\cup$ Class 2, there are exactly as many partitions with an even number of parts, as with an odd number of parts.

Finally consider Class 3: A partition in this class must have base and slope intersecting, and the base contains the same number of cells, or exactly one more, than the slope, see Figure V.4.



Figure V.4: Possibilities for partitions in Class 3

This means that, if there are k parts, it consists of a $k \times k$ rectangle and a $1, 2, \ldots, k$ triangle that might or might not overlap. In the first case there are $k^2 + k(k-1)/2 = k(3k-1)/2$ parts, respectively $k^2 + k(k+1)/2 = k(3k+1)/2 = (-k)(3(-k)-1)/2$ parts. For a given n only one of these two options is possible, and it implies that n is a pentagonal number. This proves the theorem. □

[3] Classifications become easy if we allow an "all the rest" category

To simplify notation we write $\omega(k) = k(3k-1)/2$ for $k \in \mathbb{Z}$ to state the following consequence for the inverse of the generating function of partitions.

Corollary V.5:

$$\prod_{n\geq 1}(1-t^n) = \sum_{k=-\infty}^{\infty}(-1)^k t^{\omega(k)} = 1 + \sum_{k>0}(-1)^k\left(t^{\omega(k)} + t^{\omega(-k)}\right).$$

Proof: The second equality simply relies on the now familiar two ways to describe generalized pentagonal numbers.

Let $\mathrm{even}(n)$ be the number of partitions of n into an even number of distinct parts, $\mathrm{odd}(n)$ similarly for an odd number of parts. By the Pentagonal Numbers Theorem, the right hand side of the equation is the generating function for $\mathrm{even}(n) - \mathrm{odd}(n)$. We aim to show that this is true also for the left hand side:

The coefficient for t^n will be made up by contributions from terms of the form t^{n_i}, all distinct, such that $n = n_1 + \cdots + n_l$. This partition contributes $(-1)^l$. That is, every partition into an even number of distinct parts contributes 1, every partition into an odd number of distinct parts contributes -1, which was to be shown. □

This now permits us to formulate a recurrence formula for $p(n)$:

Corollary V.6:

$$\begin{aligned} p(n) &= \sum_{k>0}(-1)^{k-1}\left(p(n-\omega(k)) + p(n-\omega(-k))\right) \\ &= p(n-1) + p(n-2) - p(n-5) - p(n-7) + p(n-12) + \cdots \end{aligned}$$

Proof: We know that $\sum_n p(n)t^n = \prod(1-t^n)^{-1}$, and therefore

$$\left(\sum_n p(n)t^n\right)\cdot\left(1 + \sum_{k>0}(-1)^k\left(t^{\omega(k)} + t^{\omega(-k)}\right)\right) = 1.$$

We now consider the coefficient of t^n is this product (which is zero for $n > 0$). This gives

$$0 = p(n) + \sum_{k>0}(-1)^k\left(p(n-\omega(k)) + p(n-\omega(-k))\right).$$

By using that $p(n) = 0$ for $n < 0$ and substituting values for the pentagonal numbers we get the explicit recursion. □

This formula can be used for example to calculate values for $p(n)$ effectively, or to estimate the growth of the function $p(n)$.

While it might not seems so we have by now wandered deep into Number theory. In this area we get for example Ramanujan's famous congruence identities

$$\begin{aligned} p(5n+4) &\equiv 0 \pmod 5 \\ p(7n+5) &\equiv 0 \pmod 7 \\ p(11n+6) &\equiv 0 \pmod{11} \end{aligned}$$

that recently have been generalized by Ono and collaborators.

Euler's theorem and its consequence are a special case of the *Jacobi triple product identity*

$$\prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n-1}t)(1+q^{2n-1}t^{-1}) = \sum_{r=-\infty}^{\infty} q^{r^2}t^r$$

which has applications in the theory of theta-functions – the territory of Number Theory – and Physics.

V.4 Tableaux

It's like the story of the drawing on the blackboard, remember?

C'est comme pour l'histoire du dessin sur le tableau noir, tu te rappelles?

Histoires inédites du Petit Nicolas, Tome 1
René Goscinny

Drawing a partition as Young diagram with boxes allows us to place numbers in them.

1	3	4	8	12
2	6	9	11	
5	7			
10				

Definition V.7: Let $\lambda \vdash n$. A (standard) Young tableau[4] is a Young diagram of shape λ into whose boxes the numbers $1, \ldots, n$ have been entered, so that every row and every column is (strictly) increasing.

Beyond their intrinsic combinatorial interest, tableaux are a crucial tool for studying permutations, and for the representation theory of symmetric groups.

An obvious question with a surprising answer is to count the number f_λ of Young tableaux of shape λ. In the case that $\lambda = n^2$ as partition of $2n$, we observe, exercise **??**, that $f_\lambda = C_{n+1}$, the counts f_λ thus can be considered as generalizing Catalan numbers.

Our aim is to derive the famous hook formula (Theorem V.13) for f_λ. While this formula is inherently combinatorial, the proof we shall give is mainly algebraic – indeed, until surprisingly recently [GNW79, Zei84] no purely combinatorial proof was known.

We start with a recursively defined function f on m-tuples of natural numbers (for arbitrary m). (As it will turn out that this function is f_λ we use the same name.)

1. $f(n_1, \ldots, n_m) = 0$, unless $n_1 \geq n_2 \geq \cdots \geq 0$.

[4] French: "(black)board". Plural: *tableaux*

2. $f(n_1, \ldots, n_m, 0) = f(n_1, \ldots, n_m)$.

3. $f(n_1, \ldots, n_m) = f(n_1 - 1, n_2, \ldots, n_m) + f(n_1, n_2 - 1, \ldots, n_m) + \cdots$
$+ f(n_1, n_2 m \ldots, n_m - 1)$, if $n_1 \geq n_2 \geq \cdots n_m \geq 0$.

4. $f(n) = 1$ if $n \geq 0$.

Properties i) and iv) are base cases, property ii) allows for an arbitrary number of arguments, Property iii) finally gives a recursion that will reduce the values of the arguments; thus f is uniquely defined.

LEMMA V.8: If $\lambda = (n_1, \ldots, n_m)$, then $f_\lambda = f(n_1, \ldots, n_m)$ is the number of Young tableaux of shape λ.

Proof: Conditions i), ii) and iv) are obviously satisfied by f_λ. For condition iii), take a tableau of shape $(n_1, \ldots, n_m)$ and consider the location of n. It needs to be at the end of a row (say row j) and the bottom of a column, and if we remove that entry, we obtain a valid tableau with $n - 1$ entries and of shape $(n_1, \ldots, n_{i-1}, n_i - 1, n_{i+1}, \ldots, n_m)$. (Note that some of the terms could be zero if $n_i - 1 < n_{i+1}$.) □

Next we shall need an indentity in the polynomial ring in variables $x_1, \ldots, x_m, y$.

DEFINITION V.9: The *root of the discriminant* of $x_1, \ldots, x_n$ is defined as

$$\Delta(x_1, \ldots, x_n) := \prod_{1 \leq i < j \leq m} (x_i - x_j)$$

The eagle-eyed reader will recognize the right hand side as a Vandermonde determinant. The next, auxiliary lemma is simply an identity about a particular multivariate polynomial.

LEMMA V.10: Define a polynomial $g \in \mathbb{Q}[x_1, \ldots, x_m, y]$ as

$$\begin{aligned} g(x_1, \ldots, x_m; y) &:= x_1 \Delta(x_1 + y, x_2, \ldots, x_m) \\ &+ x_2 \Delta(x_1, x_2 + y, \ldots, x_m) + \cdots + x_m \Delta(x_1, x_2, \ldots, x_m + y). \end{aligned}$$

Then

$$g(x_1, \ldots, x_m; y) = \left(x_1 + \cdots + x_m + \binom{m}{2} y \right) \Delta(x_1, \ldots, x_m).$$

Proof: Clearly g is a homogeneous (every monomial in g has the same total degree) polynomial of degree $1 + \deg \Delta(x_1, \ldots, x_m)$. If we interchange x_i and x_j, then g changes its sign, thus, if we set $x_i = x_j$ the polynomial evaluates to 0. Thus (consider g as univariate in x_i with all other variables part of a rational function field) $x_i - x_j$ divides g. Hence $\Delta(x_1, \ldots, x_m)$ divides g.

We now concentrate instead on the variable y. The degree of g as a polynomial in y is at most one, and for $y = 0$ the statement is obvious. We therefore only need to show that the coefficient of y in g is $\binom{m}{2}\Delta$.

If we expand g, the terms in y are $\frac{x_i y}{x_i - x_j}\Delta(x_1, \ldots, x_m)$ and $-\frac{x_j y}{x_i - x_j}\Delta(x_1, \ldots, x_m)$ for pairs i, j with $1 \le i < j \le m$. These two terms add up to $y\Delta$, thus the $\binom{m}{2}$ pairs $(i < j)$ yield in sum $\binom{m}{2}\Delta(x_1, \ldots, x_m)$. □

Using the defining recurrences, we can now verify an (otherwise unmotivated) formula for f_λ:

LEMMA V.11: If $\lambda = (n_1, \ldots, n_m)$, then

$$f_\lambda = \frac{\Delta(n_1 + m - 1, n_2 + m - 2, \ldots, n_m)n!}{(n_1 + m - 1)!(n_2 + m - 2)!\cdots n_m!}.$$

In fact this formula satisfies properties i-iv) as long as $n_1 + m - 1 \ge n_2 + m - 2 \ge \cdots n_m$.

Proof: We shall show that the right hand side satisfies conditions i)-iv). To make the recursion in iii) work we need to include the case that $n_{i+1} = n_i + 1$, but in that case $n_{i+1} + m - i + 1 = n_i + 1 + m - i + 1 = n_i + m - i$, that is two arguments of Δ become equal and the right hand side evaluates to 0.

With this, properties i) and iv) become trivially true.

Concerning property ii), if we replace m by $m + 1$ and have $n_{m+1} = 0$, the denominator changes by a factor $(n_1 + m + 1 + 1)\cdots(n_m + 1)$ which is the same factor by which the numerator changes.

Concerning property iii), this is the statement of V.10 for $x_i = n_i + m - i$ and $y = -1$. □

The Hook Formula

> Thus, although the term "hooker" did not originate during the Civil War, it certainly became popular [...] in tribute to the proclivities of General Joseph Hooker [...]
>
> Studies in Etymology and Etiology
> DAVID GOULD

A much nicer way to describe the values of f_λ is given by the following concept:

DEFINITION V.12: Given a cell in a Young diagram, its *hook* consists of the cell itself as well as the cells to the right in the same row, and those in the same column below. The *hook length* is the number of cells in the hook.

THEOREM V.13 (Hook formula, FRAME, DE ROBINSON, THRALL (1954)): The number of tableaux of shape $\lambda \vdash n$ is given by $n!$, divided by the product of all hook lengths.

Example: Consider $\lambda = 6^1 4^2 2^1$. The numbers in the diagram for λ indicate hook lengths:

9	8	6	5	2	1
6	5	3	2	•	
5	4	2	1	•	
2	1	•			

Thus

$$f_\lambda = \frac{16!}{1^3 \cdot 2^4 \cdot 3 \cdot 4 \cdot 5^3 \cdot 6^2 \cdot 8 \cdot 9} = \frac{20922789888000}{62208000} = 336336.$$

Proof: Suppose that $\lambda = (n_1 m \ldots, n_m) \vdash n$. Then the hook length in the top left corner is $n_1 + m - 1$. The hook length of the second cell is $n_1 + m - 2$, and so on, seemingly down to hook length 1 in the last cell. As in the example above, certain lengths are skipped. This happens if a lower row is shorter than the top row. This is indicated by the •-dots in the above diagram.

We are thus skipping $(n_1 + m - 1) - (n_m)$, in the column after the last row ends, $(n_1 + m - 1) - (n_{m-1} + 1)$ after the second last row ends and so on.

Going to the second row, we get hook lengths $n_2 + m - 2$, $n_2 + m - 3$, and so on, but excluding $(n_2 + m - 2) - (n_m)$ for the last row, $(n_2 + m - 2) - (n_{m-1} + 1)$ for the second last row, etc.

Combining this, we get as product of the hook lengths a fraction with numerator $(n_i + m - i)!$ for the i-th row, and the denominator combining the canceled terms to $\Delta(n_1 + m - 1, n_2 + m - 2, \ldots, n_m)$, proving the theorem with use of Lemma V.11. □

We close this section with the remark that the numbers f_λ are also the degrees of the irreducible (ordinary) representations of the symmetric group. Proving this requires a substantial amount of abstract algebra and cannot be done here..

A consequence is the nice identity that $\sum_{\lambda \vdash n} f_\lambda^2 = n!$, which we will also prove below in V.38

V.5 Symmetric Functions

Reality favors symmetries and slight anachronisms

A la realidad le gustan las simetrías y los leves anacronismos

El Sur
Jorge Luis Borges

For a commutative ring R, consider the ring $A = R[x_1, \ldots, x_N]$ of polynomials in N variables[5]

[5] This is the posh way of defining this set of polynomials. The reader will be perfectly well suited if she simply sets $R = \mathbb{Q}$ and considers A as the set of rational polynomials in n variables.

For writing polynomials, it will be convenient to introduce the following notation. A composition (see I.1) $\alpha = (n_1, n_2, \cdots, n_k)$ of a number M is a sequence of numbers such that $M = n_1 + \cdots + n_k$. We also write $\alpha_i = n_i$ to denote the i-th part of a composition α. We can consider partitions of M to be a special case of compositions. We shall use a vector notation for monomials, writing

$$\underline{x}^{\alpha} := x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k},$$

this way an arbitrary polynomial may be written as $\sum_{\alpha} c_{\alpha} \underline{x}^{\alpha}$.

Definition V.14: A polynomial $f \in A$ is a *symmetric function*, if it stays the same after any permutation of the arguments: $f(x_1, \ldots x_n) = f(x_{1^{\pi}}, \ldots, x_{n^{\pi}})$ for $\pi \in S_n$.

Some literature instead talks about *symmetric polynomials*, meaning the same property.

The set of symmetric functions clearly forms a subring of A, we denote this subring by Λ.

Note V.15: One could, in the spirit of chapter VI consider this as an action of S_n on the polynomial ring and ask what happens if we restrict to a subgroup. This is the topic of *Invariant Theory* – a beautiful subject but not the one we study here.

As a permutation of the indices leaves the degree of each term invariant we can consider a symmetric function as a sum of *homogeneous* (that is each term has the same total degree) parts. It therefore is sufficient to consider homogeneous symmetric functions of degree n, say.

As a vector space, we can consider $\Lambda = \Lambda^0 \oplus \Lambda^1 \oplus \cdots$ as a direct sum of subspaces Λ^i consisting of homogeneous symmetric functions of degree i.

The following definition describes some particular classes of symmetric functions:

Definition V.16: Let $n > 0$ and $\lambda = n_1 + n_2 + \cdots + n_k \vdash n$.

a) The *Monomial Symmetric Function* m_λ is the sum $\sum_{\alpha} \underline{x}^{\alpha}$ where α runs over all *distinct* permutations of (the entries of) λ.

b) The *Elementary Symmetric Function* e_m is the sum over all products of m (distinct) indeterminates, that is $e_m = m_{1^m}$. We define $e_\lambda = e_{n_1} \cdot e_{n_2} \cdots e_{n_k}$.

c) The *Complete Homogeneous Symmetric Function* h_m is the sum over all products of m (possibly repeated) indeterminates: $h_n = \sum_{\lambda \vdash n} m_\lambda$. We define $h_\lambda = h_{n_1} \cdot h_{n_2} \cdots h_{n_k}$.

d) The *Power Sum Function* p_m is the sum $x_1^m + x_2^m + \cdots + x_N^m$ over the m-th power of all N indeterminates. We define $p_\lambda = p_{n_1} \cdot p_{n_2} \cdots p_{n_k}$.

It is easily seen that all of these functions are indeed symmetric.

Example: Suppose that $N = 3$ and $\lambda = 2 + 1$. Then

$$\begin{aligned} m_\lambda &= x_1^2x_2 + x_2^2x_1 + x_1^2x_3 + x_3^2x_1 + x_2^2x_3 + x_3^2x_2, \\ e_\lambda &= (x_1x_2 + x_1x_3 + x_2x_3)(x_1 + x_2 + x_3), \\ p_\lambda &= (x_1^2 + x_2^2 + x_3^2)(x_1 + x_2 + x_3), \\ h_\lambda &= e_\lambda + p_\lambda \end{aligned}$$

NOTE V.17: It can be convenient to consider N arbitrary large or even infinite – the identities amongst symmetric functions still hold.

Each of these classes of symmetric functions allows to generate all:

THEOREM V.18: Suppose that $N \geq n$ and f a homogeneous symmetric function of degree n in $x_1, \ldots, x_N$. Then

a) $f = \sum_{\lambda \vdash n} c_\lambda m_\lambda$ for suitable coefficients c_λ.

b) $f = \sum_{\lambda \vdash n} c_\lambda e_\lambda$ for suitable coefficients c_λ.

c) $f = \sum_{\lambda \vdash n} c_\lambda h_\lambda$ for suitable coefficients c_λ.

d) $f = \sum_{\lambda \vdash n} c_\lambda p_\lambda$ for suitable coefficients c_λ.

In each case the coefficients c_λ are unique, furthermore in cases a,b,c), if f has integral coefficients, then all c_λ are integral.

COROLLARY V.19: Any symmetric function $f(x_1, \ldots, x_N)$ can be written as a polynomial $g(z_1, \ldots, z_N)$, where z stands for one of the symbols e,h, p.

NOTE V.20: For algebraists we remark that one can show that the ring of symmetric polynomials in N variables has transcendence degree N and that the $\{e_i\}$, the $\{h_i\}$ and the $\{p_i\}$ each form a transcendence basis of this ring.

Proof:[of Theorem V.18, part a)] If $\boldsymbol{f = \sum_\alpha c_\alpha \underline{x}^\alpha}$ is homogeneous of degree n, then $f = \sum_{\lambda \vdash n} c_\lambda m_\lambda$. □

COROLLARY V.21: The set $\{m_\lambda\}_{\lambda \vdash n}$ is a vector space basis of Λ^n, thus $\dim(\Lambda^n) = p(n)$.

We shall see below that each of the other sets, $\{e_\lambda\}$, $\{h_\lambda\}$, $\{p_\lambda\}$, also forms a basis.

V.6 Base Changes

The proof we shall give for the remaining parts of Theorem V.18 will be based on linear algebra: We shall consider the coefficients of expressing one set of symmetric

functions in terms of another set, and obtain a combinatorial interpretation of these coefficients. From this we shall deduce that the matrix of coefficients is invertible (and possibly has an integral inverse). This implies that the other sets also form a basis and thus the remaining parts of the theorem.

As we will argue with matrices, it involves an arrangement of basis vectors, we thus need to define an ordering on partitions:

DEFINITION V.22: Let $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$ and Let $\mu = (\mu_1, \mu_2, \ldots, \mu_l)$ be partitions of n. We assume WLOG that $k = l$ by allowing cells of size 0.

We say that $\mu \leq \lambda$ in the *natural order*, (also called *dominance order* or *majorization order*) if $\mu \neq \lambda$ and for any $i \geq 1$ we have that

$$\mu_1 + \mu_2 + \cdots + \mu_i \leq \lambda_1 + \lambda_2 + \cdots + \lambda_i.$$

Following Theorem III.3, this partial order has a total order as a linear extension, one such total order is given in exercise **??**.

DEFINITION V.23: Let $A = (a_{ij})$ a matrix. We consider row sums $r_i = \sum_j a_{ij}$ and column sums $c_j = \sum_i a_{ij}$ and call $\mathrm{row}(A) = (r_1, r_2, \ldots)$ the *row sum vector*, respectively $\mathrm{col}(A) = (c_1, c_2, \ldots)$ the *column sum vector*.

Since the m_λ form a basis there are integral coefficients $M_{\lambda\mu}$, such that

$$e_\lambda = \sum_{\mu \vdash n} M_{\lambda\mu} m_\mu.$$

In fact $M_{\lambda\mu}$ is simply the coefficient of $\underline{\boldsymbol{x}}^{\boldsymbol{\mu}}$ in e_λ.

LEMMA V.24: $M_{\lambda\mu}$ equals the number of $(0,1)$-matrices (that is, matrices whose entries are either 0 or 1) $A = (a_{ij})$ satisfying $\mathrm{row}(A) = \lambda$ and $\mathrm{col}(A) = \mu$.

Proof: Let $\lambda = (\lambda_1, \lambda_2, \ldots)$. Any term of e_λ is a product of a term of e_{λ_1}, e_{λ_2}, etc. and any term of e_{λ_i} is the product of i different variables. We shall describe any term using a matrix. Let

$$X = \begin{pmatrix} x_1 & x_2 & x_3 & \cdots \\ x_1 & x_2 & x_3 & \cdots \\ \vdots & & \vdots & \end{pmatrix}$$

Then each monomial $\underline{\boldsymbol{x}}^{\boldsymbol{\alpha}}$ of e_λ is the product of λ_1 entries from the first row, λ_2 entries from the second row and so on. We describe the selection of the factors by a $(0,1)$-matrix A with 1 indicating the selected factors.

The products contributing to a particular monomial $\underline{\boldsymbol{x}}^{\boldsymbol{\alpha}}$ correspond exactly to the matrices with row sums λ and column sums α. The statement follows. □

We denote by $M = (M_{\lambda\mu})$ the matrix of these coefficients.

As the transpose of a matrix exchanges rows and columns we get

COROLLARY V.25: The matrix M is symmetric, that is $M_{\lambda\mu} = M_{\mu\lambda}$.

We now establish the fact that the matrix M is, for a suitable arrangement of the partitions, triangular with diagonal entries 1. This shows[6] that $\det M = 1$, that is M is invertible over $\mathbb{Z}$. This in turn implies Theorem V.18, b).

PROPOSITION V.26: Let $\lambda, \mu \vdash n$. Then $M_{\lambda\mu} = 0$, unless $\mu \leq \lambda^*$. Also $M_{\lambda\lambda^*} = 1$. Thus, if the λ are arranged in (a linear extension of) the natural ordering, and the μ according to ordering of its duals, then M is upper triangular with diagonal 1.

Proof: Suppose that $M_{\lambda\mu} \neq 0$, so by Lemma V.24 there is a $(0,1)$-matrix A with $\mathrm{row}(A) = \lambda$ and $\mathrm{col}(A) = \mu$. Let A^* be the matrix with $\mathrm{row}(A^*) = \lambda$ and the 1's left justified, that is $A^*_{ij} = 1$ if and only if $1 \leq j \leq \lambda_i$. By definition of the dual, we have that $\mathrm{col}(A^*) = \lambda^*$. On the other hand, for any j the number of 1's in the first j columns of A is not less that the number of 1's in the first j columns of A, so in the natural order of partitions we have that

$$\lambda^* = \mathrm{col}(A^*) \geq \mathrm{col}(A) = \mu.$$

To see that $M_{\lambda\lambda^*} = 1$ we observe that A^* is the *only* matrix with $\mathrm{row}(A^*) = \lambda$ and $\mathrm{col}(A^*) = \lambda^*$. □

NOTE V.27: This approach of proving that the e_λ form a basis — expressing the e_λ in terms of the m_μ, giving a combinatorial interpretation of the coefficients in this expression, and deducing from there that the coefficient matrix must be invertible — could also be applied to the other two bases, h_λ and p_λ. We shall instead give other proofs, in part because they illustrate further aspects of symmetric functions.

For the reader who feels that we have moved far away from enumerating combinatorial structures, we close this section with the following observation (whose proof is left as exercise **??**):

Suppose we have n balls in total, of which λ_i balls are labeled with the number i. We also have boxes 1, 2,.... Then $M_{\lambda\mu}$ is the number of ways to place the balls into the boxes such that box j contains exactly μ_j balls, but no box contains more than one ball with the same label.

Complete Homogeneous Symmetric Functions

For the complete homogeneous symmetric functions h_λ we get the following analog to Lemma V.24:

LEMMA V.28: Define $N_{\lambda\mu}$ as the coefficient of $\underline{\boldsymbol{x}}^{\boldsymbol{\mu}}$ in h_λ, that is

$$h_\lambda = \sum_{\mu \vdash n} N_{\lambda\mu} m_\mu.$$

Then $N_{\lambda\mu}$ equals the number of $\mathbb{N}_0$-matrices $A = (a_{ij})$ satisfying $\mathrm{row}(A) = \lambda$ and $\mathrm{col}(A) = \mu$.

[6] A consequence of the Cayley-Hamilton theorem is that the inverse of a matrix A is a polynomial in A with denominators being divisors of $\det(A)$.

Proof: Exercise **??**

We now establish a duality between the e-polynomials and the h-polynomials.

DEFINITION V.29: We define a map $\omega\colon \Lambda \to \Lambda$ by setting

$$\omega\colon \begin{array}{l} e_n \to h_n \\ e_\lambda \to h_\lambda \end{array}$$

and extending linearly.

As the $\{e_\lambda\}_{\lambda\vdash n}$ for a basis of Λ^n this defines ω uniquely and shows that ω is a linear map. Since the e_i are also algebraically independent (a fact we shall take as given), ω is also an algebra endomorphism, that is it preserves products.

PROPOSITION V.30: The endomorphism ω is an involution, that is ω^2 is the identity. In particular $\omega(h_n) = e_n$ and $\omega(h_\lambda) = e_\lambda$.

We note that Theorem V.18, c) is an immediate consequence of this proposition.
Proof: We go in the ring $\Lambda[[t]]$ of formal power series over Λ and set

$$\begin{aligned} E(t) &:= \sum_{n\geq 0} e_n t^n, \\ H(t) &:= \sum_{n\geq 0} h_n t^n. \end{aligned}$$

We have that

$$\begin{aligned} E(t) &= \prod_{r=1}^{N}(1 + x_r t), \\ H(t) &= \prod_{r}(1 + x_r t + x_r^2 t^2 + \cdots) = \prod_{r=1}^{N}(1 - x_r t)^{-1}, \end{aligned}$$

as can be seen by expanding the products on the right hand side. This implies that $H(t)E(-t) = 1$. Comparing coefficients of t^n on both sides yields

$$0 = \sum_{i=0}^{n}(-1)^i e_i h_{n-i}, \qquad n \geq 1.$$

We now apply ω, use $\omega(e_i) = h_i$, and reindex and thus get

$$0 = \sum_{i=0}^{n}(-1)^i h_i \omega(h_{n-i}) \qquad \text{and} \qquad 0 = \sum_{i=0}^{n}(-1)^i h_{n-i}\omega(h_i)$$

We consider this as a linear system of equations with the $\omega(h_i)$ as variables. The coefficient matrix of this system is lower triangular with diagonal entries $h_0 = 1$, thus has full rank. The solution therefore must be unique, but we know already

that the e_i form a solution, proving the theorem. □

We finally come to the power sum functions p_λ. Define

$$P(t) := \sum_{n\geq 1} p_n t^{n-1}$$

PROPOSITION V.31: $\frac{\mathrm{d}}{\mathrm{d}t}H(t) = P(t)H(t)$ and $\frac{\mathrm{d}}{\mathrm{d}t}E(t) = P(-t)E(t)$.

Proof:

$$\begin{aligned} P(t) &= \sum_{r\geq 1} p_r t^{r-1} = \sum_{r\geq 1}\sum_{i=1}^{N} x_i^r t^{r-1} \\ &= \sum_{i=1}^{N} \frac{x_i}{1-x_i t} = \frac{\mathrm{d}}{\mathrm{d}t}\sum_{i=1}^{N} \log(1-x_i t)^{-1} \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\log\left(\prod_{i=1}^{N}(1-x_i t)^{-1}.\right) \end{aligned}$$

The result for H follows by logarithmic differentiation. The argument for E is analog. □

By considering the coefficients of the power series, we can write this result in the form $nh_n = \sum_{r=1}^{n} p_r h_{n-r}$. This allows us to express the h_i in terms of the p_i, albeit at the cost of introducing denominators. For example, $h_2 = \frac{1}{2}(p_1^2 + p_2)$.

Therefore the p_λ generate as vector space and we get Theorem V.18 d); albeit not necessarily with integral coefficients.

We close this section with the mention of another basis of symmetric polynomials which is of significant practical relevance, but more complicated to define. Given a partition $\lambda = (\lambda_1, \lambda_2, \ldots)$, define

$$snum_\lambda(x_1,\ldots,x_n) = \det\begin{pmatrix} x_1^{\lambda_1+n-1} & x_2^{\lambda_1+n-1} & \cdots & x_n^{\lambda_1+n-1} \\ x_1^{\lambda_2+n-2} & x_2^{\lambda_2+n-2} & \cdots & x_n^{\lambda_2+n-2} \\ \vdots & \vdots & \ddots & \vdots \\ x_1^{\lambda_n} & x_2^{\lambda_n} & \cdots & x_n^{\lambda_n} \end{pmatrix}.$$

By the properties of the determinant, *snum* is invariant under all even permutations of the variables, but is not symmetric. It thus must be divisible by the Vandermonde determinant $\prod_{i<j}(x_i - x_j)$, and the quotient

$$s_\lambda(x_1,\ldots,x_n) = \frac{snum_\lambda(x_1,\ldots,x_n)}{\prod_{i<j}(x_i - x_j)}$$

is symmetric. We call the set of s_λ the *Schur polynomials*. They form another basis of Λ^n, but we will not prove this here.

Amongst the many interesting properties of Schur functions, one can show, for example, that in the expression $s_\lambda = \sum_\mu K_{\lambda\mu} m_\mu$, the coefficient $K_{\lambda\mu}$ (called a *Kostka Number*) is given by the number of semistandard tableaux (that is, we allow duplicate entry and enforce strict increase in columns, but allow equality or increase in rows) of shape λ. Also the base change matrix from $\{p_\lambda\}$ to $\{s_\lambda\}$ is the character table of the symmetric group S_n.

V.7 The Robinson-Schensted-Knuth Correspondence

From ghoulies and ghosties
And long-leggedy beasties
And things that go bump in the night,
Good Lord, deliver us!

Scottish Prayer
TRADITIONAL

We now investigate a miraculous connection between tableaux and permutations. For this we relax the definition of a standard tableau to not requires entries from $\{1, \ldots, n\}$, but allow larger entries (thus skipping some smaller numbers). It also will be convenient to consider permutations as sequences of images.

The crucial step is an *insert* routine that takes a tableau and a number and inserts the number in the tableau, adding one cell and moving some entries, so that the tableau property is preserved.

The routine proceeds by inserting the entry a in the first row. Let x be the smallest entry in this row for which $x \geq a$. If no such element exists, add a (in a new cell) at the end of the row. Otherwise a "bumps out" x and takes its place in this row. Then x is inserted into by the same process into the second row, possibly bumping out another number which then is inserted into the third row and so on.
Example: Suppose we want to insert 8 into the tableau

1	3	5	9	12	16
2	6	10	15		
4	13	14			
11					
17					

The smallest entry of the first row larger than 8 is 9, which gets bumped out, resulting in the row 1, 3, 5, 8, 12, 16 and entry 9 being inserted into the second row.

There 10 is the smallest entry larger than 9 and gets bumped out, resulting in row 2, 6, 9, 15.

Inserting 10 into row three bumps out 13. Finally 13 is added at the end of row four, obtaining the tableau

1	3	5	8	12	16
2	6	9	15		
4	10	14			
11	13				
17					

To be able to prove statements, we now describe this process in a formal way for a tableau $T = (T_{ij})$. For convenience we assume that the tableau is bordered by 0's to the top and left, and by ∞ to the bottom and right, so that T_{ij} is defined for all $i, j \geq 0$.

We also define a relation $\prec$ on entries of T by

$$a \prec b \text{ if and only if } a < b \text{ or } a = b = 0 \text{ or } a = b = \infty,$$

with this the property of being a tableau is given by the following characterization

$$T_{ij} = 0 \text{ if and only if } i = 0 \text{ or } j = 0; \tag{V.32}$$

$$T_{ij} \prec T_{i(j+1)} \text{ and } T_{ij} \prec T_{(i+1)j}, \text{ for all } i, j \geq 0. \tag{V.33}$$

We write $x \notin T$ if $x = \infty$ or $x \neq T_{ij}$ for all $i, j \geq 0$.

ALGORITHM V.34 (Insert): Let $T = (T_{i,j})$ a tableau and $x \notin T$ a positive integer. This algorithm transforms T into another tableau that contains x in addition to the elements of T and adds one cell in a position (s, t) determined by the algorithm.

1. [Input x.] Set $i := 1$, $x_1 := x$, and j smallest such that $T_{1j} = \infty$.

2. [Find x_{i+1}.] *(At this point $T_{(i-1)j} < x_i < T_{ij}$ and $x_i \notin T$.)* If $x_i < T_{i(j-1)}$ then decrease j and repeat the step. Otherwise set $x_{i+1} := T_{ij}$ and $r_i := j$.

3. [Replace by x_i.] *(Now $T_{i(j-1)} < x_i < x_{i+1} = T_{ij} \prec T_{i(j+1)}$, $T_{(i-1)j} < x_i < x_{i+1} = T_{ij} < T_{(i+1)j}$, and $r_i = j$.)* Set $T_{ij} := x_i$.

4. [At end?] *(Now $T_{i(j-1)} < T_{ij} = x_i < x_{i+1} \prec T_{i(j+1)}$, $T_{(i-1)j} < T_{ij} = x_i < x_{i+1} \prec T_{(i+1)j}$, $r_i = j$, and $x_{i+1} \notin nT$.)* If $x_{i+1} \neq \infty$ then increase i by 1 and return to step 2.

5. return $s := i$, $t := j$ and terminate. *(At this point $T_{st} \neq \infty$ and $T_{(s+1)t} = T_{s(t+1)} = \infty$.)*

Proof: The parenthetical remarks in the algorithm ensure that T remains a tableau at each step. □

The algorithm implicitly defines a "bumping sequence"

$$x = x_1 < x_2 < \cdots < x_s < x_{s+1} = \infty,$$

as well as column indices

$$r_1 \geq r_2 \geq \cdots \geq r_s = t.$$

We now consider a slight generalization of permutations, namely *two-line arrays* of integers A=

$$\begin{pmatrix} q_1 & q_2 & \cdots & q_n \\ p_1 & p_2 & \cdots & p_n \end{pmatrix} \tag{V.35}$$

such that $q_1 < q_2 \cdots < q_n$ and that all p_i are different. Permutations thus are the special case of $q_i = i$ and $\{p_1, p_2, \ldots, p_n\} = \{1, \ldots, n\}$.

The following algorithm takes such an array and produces a set of two tableaux of the same shape

ALGORITHM V.36 (Robinson-Schensted-Knuth): Given a two-line array of the form (V.35), the following algorithm produces two tableaux P, Q of the same shape with entries from the p_i, respectively the q_i:

1. Initialize P and Q to the empty tableau. Let $i := 1$.
2. Insert p_i into P, obtaining a position (s, t).
3. Place q_i into a new cell at position (s, t) in Q.
4. Increment i. If $i \leq n$ go to step 2. Otherwise terminate and return P and Q.

Proof: The proof of Algorithm V.34 shows that P is a tableau. Furthermore, the position s, t returned is always a valid position to add a cell to the Young diagram. As the entries of q_i are increasing this shows that Q always is a tableau as well. □

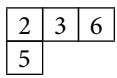
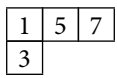
We abbreviate this process as RSK and note that if the array is permutation both P and Q are standard tableaux with entries $1, \ldots, n$.

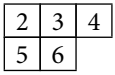
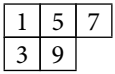
Example: We illustrate the RSK process with the example of the array $\begin{pmatrix} 1 & 3 & 5 & 7 & 9 \\ 5 & 2 & 3 & 6 & 4 \end{pmatrix}$.

Insert 5 and 1 $P =$  , $Q =$ 

Insert 2 and 3 $P =$  , $Q =$ 

Insert 3 and 5 $P =$  , $Q =$ 

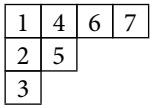
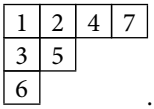
Insert 6 and 7 $P =$  , $Q =$ 

Insert 4 and 9 $P =$  , $Q =$ 

Amazingly this process is a proper bijection

THEOREM V.37 (RSK Correspondence): There is a bijection between the set of permutations of $\{1, \dots n\}$ and the set of ordered pairs (P, Q) of tableaux of the same shape, formed from $\{1, \dots, n\}$. Under this bijection, if g corresponds to the part (P, Q), then g^{-1} corresponds to (Q, P).

Example: The permutation $(1, 3, 2, 5, 4, 6) \in S_7$ has an array with second row $3, 5, 2, 6, 4, 1$ and gives the tableaux

$P =$  , $Q =$  .

Before proving this theorem we notice a number of consequences concerning the number f_λ of tableaux:

COROLLARY V.38:

$$\sum_{\lambda \vdash n} f_\lambda^2 = n!$$

Example: For $n = 4$ there are 5 partitions and respective tableau counts of $f_{1^4} = 1$, $f_{1^2 2^1} = 3$, $f_{2^2} = 2$, $f_{1^1 3^1} = 3$, $f_{4^1} = 1$. Also $1^2 + 3^2 + 2^2 + 3^2 + 1^2 = 24 = 4!$.

COROLLARY V.39: The number of involutions (permutations that are self-inverse, see II.6 for another formula) is given by $\sum_{\lambda \vdash n} f_\lambda$.

Proof: A permutation g is an involution if and only if $g = g^{-1}$. That means that involutions correspond to pairs with $P = Q$. □

Example: S_4 has one identity, three elements of shape $(1, 2)(3, 4)$ and $\binom{4}{2} = 6$ elements of shape $(1, 2)$, thus $10 = 1 + 3 + 2 + 3 + 1$ involutions.

Proof of the RSK correspondence

> The unusual nature of these coincidences
> might lead us to suspect that some sort
> of witchcraft is operating behind the scenes.
>
> The Art of Computer Programming,
> Section 5.1.4: Tableaux.
> DONALD E. KNUTH

The first part of the proof is to observe that Algorithm V.34 can be reversed:

We say that a cell of a tableau is "*at the edge*", if there is no cell to the right or below. The insertion algorithm adds a cell which is on the edge of the new tableau.

Now suppose we have a tableau into which an element x has been inserted, resulting in a new cell at the border of the resulting tableau P, say at coordinates (s,t). If we are in row $s = 1$, the entry of the cell is simply the element x.

Otherwise the entry x_s must have been bumped out of the previous row. It was bumped out by an element x_{s-1} and x_s was the smallest number in the row that was larger than x_{s-1}. That is, we can identify x_{s-1} in P as the largest number in row $s-1$ that is smaller than x_s. We thus can simply put x_s back into its old position and now have to deal with x_{s-1}. If $s - 1 = 1$ it was the element x inserted into the tableau, otherwise we repeat the process for $s - 2$ and so on.

We can consider this process as an "uninsert" algorithm that takes an (arbitrary) tableau P, a position (s,t) and returns a tableau P' missing a cell in position $s,t)$ as well as an element x. (It is easily see that at all steps the tableau properties are maintained.)

The description of the uninsert process also ensures that, after uninserting, we could obtain P back by inserting x into P' – insert and uninsert are mutual inverses.

Proof:(of Theorem V.37, part 1) To establish a bijection, we describe an inverse process to Algorithm V.36: Given two tableaux P, Q, we produce a two-line array A.

1. Let A be the empty two-line array.

2. Identify the position (s,t) of the largest entry q of Q. *((s,t) must be at the edge of Q.)* Remove that entry q from Q.

3. Apply the uninsert process to P for position (s,t), resulting in the removal of an element x.*(and reducing P to a tableau as the same shape as the reduced Q.)*

4. Add the column $\binom{q}{x}$ *at the front* of A.

5. Unless P and Q are empty, go back to step 2.

The result of this process is a two-line array A with the first row consisting of the numbers $1, 2 \ldots, n$ in canonical order, and the second row containing each of these numbers exactly once, that is a permutation.

It is easily seen that this process and the RSK algorithm are mutually inverse, thus establishing the bijection. □

To prove the second part, inversion corresponding to swap of the tableaux, we need further analysis of the RSK process:

If we consider the first row of the P-tableau, the construction process inserts some numbers, and bumps some out. Once a number has been bumped it will never affect row 1 again.

Furthermore, the lower rows of P and Q simply are tableaux corresponding to the "bumped" two-line arrays, that is two-line arrays that consist of the entries

bumped out of the first row and the associated q-entries. (This bumped array is the reason why we do not require the first row to be the canonical sequence 1, 2, ...)

In the example of the permutation $\begin{pmatrix} 1 & 3 & 5 & 7 & 9 \\ 5 & 2 & 3 & 6 & 4 \end{pmatrix}$ used above, we inserted 2, 3, 4 in the top row and bumped 5 and 6, resulting in the two-line array $A_{\text{bump}} = \begin{pmatrix} 3 & 9 \\ 5 & 6 \end{pmatrix}$ describing the second row and below.

This shows that the RSK algorithm can be split into two parts – constructing the first row only, and constructing the bumped two-line array. We consider both parts:

Construction of the first row To study the construction of the first row – remember that every entry of P is inserted first into this to row – we define:

Definition V.40: Consider a two-line array as given in (V.35). A column (q_i, p_i) of this array is *in class t*, if after inserting $p_1, \ldots, p_i$ into an empty tableau, p_i is in column t of the tableau.

Example: Looking at the example above, $(3, 2)$ and $(1, 5)$ are in class 1; $(5, 3)$ is in class 2; and $(7, 6)$ and $(9, 4)$ are in class 3. We can easily characterize the class of a column:

Lemma V.41: The column (q_i, p_i) belongs to class t if and only if t is the length of the longest increasing subsequence $p_{i_1} < p_{i_2} < \ldots < p_{i_t} = p_i$ ending at p_i.

Proof: For p_i to be in class t, that is inserted into column t, there must be $t-1$ elements in the first row that are smaller. These elements must be p_j's for $j < i$ showing that these p_j's together with p_i give an increasing subsequence of length t. We thus need to show the converse and will do so by induction on the length t of the longest increasing subsequence ending at p_i:

For $t = 1$, p_i is the smallest element and clearly will be inserted in column 1. Now assume that $t > 1$ and that the theorem holds for class $t-1$. Let p_j the predecessor of p_i in an increasing subsequence of length j. Then by induction p_j belongs to class $t-1$, that is p_j was inserted into the $t-1$-st column. That means that at the point of inserting p_i, the entry in this position is not larger than p_j, implying that p_i must be inserted in column t or higher. But if it was higher there would be an increasing subsequence of length $> t$, contradiction. □

Corollary V.42: The length of row 1 of the tableau is the length of a longest increasing subsequence of $\{p_i\}$.

Classes and bumped tableaux We observe that the tableaux resulting from the RSK process are completely determined by the columns of the two-line array and their classes, regardless of the order in which these columns are given:

First observe, that if $(q_{i_1}, p_{i_1}), \ldots, (q_{i_k}, pi_k)$ form the set of columns in one particular class, the way the insertion process and bumping works means that we can assume without loss of generality that the indices have been arranged so that $q_{i_1} < q_{i_2} < \cdots < q_{i_k}$ and simultaneously $p_{i_1} > p_{i_2} > \cdots > p_{i_k}$.

If we consider the first row of tableau P, the entry in column t must be the p-part of a column (q, p) in class t. Because of bumping, it will be the (smallest) value p_{i_k}. Similarly the corresponding entry in Q must be the q-part of such a column, but it is the q-part used when the entry is created for the first time, that is the minimal entry q_{i_1}. This describes the first row of P and Q fully.

For the further rows consider the bumped tableaux. To describe these it is sufficient to see what becomes of columns within class t (as entries from different classes never bump one another). The entry p_{i_1} gets bumped by p_{i_2} which comes with q-part q_{i_2}, resulting in a column (q_{i_2}, p_{i_1}). Similarly p_{i_2} gets paired with q_{i_3} and so on, giving the (partial) two-line array arising from class t as

$$A_{\text{bump}} = \begin{pmatrix} q_{i_2} & q_{i_3} & \cdots q_{i_k} \\ p_{i_1} & p_{i_2} & \cdots p_{i_{k-1}} \end{pmatrix}. \tag{V.43}$$

We then proceed from the bumped array anew to obtain new classes and from these the second rows of P and Q and so on.

With this observation, we are now able to prove the second part of the theorem, concerning the swap of P and Q. As the two tableaux are constructed by very different methods, this is a most surprising symmetry.

Proof:(of Theorem V.37, part 1) On the level of two-line arrays, inversion means to swap the two rows and sort the columns according to the entries in the first row. Denote this array by

$$A^- = \begin{pmatrix} p'_1 & p'_2 & \cdots p'_n \\ q'_1 & q'_2 & \cdots q'_n \end{pmatrix}. \tag{V.44}$$

If we ignore the arrangement of columns, Lemma V.41 states that a column (q_i, p_i) is in class t if and only if t is the maximal size of an index set $\{i_1, \ldots, i_t\}$ such that

$$q_{i_1} < q_{i_2} < \cdots < q_{i_t} \quad \text{and}$$
$$p_{i_1} < p_{i_2} < \cdots < p_{i_t}.$$

This means that (q, p) is of class t in (V.35), if and only if (p, q) is of class t in (V.44).

That implies that the RSK algorithm applied to A^- produces tableaux whose first rows are exactly swapped from the tableaux produced for the original two-line array A.

Furthermore, following the formula (V.43) for the bumped array – within one class, remove the smallest p and the smallest q entries and "shift the entries together" – we have that A^-_{bump} is obtained from A^-_{bump} by swapping the two rows and sorting according to the first row.

Applying the same argument again thus gives us that the second rows etc. also are swapped, as was to be proven. □

There is much more to the RSK construction. For example *reverting* the permutation (as a sequence of numbers) results in a transposition of P (but a somewhat more complicated transformation of Q namely the transpose of a transformed tableau.)

Chapter

VI

Symmetry

Tyger Tyger, burning bright,
In the forests of the night;
What immortal hand or eye,
Could frame thy fearful symmetry?

The Tyger
William Blake

The use of symmetries is one of the tools of science that crosses into many disciplines. So far we have only touched upon this, in a rather perfunctory way, by considering objects as identical or non-identical, or by counting labeled versus non-labeled objects. In this chapter we shall go into more detail. In combinatorics, there are at least three uses of symmetry:

- We might want to consider objects only up to symmetry, thereby reducing the total number of objects. For example when considering bracelets made from 10 green and one golden pearl, we want to consider this is one bracelet, rather than 11 possible positions of the golden pearl. We will look at counting up to symmetry in Section VI.5.
- Symmetries can be used to classify what is sometimes a huge number of objects into equivalence classes. For example, relabeling points in a permutation preserves the cycle structure.
- Finally, in a related way, we might want to use symmetries as a proxy for objects being interesting. When combining parts, there might be a huge number of possible objects. For example there are

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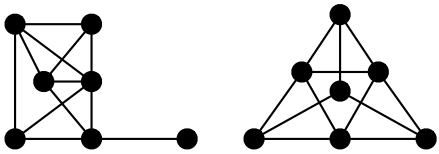


Figure VI.1: Two graphs with different symmetries

different graphs on 20 vertices. But, be it graphs or other objects, it turns out that many of them are just stuck together from smaller objects and do not amount to more than the sum of their parts. On the other hand symmetries that do not keep the parts fixed often indicate something interesting. Figure VI.1 shows two graphs with 7 vertices and 12 edges. Clearly the graph on the right (which has 5 nontrivial symmetries) is more interesting than the left one (with only trivial symmetries).

Of the formal setting in mathematics for symmetries is a group. Instead of going all abstract, we will consider first, what we want symmetries to be, and then get the formal axioms as an afterthought. So, what are symmetries? If we have an object (which can be a set of smaller objects), a symmetry is a map from the object to itself, may be mixing up the constituent parts, that preserves some structural properties.

VI.1 Automorphisms and Group Actions

> Nothing will make me believe that a man who arranged all the rest of that room with that exaggerated symmetry left that one feature of it lopsided.
>
> The Worst Crime in the World
> G. K. Chesterton

We call such a map an automorphism, from the greek autos for self and morphe for shape. There always is at least one such map, namely the identity which doesnfbt do anything. In an abuse of notation we shall write 1 for the identity map. By preserving the larger collection, the maps have to be bijective, and thus afford an inverse. Finally we can compose these maps, that is apply one after the other. As always, composition of maps is associative. we thus get the formal axioms of a group, though this axiomatic theory shall be of little concern to us. Following conventions introduced in [Wie64], we denote groups by upper case letters, group elements by lower case letters, and objects the group acts on by greek letters. If the objects we act on have numbered parts or are themselves numbered, it can be convenient to

represent maps as permutations. Indeed, if we act on a set of objects with numbered parts, permutations of these numbers yield automorphisms.

Formally, we can distinguish maps and the permutations induced by them, in this case an action of a group G on n numbered objects yields a group homomorphism $G \to S_n$, assigning to every map the permutation induced by it.

We shall write the image of a point ω under a map g as ω^g. That is our groups act on the right. That way the image of Omega under the product of g with h is simply gh. Similarly permutations multiply as $(1,2,3)\cdot(2,3) = (1,3)$. (While this is clearly the natural way for languages that are written from left to right, the reader who had abstract algebra might notice that in some classes the convention used is an action from the left, which then requires the introduction of a special product operation for composition of maps.)

In some situations we might want to consider what maps do to parts, collections, or structures derived from the original objects. For this, it is most convenient to not consider new maps, but a new action of the original maps. The rules for this are simply that the identity must act trivially, and that products act from left to right, that is

$$\omega^1 = \omega \quad \forall \omega \in \Omega, \qquad \text{and} \qquad \omega^{(gh)} = \left(\omega^g\right)^h \quad \forall \omega \in \Omega \forall g, h \in G$$

Such an action, given by a group G, a domain Ω, and a map $\Omega \times G \to \Omega$, $(\omega, g) \mapsto \omega^g$, is called a group action. Group actions are the basic objects we shall consider.

Examples

Before delving into further details of the theory, lets consider a few examples.

The set of all permutations of n points form a group, called the *symmetric group* S_n. It acts on the points $\{1, \dots n\}$, but we also can have it act on opther structures that involve these numbers, for example all subsets of size k, sequences of length k, or partitions with k cells.

If we consider a bracelet with 6 pearls as depicted on the left in Figure VI.2, then automorphisms are given for example by a rotation along the bracelet (which we shall denote by the permutation $(1,2,3,4,5,6)$), or by flips along the dashed reflection axes. For example, the flip along the axis going between 1 and 2 is given by the permutation $(1,2)(3,6)(4,5)$. Together with the trivial rotation this gives $1+5+6 = 12$ symmetries. We also observe that there are 6 positions where 1 can go, and then 2 positions where 2 goes with which an automorphism is fully determined. The whole symmetry group thus has order 12. (For those who had abstract algebra, it is the dihedral group of order 12.)

Next consider symmetries of a cube that are physically possible without disassembling, that is only rotations and no reflections. (However in other cases one might make another choice, indeed when talking about automorphisms one needs to decide what exacty is to be preserved.) Besides the identity there are 3 rotations each for the three pairs of opposite faces, 2 rotations for each of the 4 diagonal axes,

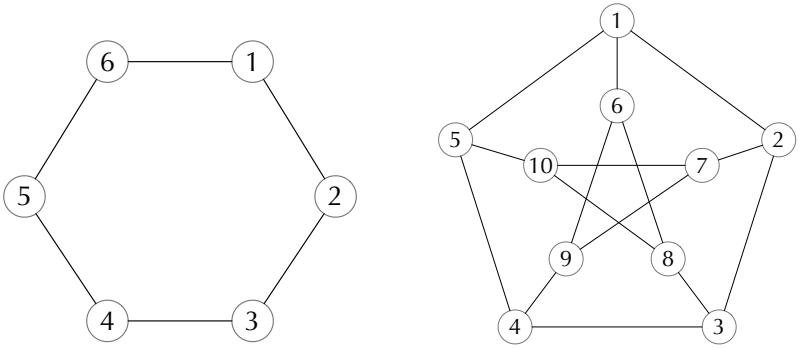


Figure VI.2: A circle graph and the Petersen graph

as well as 1 rotation on axes that go through the middle points of 6 pairs of opposite faces, for a total of $1+9+8+6 = 24$ symmetries. (Again, the images of neighboring faces gives us at most $6 \cdot 4 = 24$ possibilities, thus we found all.)

The set of automorphisms of an object X is called its *automorphism group* and denoted by $\mathrm{Aut}(X)$. (As noted that might depend on what structure is to be preserved. The automorphism group of the 6 faces would be S_6 if we did not want to preserve the cube structure.)

It is not always that easy to determine the number of automorphisms (also called the order of the automorphism group). Take the graph on the right side of Figure VI.2 (which is called the Petersen-graph). An automorphism of a graph is a permutation of the vertices that preserves adjacency. By checking this property, it is easy to verify that the permutation $(1,2,3,4,5)(6,7,8,9,10)$ is an automorphism, as is (with a bit more effort) $(3,7)(4,10),(8,9)$. But is is not trivial to show that there are 120 automorphisms in total, and that any automorphism can be formed as a suitable product of the two permutations given. We will look at this in Exercise **??**.

For an example with a larger set of objects, consider the order-signifcant partitions of n into k parts and S_k acting by permuting the parts.

Finally, we consider the set of all possible graphs on three vertices, labelled with 1, 2, 3, 4, as depticted in Figure VI.3. There are $\binom{3}{2} = 3$ possible edges and thus $2^3 = 8$ different graphs. The groul S_3 acts on this set of graphs by permuting the vertices. (When doing so, the *number of edges* (and more — On four vertices we cannot map a triangle to a Y) is preserved, so not every graph may be mapped to any other.

In general, for two given graphs on n (labelled) vertices, a permutation in S_n that maps one graph to the other is called a *graph isomorphism*, if such a map exists the graphs are called *isomorphic*.

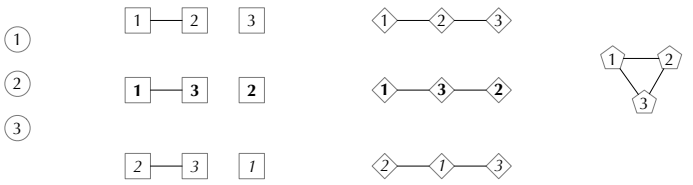


Figure VI.3: The labelled graphs on 3 vertices.

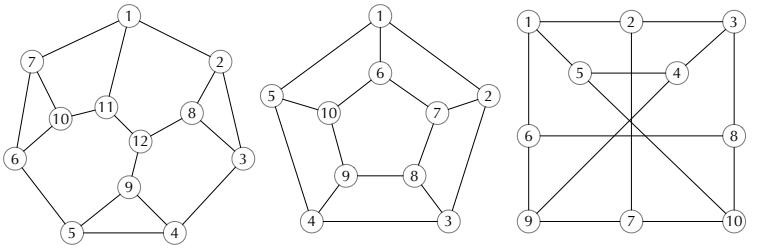


Figure VI.4: Three nonisomorphic graphs that are regular of degree 3.

Note VI.1: The question of determining graph isomorphisms is a prominent [**?**] and hard problem. While local properties — number of edges or degrees of vertices — can eliminate the existence of automorphisms, one can show that for any set of local properties, there are graphs in different orbits which agree on all of the invariants.

Figure VI.4 shows three graphs, each of them regular of degree 3. The left graph has trivial automorphism group, the middle graph an automorphism group of order 20. The right graph is in fact, with the given labelling, isomorphic to the Petersen graph (with automorphism group of order 120), though this isomorphism is not obvious without the labelling – there are 10! possible maps, only 120 of them, that is 1 in 30240 induce an isomorphism.

We close this section with the definition of an equivalence of actions:

Definition VI.2: Let G be a group acting on a set Ω and H a group acting on a set Δ. We call these actions *equivalent (actions)*, if there is a group isomorphism $\alpha\colon G \to H$ and a bijection $\psi\colon \Omega \to \Delta$, such that

$$\psi(\omega^g) = \psi(\omega)^{\alpha(g)}$$

That means if a group G acts on a set Ω of size n, inducing a homomorphism $\varphi\colon G \to S_n$, the action of G on Ω and the action of $\varphi(G)$ on $\{1, \ldots, n\}$ are equivalent.

Orbits and Stabilizers

> Ever he would wander, selfcompelled, to the extreme limit of his cometary orbit.
>
> Ulysses
> James Joyce

The two main concepts for working with group actions are orbits and stabilizers.

Definition VI.3: Let G act on Ω and $\omega \in \Omega$.
a) The *Orbit* of ω is the set of all possible images:

$$\omega^G = \{\omega^G \mid g \in G\}.$$

b) The *stabilizer* of ω is

$$\mathrm{Stab}_G(\omega) = \{g \in G \mid \omega^G = \omega\}.$$

Since group elements are invertible, the orbits of a group form equivalence classes. They represent "equivalence up to symmetry". If we want to describe objects up to symmetry, we really ask to describe the orbits of an appropriate symmetry group.
Example: Determine orbits in the examples of the previous section?

Definition VI.4: The action of G on Ω is *transitive* if there is only one orbit, that is $\Omega = \omega^G$ for (arbitrary) $\omega \in \Omega$. An element $\omega \in \Omega$, that is the only one in its orbit — i.e. $\omega^G = \{\omega\}$ — is called a fixed point of G.

The stabilizer of a point is closed under products and inverses and thus always forms a group itself. Indeed,the automorphism group of an object can be considered as a stabilizer of the object in a larger "full" symmetry group: In the case of the bracelet we can consider the set of all permutations of the pearls and want only those that preserve the neighbor relations amongst pearls. The automorphism group of a cube (centered at the origin) is the stabilizer of the cube in the orthogonal group of 3-dimensional space.

The key theorem about orbits and stabilizers is the bijection between orbit elements and stabilizer cosets. Two elements g, h map ω in the same way if and only if their quotient lies in the stabilizer:

$$\omega^g = \omega^h \Leftrightarrow \omega = \omega^{(g \cdot g^{-1})} = (\omega^g)^{g^{-1}} = (\omega^h)^{g^{-1}} = \omega^{h/g} \Leftrightarrow h/g \in \mathrm{Stab}_G(\omega).$$

This means that if g is one element such that $\omega^g = \delta$ then the set of all elements that map ω in the same way is (what is called a *coset*):

$$\mathrm{Stab}_G(g) \cdot g = \{s \cdot g \mid s \in \mathrm{Stab}_G(\omega)\}.$$

For any orbit element $\delta \in \omega^G$ there therefore are exactly $|\mathrm{Stab}_G(\omega)|$ elements of G that map ω to δ. We therefore have

THEOREM VI.5 (Orbit-Stabilizer): There is a bijection between elements of ω^G and cosets of $\mathrm{Stab}_G(\omega)$. In particular:

$$|G| = \left|\omega^G\right| \cdot |\mathrm{Stab}_G(\omega)|.$$

In particular, this means that the length of an orbit must divide the order of the acting group.

Example: A group of order 25, acting on 12 points thus must have fixed points.

We note some consequences from this bijection. They show that the possible transitive actions of a group G are prescribed by its structure:

LEMMA VI.6: Let G act on Ω and $\omega, \delta \in \Omega$ with $g \in G$ such that $\omega^g = \delta$. Then

a) $\mathrm{Stab}_G(\omega)^g = \mathrm{Stab}_G(\delta)$.

b) The set of elements of G mapping ω to δ is

$$\mathrm{Stab}_G(\omega) \cdot g = g \cdot \mathrm{Stab}_G(\delta) = \mathrm{Stab}_G(\omega) \cdot g \cdot \mathrm{Stab}_G(\delta).$$

c) The action of G on the cosets of $S = \mathrm{Stab}_G(\omega)$ by $(Sx, g) \mapsto S(xg)$ is compatible with the bijection: If $\omega^x = \delta$, $\omega^y = \gamma$ and $\delta^g = \gamma$ then $S(xg) = Sy$.

In cases such as c), of a bijection between sets being compatible with group actions (maybe even involving an isomorphism of groups) we talk about an equivalence (or isomorphism) of group actions.

We now define a directed graph that describes the action of certain elements $A \subset G$ on Ω. (Typically A will be a generating set for G, that is every element of G is a product of elements of A and their inverses.)

DEFINITION VI.7: Let $\mathrm{Sch}_A(\Omega) = (\Omega, E)$ the digraph with vertex set Ω and edges $E \subset \Omega \times \Omega$ labelled by elements of A: We have that $(\omega, \delta) \in E$ if and only if there exists an $a \in A$ such that $\omega^a = \delta$. In this case we label the edge (ω, δ) with a. We call $\mathrm{Sch}_A(\Omega)$ the *Schreier graph*[1] for the action of A on Ω.

For example, Figure VI.5 gives the Schreier graph for the action of S_4, generated by the permutations $a = (1, 2)$ and $b = (1, 2, 3, 4)$ on sets of order 2 (which form just one orbit).

By tracing paths between vertices and multiplying up the elements of A on the edges (inverses, if we go an arrow in reverse), we obtain a factorization of an element, mapping one point to another, as a word in A. In the example above, we can read off that b^2 will map $\{2, 3\}$ to $\{1, 4\}$, as would aba, b^{-2}, bab, and so on.

Puzzles, such as Rubik's cube are essentially just about finding such factoriations, in examples in which the group is far too large to draw the Schreier graph.

[1] after OTTO SCHREIER, 1901-1929

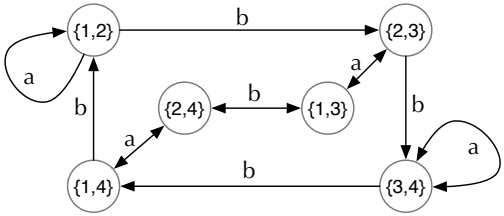


Figure VI.5: Schreier graph for the action of $\langle a = (1,2), b = (1,2,3,4)\rangle$ on pairs.

We can determine the orbit of a point as the connected component of the Schreier graph, starting from a vertex and continuing to add images under group generators, until no new images are found.images under group generators, until no new images are found. (One needs to store only vertices encountered this way, not the edges.) A spanning tree of a connected Schreier graph is called a Schreier tree, it is easy to see that often there will be many possible Schreier trees, of different depth.

We note that loops in the Schreier graph images yield stabilizer elements, that is if $\omega^g = \delta$, $\delta^a = \gamma$, and $\omega^h = \gamma$, then ga and h are in the same coset of $S = \mathrm{Stab}_G(\omega)$ and $ga/h \in S$. Schreier's theorem (which we shall not prove here) shows that if we do so systematically for all loops that get created from elements of A, we obtain a generating set for $\mathrm{Stab}_G(\omega)$.

This in fact provides the tool — called the Schreier-Sims algorithm — by which computers calculate the order of a permutation group: They calculate the orbit of a point and generators for its stabilizer, then recurse to the stabilizer (and another point). By the Orbit-Stabilizer theorem the group order then is the product of the orbit lengths.

Example: An icosahedron has 20 faces that are triangle shaped. Rotations can map any face to any other face, yielding an orbit of length 20. The stabilizer of a face can clearly have only 3 rotations, yielding a rotational automorphism group of order 60. Similarly we obtain a rotation/refelction automorphism group of order 120.

VI.2 Cayley Graphs and Graph Automorphisms

> Spread oot, lads, and pretend ye're enjoying the cailey
>
> The Wee Free Men
> Terry Pratchett

Definition VI.8: An action of G on Ω is *semiregular* if the stabilizer of any $\omega \in \Omega$ is trivial. If the action is also transitive, it is called *regular*.

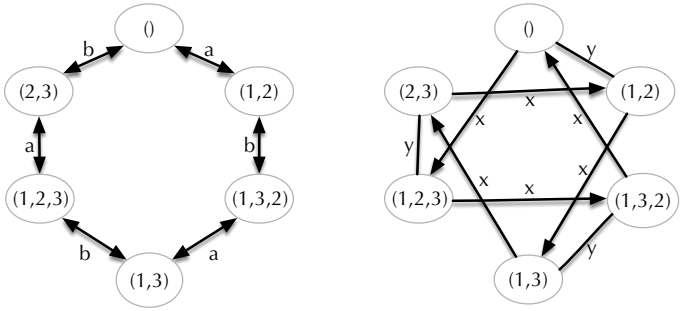


Figure VI.6: Two Cayley graphs for S_3.

By Lemma VI.6, a regular action of G is equivalent to the action on the cosets of the trivial subgroup, that is the action of G on its elements by right multiplication: $(x, g) \mapsto xg$. If we have a generating set A of G, the Schreier graph for this action gets a special name, it is called the *Cayley graph* of G (with respect to A).

Figure VI.6 shows the Cayley graphs for S_3 for the generating set $a = (1, 2)$, $b = (2, 3)$, as well as for the generating set $x = (1, 2, 3)$, $y = (1, 2)$.

We notice (Exercise **??**) that G induces automorphisms of the Cayley graph $\mathrm{Cay}_A(G)$. To be consistent with the way we defined actions and with the labelling of edges, the action of $g \in G$ is defined by $(v, g) \mapsto g^{-1}v$ for a vertex $v \in G$. We denote this automorphism by μ_g.

However Cayley graphs can, as graphs that ignore labels and directions, have other automorphisms. For example, the graph in Figure VI.6, left, has a dihedral group of order 12 of symmetries, no all of which make sense for S_3 (a rotation for example would change element orders).

Such extra automorphisms however will not preserve edge labels:

LEMMA VI.9: Let $\Gamma = \mathrm{Cay}_A(G)$ and $\alpha \in \mathrm{Aut}(\Gamma)$. Then α preserves the labels of all edges, if and only if there exists $g \in G$ such that $\alpha = \mu_g$, that is $v^\alpha = g^{-1}v$ for all $v \in G$.

Proof: If $x \cdot a = y$ then $g^{-1} \cdot (x \cdot a) = g^{-1} \cdot y$. Thus all μ_g preserve the edge labels. Let 1 be the identity of G and $g := 1^\alpha$. Then $\alpha \cdot \mu_g$ is an automorphism that maps 1 to 1. Furthermore (as μ_g preserves edge labels), $\alpha \cdot \mu_G$ preserves edge labels if and only if α does. We thus may assume, without loss of generality, that $1^\alpha = 1$. (We shall show that in this case α must be the identity map that is μ_1. As $\alpha \cdot \mu_g = \mu_1$ we get that $\alpha = \mu_{g^{-1}}$.)

By definition, for every $a \in A$, there exists a unique vertex (namely the vertex a) that is connected by an edge from 1 with label a. That means that this edge must be fixed by α and thus $a^\alpha = a$. The same argument for edges *to* 1 shows that a^{-1} must be fixed by α for any $a \in A$. This shows that every vertex at distance one from

1 must be fixed by α.

The same argument now can be used by induction to show that all elements at distance $2, 3 \ldots$ from 1 must be fixed by α. As A is a generating set of G, this is all elements, thus α must be the identity, completing the proof. □

This result implies that all finite groups can be considered as the automorphism group of a suitable graph.

THEOREM VI.10 (FRUCHT): Let G be a finite group. Then there exists a finite graph Γ such that $\mathrm{Aut}(\Gamma) \cong G$.

Proof: We have seen already that every group is the automorphism group of a directed graph with labelled edges, for example its Cayley graph. What we will simply do is to simulate direction and labelling by extra vertices and edges. First for direction: For each directed edge $a \to b$, we introduce three new vertices x, y, z and replace the directed edge by undirected edges $a - x - y - b$, as well as $y - z$. That is, a and b are still connected, and z indicates the arrow direction. The z-vertices of this new graph are the only ones of degree 1, so the automorphisms must map a z-vertex to a z-vertex, and thus an y-vertex to a y-vertex. We also add a "universal" vertex $\underline{X}$ which we connect to all x-vertices. Then (ensured if necessary by adding further vertices we connect to $\underline{X}$) the vertex $\underline{X}$ is the only vertex of its degree and thus must remain fixed under any automorphism, implying that the x-vertices must be permuted amongst each other as well. Thus the "original" vertices of the graph cannot be mixed up with any of the newly introduced vertices representing edges. Also distance arguments show that each group of $x/y/z$ vertices must stay together as a group. The automorphisms of this new undirected graph thus are the same as the automorphisms of the original directed graph.

To indicate edge label number i, we insert $i - 1$ extra vertices into the $y - z$ edges, replacing $y - z$ by $y - z_1 - z_2 - c \cdots - z_{i-1} - z$. Automorphisms then need to preserve these distances and thus are restricted to preserving the prior labelling. □

(Frucht in fact proved a stronger theorem, namely that every finite group is the automorphism group of a 3-regular graph, i.e. a graph in which every vertex has degree 3.)

VI.3 Permutation Group Decompositions

Permutation:
The theory of how hairdos evolved

The New Uxbridge Dictionary
BROOK-TAYLOR ET.AL.

Finding the automorphism group of an object can be hard. In many practically relevant cases, however the objects have substructures that need to be preserved

by automorphisms. In this case we can relate automorphisms of the structure with automorphisms of the substructures in a meaningful way.

In this section we shall look at two kinds of decompositions that lend themselves to nice group theoretic structures.

By labelling relevant parts of the object acted on by G we can assume without loss of generality that the automorphism group group on a set Ω.

We start with the situation that Ω can be partitioned into two subsets that must be both preserved as sets, that is $\Omega = \Delta \cup \Lambda$ with $\Delta \cap \Lambda = \varnothing$. This could be distinct objects — say vertices and edges of a graph — or objects that structurally cannot be mapped to each other – for example verticed of a graph of different degree. (The partition could be into more than two cells, in which case one could take a union of orbits and iterate, taking the case of two as base case.)

In this case G must permute the elements of Ω amongst themselves, as well as the elements of Λ. We thus can write every element of G as a product of a permutation on Δ and a permutation on Λ. Since the sets are disjoint, these two permutations commute.

Group theoretically, this means that we get a homomorphism $\varphi\colon G \to S_\Delta$, as well as a homomorphism $\psi\colon G \to S_\Lambda$. The intersection $\operatorname{Kern}\varphi \cap \operatorname{Kern}\psi$ is clearly trivial, as elements of G are determined uniquely by their action on $\Omega = \Delta \cup \Lambda$. We thus can combine these maps to an embedding (an injective group homomorphisms)

$$\iota\colon G \to S_\Delta \times S_\Lambda, \qquad g \mapsto (\varphi(g), \psi(g))$$

If we consider the direct product as acting on the disjoint union of Δ and Λ, this map is simply the identity map on permutations.

The images of φ and ψ are typically not the whole symmetric groups. Thus, setting $A = \varphi(G)$ and $B = \psi(G)$, we can consider G as a subgroup of $A \times B$, and call it a *subdirect product* of A and B. All intransitive permutation groups are such subdirect products.

Note that a subdirect product does not need to be a direct product itself, but can be a proper subgroup, see Exercise **??**. There is a description on possible groups arising this way, based on isomorphic factor groups. Doing this however requires a bit more of group theory than we are prepared to use here.

We also note that it is not hard to classify subdirect products abstractly, and that this indeed has been done to enumerate intransitive permutation groups.

Block systems and Wreath Products

The second kind of decomposition we shall investigate is the case of a group G that is transitive on Ω, but permutes a partition into subsets (which then need to be of equal size).

For example, in the two graphs in Figure VI.7, one such partition would be given by vertices that are diagonally opposed. Neither diagonal is fixed, but a diagonal must be mapped to a diagonal again.

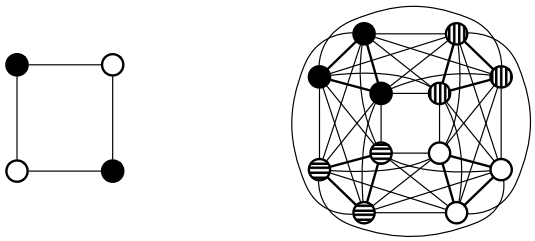


Figure VI.7: Two graphs with an imprimitive automorphism group.

In the graph on the right, constructed from four triangles that are connected in all possible ways with the two neighbors, such a partition would be given by the triangles.

We formalize this in the following definition:

Definition VI.11: Let G act transitively on Ω. A *block system* (or *system of imprimitivity*) for G on Ω is a partition $\mathcal{B}$ of Ω that is invariant under the action of G.

Example: For example, taking $G = \mathrm{GL}_n(F)$ for a finite field F, acting on the nonzero vectors of F^n, the sect of vectors that span the same 1-dimensional space, that is those that are equivalent under multiplication by F^*, form a block system.

We note a few basic properties of block systems, the proof of which is left as an exercise.

Lemma VI.12: Let G act transitively on Ω with $|\Omega| = n$.

a) There are two *trivial* block systems, $\mathcal{B}_0 = \{\{\omega\}\}_{\omega\in\Omega}$, as well as $\mathcal{B}_\infty = \{\Omega\}$.

b) All blocks in a block system must have the same size.

c) If $\mathcal{B}$ is a block system, consisting of a blocks of size b, then $n = ab$.

d) If $\mathcal{B}$ is a block system, then G acts transitively on the blocks in $\mathcal{B}$.

e) A block system is determined uniquely by one of its blocks.

We note — See section **??** — that part a) of the lemma is the best possible – there are groups which only afford the trivial block systems.

A connection between blocks and group structure is given by the following proposition which should be seen as an extension of Theorem VI.5:

Proposition VI.13: Let G act transitively on Ω and let $\omega \in \Omega$ with $S = \mathrm{Stab}_G(\omega)$. There is a bijection between block systems of G on Ω and subgroups $S \leq U \leq G$.

Proof: We will establish the bijection by representing each block system by the block B containing ω. For a block B with $\omega \in B$, let $\mathrm{Stab}_G(B)$ be the set-wise stabilizer

of B. We have that $S \leq \mathrm{Stab}_G(B)$, as S maps ω to ω and thus must fix the block B. Since G is transitive on Ω there are elements in G that map ω to an arbitary $\delta \in B$, as B is a block this means that these elements must lie in $\mathrm{Stab}_G(B)$. This shows that $\mathrm{Stab}_G(B)$ acts transitively on B and that $B = \omega^{\mathrm{Stab}_G(B)}$. The map fro blocks to subgroups therefore is injective.

Vice versa, for a subgroup $U \geq S$, let $B = \omega^U$. Clearly B is, as a set, stabilized by U. We note that in fact $U = \mathrm{Stab}_G(B)$, as any element $x \in \mathrm{Stab}_G(B)$ must map ω to $\omega^x inB$ and by definition there is $u \in U$ such that $\omega = (\omega^x)^u = \omega^{xu}$, so $xu \in S \leq U$ and therefore $x \in U$. Thus the map from subgroups containing S to subsets containing ω is injective.

We claim that B is the block in a block system. Since G is transitive, the images of B clearly cover all of Ω. We just need to show that they form a partition. For that, assume that $B^g \cap B^h$. This means that $\delta^{g^{-1}}, \delta^{h^{-1}} \in B = \omega^U$ and thus there exists $u_g, u_h \in U$ such that $\omega^{u_g g} = \delta = \omega^{u_h h}$. But then $u_g g (u_h h)^{-1} = u_g g h^{-1} 1 u_h^{-1} \in \mathrm{Stab}_G(\omega) \leq U$, the $gh^{-1} \in U$ and (as $U = \mathrm{Stab}_G(B)$) we have that $B^g = B^h$. This shows that the images of B form a partition of Ω.

The properties shown together establish the bijection. □

Corollary VI.14: The blocks systems for G on Ω form a lattice under the "subset" (that is blocks are either subsets of each other or intersect trivially) relation. Its maximal element is $\mathcal{B}_\infty$, its minimal element is $\mathcal{B}_0$.

We call a transitive permutation group *imprimitive*, if it affords a nontrivial block system on Ω. We want to ontain an embedding theorem for imprimitive groups, similar to what we did for direct products. That is, we want to describe a "universal" group into which imprimitive groups embed.

The construction for this is called the wreath product"

Definition VI.15: Let A be a group, b an integer, and $B \leq S_b$ a permutation group. The *wreath product* $A \wr B$ is the semidirect product of $N = A^b = \underbrace{A \times \cdots \times A}_{b}$ with B where B acting on N by permuting the components.

If $A \leq S_a$ is also a permutation group, we can represent $A \wr B$ as an imprimitive group on $a \cdot b$ points: Consider the numbers $1, \ldots, ab$, arranged as in the following diagram.

1	2	$\cdots$	a
$a+1$	$a+2$	$\cdots$	$2a$
		$\vdots$	
$a(b-1)+1$	$a(b-1)+2$	$\cdots$	ab

Then the direct product $N = A^b$ can be represented as permutations of these numbers, the i-th copy of A acting on the i-th row. Now represent the permutations of B by acting *simultaneously* on the columns, permuting the b rows. The resulting group W is the wreath product $A \wr B$. Points in the same row of the diagram will

be mapped by W to points in the same row, thus W acts imprimitively on $1, \ldots, ab$ with blocks according to the rows of the diagram. W is therefore called the *imprimitive action* of the wreath product.

LEMMA VI.16: Let G act imprimitively on $1, \ldots n = ab$ with b blocks of size a. Then G can be embedded into a wreath product $S_a \wr S_b$ in

Proof: By renumbering the points we may assume without loss of generality that the blocks of G are exactly $\{1, \ldots, a\}$, $\{a+1, \ldots, 2a\}$ and so on, as given by the rows of the above diagram. Let $g \in G$. Then g will permute the blocks according to a permutation $b \in S_b$. By considering S_b as embedded into $S_a \wr S_b$, we have that g/b fixes all rows of the diagram as sets and thus is in $N = (S_a)^b$. □

(Again, it is possible — for example under the name of *induced representations* — to give a better description, reducing to a wreath product of the block stabilizers action on its block with the groups action on all the blocks, but doing so requires a bit more work.)

VI.4 Primitivity and Synchronization

In action be primitive; in foresight, a strategist.

Agir en primitif et prévoir en stratège.

Feuillets d'Hypnos #72
RENÉ CHAR

A permutation group $G \leq S_\Omega$ is called *primitive*, if the only block systems it affords are the trivial ones $\mathcal{B}_0$ and $\mathcal{B}_\infty$. Since block sized need to divide the degree, every transitive group of prime degree is primitive.

Example in arbitrary degree are given by symmetric groups:
Example: For $n > 1$, the symmetric group S_n is primitive: Assume that B is a block in a nontrivial block system, then $2 \leq |B| \leq n-1$. Thus there are $\omega, \delta \in B$ and $\gamma \in \Omega \setminus B$. But there is $g \in S_n$ such that $\omega^g = \omega$ and $\delta^g = \gamma$, thus B is mapped by g to a set that has proper, nontrivial, intersection with B contradicting the block property.

This argument in fact only requires that the action on pairs of points is transitive — map $\{\omega, \delta\}$ to $\{\omega, \gamma\}$, giving the following definition and lemma.

DEFINITION VI.17: Let G be a permutation group that is transitive on Ω.
a) If G acts transitively on all k-tuples of distinct elements of Ω, then G is called k-transitive.
b) If G acts transitively on all k-sets of elements of Ω, then G is called k-homogeneous.

LEMMA VI.18: a) If G is k-transitive for $k > 1$ then G is also $k-1$-transitive

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Some Counting Sequences

Bell OEIS A000110, page 27
B_0=1, B_1=1, B_2=2, B_3=5, B_4=15, B_5=52, B_6=203, B_7=877, B_8=4140, B_9=21147, B_{10}=115975

Catalan OEIS A000108, page 22
C_0=1, C_1=1, C_2=2, C_3=5, C_4=14, C_5=42, C_6=132, C_7=429, C_8=1430, C_9=4862, C_{10}=16796

Derangements OEIS A000166, page 24
$d(0)$=1, $d(1)$=0, $d(2)$=1, $d(3)$=2, $d(4)$=9, $d(5)$=44, $d(6)$=265, $d(7)$=1854, $d(8)$=14833

Involutions OEIS A000085, page 31
$s(0)$=1, $s(1)$=1, $s(2)$=2, $s(3)$=4, $s(4)$=10, $s(5)$=26, $s(6)$=76, $s(7)$=232, $s(8)$=764

Fibonacci OEIS A000045, page 16
F_0=1, F_1=1, F_2=2, F_3=3, F_4=5, F_5=8, F_6=13, F_7=21, F_8=34, F_9=55, F_{10}=89, F_{11}=144

Partitions OEIS A000041, page 67
$p(0)$=1, $p(1)$=1, $p(2)$=2, $p(3)$=3, $p(4)$=5, $p(5)$=7, $p(6)$=11, $p(7)$=15, $p(8)$=22, $p(9)$=30